Reputation-Based Consensus in Decentralised Systems

A thesis submitted in fulfilment of the requirements for the degree of Doctor of Philosophy at King's College London

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One can have, some claim,
as many electronic personas as one has time and energy to create.

– Donath, J. S. (1999). Identity and deception in the virtual community. In P. Kollock & M. Smith (Eds.), *Communities in Cyberspace*. Routledge, p. 27

Abstract

The emergence of permissionless decentralised systems, as exemplified in numerous public blockchains, presents a novel challenge in the field of distributed systems research, since these systems operate without a central control mechanism. Because such systems originate from circles intent on reversing the power relationship between government and citizens, some have assumed that they are better able than centralised systems to support the individual freedom of their participants, and may therefore be considered more democratic. While most permissionless decentralised systems achieve consensus via mechanisms resembling voting, the threat of Sybil attacks, in which attackers cast large numbers of bogus votes, makes the application of consensus mechanisms to approximate majority rule necessary. Due to the requirement of common consensus mechanisms to prove ownership of some resource (e.g. compute power or cryptocurrency) to participate in governance, they have to be considered plutocratic.

This thesis explores whether community-oriented and democratic systems governance following the principle of *one person, one vote* is possible in open networks, where no authorities are present to control network permissions. This is done through a combination of methods. Initially, a comprehensive literature survey is presented to identify evidence of such approaches to system governance in the wider literature. Then, novel governance approaches (e.g. Proof-of-Work (PoW), Proof-of-Stake (PoS)) are investigated to evaluate whether they can measure up to centralised control mechanisms in terms of energy efficiency and whether their users are aware of how they compare. Subsequently, a consensus mechanism is proposed that aims to provide democratic system governance while consuming negligible electricity. To evaluate this novel consensus mechanism, Sybil attacks are simulated using agent-based modelling. Finally, a model for blockchain systems is defined and the conditions are evaluated under which decentralised systems can implement democratic system governance.

This thesis shows that, at the time of writing, only a small set of *archetypal* consensus mechanisms exist; namely, PoW, PoS, and variants of these. It then demonstrates that it is possible to establish a novel democratic consensus mechanism that approximates majority voting by combining primitives of PoS with a reputation system. The study further shows that, under relaxed conditions (e.g., when an attack is discoverable and the attackers can be identified), Sybil attacks on this novel democratic consensus mechanism can be prevented. However, ultimately, such a mechanism is shown to be unable to withstand Sybil attacks under generic conditions.

It can be concluded that, while blockchains may be effective as operating systems for democratic systems under certain premises (such as central admission), they are not universally suited to this purpose. This finding has consequences for those who design, build, and apply blockchain technology for self-governance.

Declarations

Statement of Originality

I declare that all the work presented in this thesis, unless otherwise acknowledged, is my own. Parts of this thesis have been previously published, and the extent of my contribution is detailed in chapter 1, section 1.6.

Statement Concerning the Use of AI Tools

I declare that no part of this thesis has been generated by artificial intelligence (AI) software. These are my own words, and all references are cited accordingly.

Acknowledgements

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[*] In alphabetical order

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Introduction | 1

Many modern decentralised systems rely on resource commitments, such as the requirement to use electricity, to solve computationally expensive puzzles in Proof-of-Work (PoW), to achieve system stability. At the same time, decentralised systems are posited as suitable environments for self-governing communities. Identifying the conflict between the need for resource commitments and democratic governance of decentralised systems allows us[1] to define narrow research questions that focus on the ability of reputation systems to achieve democratic system governance. With this thesis, we investigate the suitability of decentralised systems for facilitating democratic self-governance without resource commitments.

1: Throughout this thesis, irrespective of the number of authors, the personal pronoun 'we' is used for consistency in expository style.

Our main contribution lies in applying a multitude of research methods (systematic literature review, survey research, agent-based-modelling, and logical arguments) to this research question. This is done with the objective of identifying shortcomings in existing systems, suggesting solutions that we presume not to suffer from these shortcomings, and, subsequently, formally investigating the suitability of these solutions. Ultimately, this thesis concludes with a negative result: no decentralised system can simultaneously exhibit permissionlessness, Sybil attack resistance, and freeness–three properties that we deem prerequisites for self-governing decentralised systems without resource commitments.

1.1 Topic and Aims

Decentralised blockchain systems, characterised as symmetric, admin-free, ledgered, and time-consensual (Tai *et al.*, 2017) have become a focus of the distributed systems community. This technology originated in circles dismissive of governments, corporations, and large organisations, whose involvement was seen as a barrier to privacy in communications and an enabler of censorship. Consequently, decentralised systems, especially in their early days following the inception of Bitcoin in 2008, were proposed as a suitable foundation for the realisation of individual freedom, autonomy, and democracy (Atzori, 2017). With increasing popularity, the technology moved out of its niche existence and was subjected to broader scrutiny: decentralised blockchain systems, particularly in conjunction with cryptocurrencies, were often criticised for their role in crime, evasion of regulatory oversight, substantial energy consumption, and exacerbation of economic disparities.

Atzori, M. (2017). Blockchain technology and decentralized governance: Is the state still necessary? *Journal of Governance and Regulation*, *6*(1)

It is against this backdrop that we confront the ideals from the early days of this technology with its modern critique. We recognise self-governance, defined as a group's ability to regulate itself without external intervention (Janowitz, 1975), as an archetype of democratic activity. Furthermore, our focus is primarily on use cases that lack a financial element: these are scenarios that do not require participants to incur significant explicit monetary costs, as we recognise such participation costs as exclusionary and, therefore, contrary to the goals of democratic self-governance. Ultimately, our aim is, formally and conclusively, to analyse permissionless decentralised blockchain technology with the goal of proving or disproving its suitability for democratic self-governance.

Janowitz, M. (1975). Sociological theory and social control. *American Journal of Sociology*, *81*(1)

1.2 Key Literature and Terms

Research into the resiliency of distributed systems has a rich history, dating back to the inception of multiprocess computing in the 1960s (Adair *et al.*, 1966; Kleinrock, 1961). Subsequent work by Pease *et al.* (1980) and, later, Lamport *et al.* (1982) formalised the problem of faults in distributed systems and quantified the proportion of faulty processes a system can withstand as 1/3. Early works assumed central admission processes that limited the set of potentially faulty participants to those admitted to the system. Later works targeted open systems: in his 2002 paper 'The Sybil Attack', Douceur (2002) debated the vulnerability of systems without central admission procedures to scenarios in which an attacker creates many faulty processes. Sybil attacks gained further significance with the advent of blockchain systems, which emerged with Bitcoin (Nakamoto, 2008). Those were the first large-scale implementations of public and permissionless decentralised systems in which no central admission procedures were applied. Consequently, the majority of literature relevant to the goals and aims of this thesis has been created since the inception of Bitcoin, the first implementation of blockchain technology: in the seminal 2008 paper, Nakamoto (2008)[2] initially presented the idea of employing a computationally expensive PoW function to ensure the blockchain underlying Bitcoin would act as a distributed data storage layer that was tamper-proof, censorship resistant, and could be used by anyone.

This section serves as an overview. A more comprehensive literature review is undertaken in chapter 2.

Douceur, J. R. (2002). The Sybil attack. P. Druschel, F. Kaashoek & A. Rowstron (Eds.), *Proceedings of the 1st workshop on peer-to-peer systems*. Springer

Nakamoto, S. (2008, November). *Bitcoin: A peer-to-peer electronic cash system.*

2: *Satoshi Nakamoto* is the pseudonym adopted by an anonymous, to date, author or group of authors (Lemieux, 2013).

As most public decentralised systems, including Bitcoin and Ethereum, have been conceptualised and implemented by practitioners, the application of modelling and verification during the design of their protocols is rare. Some work investigating the correctness of common protocols, such as the formalisation of Bitcoin by Chaudhary *et al.* (2015) or the formalisation of the Ethereum virtual machine by Hildenbrandt *et al.* (2018), has been undertaken post hoc. The security characteristics of PoW have been extensively studied and been found to be incentive-compatible (Chiu & Koeppl, 2019); that is, rational participants maximise their utility by adhering to the protocol, provided that others also conform, establishing a coalition-safe equilibrium (Stouka, 2021). However, this incentive compatibility comes at the cost of significant electricity consumption – an argument that Sedlmeir *et al.* (2020b) substantiate by showing a correlation between the market price of an asset on a PoW blockchain and the electricity consumption of that chain. Despite extensive research on the security, correctness, and efficiency of blockchain systems, the broader societal implications, particularly regarding democratic self-governance, remain underexplored.

Sedlmeir, J., Buhl, H. U., Fridgen, G., & Keller, R. (2020b). The energy consumption of blockchain technology: Beyond myth. *Business & Information Systems Engineering*, *62*(6)

In this section, essential terminology relevant to this thesis is defined.

Distributed and decentralised systems We distinguish between distributed and decentralised systems as follows: *distributed* systems, a well-established class characterised by geographically dispersed and loosely connected computing facilities (Davenport, 1979), have been extensively studied (J. Wu, 1999). These systems typically exhibit central governance (i.e., a central admissions process). They may rely on a-priori knowledge of participants and their trustworthiness (Satyanarayanan, 1989). In contrast, *decentralised* systems represent a novel paradigm, building upon distributed systems, by adding elements of publicness and permissionlessness, where participants are not necessarily known and no admission process is needed for system access, interaction, or maintenance (Nakamoto, 2008).

Davenport, R. (1979). The state of the art of distributed databases. *Information & Management*, *2*(6)

Public systems In public decentralised systems, participants' anonymity or pseudonymity is crucial. This property extends to both *read* access (enabling visibility of transactions) and *write* access (facilitating transaction proposals), but not to *system maintenance*. Consequently, access controls cannot be implemented in public systems due to the inherent impossibility of differentiating between various anonymous or pseudonymous participants.

Permissionless systems Permissionlessness in decentralised systems is concerned with *system maintenance*, i.e., the development and governance of a system. Permissionlessness allows public participation in activities such as finalising transaction history through block validation, without requiring permission from a central authority, simply by adhering to publicly stated procedures (Nabben & Zargham, 2022). As system maintenance is a critical and failure-sensitive activity, permissionless systems are vulnerable to Sybil attacks, making mechanisms to combat them particularly relevant in this context.

Free systems Freeness is concerned with the cost of participation in a decentralised system. We consider a system free if there is no 'explicit monetary cost' (Saleh, 2020, p. 1157) to participate in it, whether as a reader, writer, or for system maintenance.

Sybil attacks Following Douceur (2002), we define a Sybil[3] attack as the presentation of arbitrary entities with malice, threatening decentralised systems by disrupting the determination of a canonical system state. Such attacks can be particularly successful in public/permissionless systems, where anonymous/pseudonymous participants are not subject to accountability for their actions (Haeberlen, 2009). Successful Sybil attacks can stall decentralised systems, necessitating drastic measures such as hard forks, which are controversial due to their economic and ethical implications (V. Dhillon *et al.*, 2017).

3: Douceur used the name 'Sybil' as an homage to a novel of the same title written by Schreiber (1973), which thematises schizophrenia (Douceur, 2002, p. 259).

Reputation systems A reputation system is a mechanism that entities use to assess the trustworthiness of other entities based on past behavioural feedback (Hendrikx *et al.*, 2015). In decentralised computing, such systems typically rely on observations of past adherence to protocol rules. Observations around the trustworthiness of presented identities can be used to mitigate Sybil attacks if these are shared among participants and these participants adapt their behaviour accordingly.

Hendrikx, F., Bubendorfer, K., & Chard, R. (2015). Reputation systems: A survey and taxonomy. *Journal of Parallel and Distributed Computing*, *75*

Democraticness Democraticness refers to the degree to which a decentralised system approximates the principle of 'one person, one vote'. This concept is crucial for evaluating the fairness and inclusivity of governance mechanisms within a given system. Variables influencing democraticness include voting mechanisms, i.e., the design of voting processes and how votes are weighted. Participation barriers, i.e., whether monetary costs must be incurred to participate in network consensus. Power distribution, i.e., ensuring that power and influence are evenly distributed among participants, preventing dominance by a minority.

1.3 Current State of the Literature

The energy consumption characteristics of decentralised systems have been thoroughly researched (Agur *et al.*, 2022, 2023; de Vries, 2018, 2021, 2022; George, 2021; Gola & Sedlmeir, 2022; Kohli *et al.*, 2023; Maiti *et al.*, 2023; Rieger *et al.*, 2022; Schinckus, 2021; Sedlmeir *et al.*, 2020a, 2020b; Stinner, 2022; Truby *et al.*, 2022; D. Zhang *et al.*, 2023). Consensus has been reached on the relatively high electricity consumption of popular PoW blockchains (Gallersdörfer *et al.*, 2020; Krause & Tolaymat, 2018; D. Zhang *et al.*, 2023) and on the fact that other consensus mechanisms, such as Proof-of-Stake (PoS), are significantly more energy efficient (Agur *et al.*, 2022, 2023; de Vries, 2021; Platt, Sedlmeir *et al.*, 2021; Sedlmeir *et al.*, 2020a, 2020b). Furthermore, it is widely accepted that the energy footprint of a given PoW blockchain correlates with the monetary value of the rewards that a block producer can reap (de Vries, 2021; Sapra *et al.*, 2023; Schinckus *et al.*, 2020; Söylemez & Gürsoy, 2022). However, there is no scientific agreement on how different consensus mechanisms influence overall system stability. This question is the subject of some controversy (Brown-Cohen *et al.*, 2019; Gans & Gandal, 2019; S. Lee & Kim, 2020; Lundin & Rahm, 2018; Mueller-Bloch *et al.*, 2022; Nair & Dorai, 2021; Zięba, 2024).

The application of blockchain technology for self-governance has been debated on various occasions. The debate focuses mainly on the risks and benefits of algorithmic governance (Y. Liu, Lu *et al.*, 2022; Y. Liu *et al.*, 2023): some works highlight the potential of decentralised commons governance (Rozas, Tenorio-Fornés & Hassan, 2021). Others emphasise the challenges that come with algorithmic decision-making, such as the potential for citizen disenfranchisement as a consequence of the rise of a stateless global community driven by algorithmic consensus (Atzori, 2017), or the legal and regulatory challenges posed by decentralised autonomous organizations (DAOs) (Llamas Covarrubias & Llamas Covarrubias, 2021; Zwitter & Hazenberg, 2020), as well as the risk of DAO decision-making processes becoming dysfunctional (Liebau, 2024). However, work in this area focuses predominantly on the possibility of self-governance in existing decentralised systems, often focusing on algorithmic governance through DAOs. The characteristics of the underlying systems, and specifically their consensus mechanisms, which are of concern for this thesis, are commonly not discussed in detail.

The problem of Sybil attacks in such contexts is under-researched beyond foundational works that formalise consensus in the presence of malicious actors (Roscoe *et al.*, 2023). Douceur (2002) shows that, without a logically centralised authority to verify identities, and under realistic environmental conditions, Sybil attacks on large-scale peer-to-peer systems cannot be prevented. However, since his work precedes Bitcoin (Nakamoto, 2008) and, therefore, the PoW consensus mechanism, the notion of a logically centralised authority is not conclusive for the research question of this thesis. Similarly to Douceur (2002), Seuken and Parkes (2011) explore the limitations of Sybil attack resistance. Following the work of Friedman *et al.* (2007) on manipulation of reputation systems, Seuken and Parkes (2011) show that achieving absolute robustness against Sybil attacks in distributed work systems is impossible, but that protocols may provide limited robustness, highlighting a fundamental challenge for democratic self-governing systems that rely on user reputation. Poupko *et al.* (2019) have contributed one of the few research works that explicitly focus on resistance to Sybil attacks in the context of community growth, proposing a trust-based graph system to combat Sybil attacks. Building on the work of Alvisi *et al.* (2013), Poupko *et al.* (2019) demonstrate that under the assumption of a limited number of attack edges and a small fraction of attackers in the initial community, self-governing systems can be Sybil-attack resistant, a concept similar to *K*-Sybil-proofness, as established by Seuken and Parkes (2011).

The research gap that can be identified based on the current state of the literature is threefold: first, a discussion of Sybil-attack resistance in the context of self-governance that includes modern consensus mechanisms like PoW and PoS is absent. Second, the suitability of PoS as a foundation for a reputation-based consensus mechanism has not been evaluated. Third, no formalisation of the desirable characteristics of decentralised systems (including their electricity demand) and their relationship to Sybil-attack vulnerability has been undertaken.

This research gap is particularly relevant in the context of the recent surge in interest around blockchain technology, which has led to the application of decentralised protocols in many domains and industries (Notheisen *et al.*, 2017), including self-governance (Rozas, Tenorio-Fornés, Díaz-Molina & Hassan, 2021), often without a clear understanding of problem-solution fit (Platt, Bandara *et al.*, 2021). Closing this research gap can help designers of decentralised systems make more appropriate choices around the features their systems offer.

1.4 Research Contribution

This thesis addresses a research question concerning the feasibility of democratic self-governance in decentralised systems without resource commitments. Specifically, it investigates whether reputation systems can operate democratically in permissionless environments. The thesis, furthermore, explores the differences in energy consumption between different consensus mechanisms and how these differences impact user choice. With this approach we seek to address the following fundamental research question:

> Can reputation systems enable 'one person, one vote' characteristics in permissionless decentralised systems?

To address this question, we hypothesise as follows:

Hypothesis 1 PoW consumes significantly more electricity than PoS.
Hypothesis 2 Users who are aware of the electricity consumption of PoW systems support measures to reduce their energy usage.
Hypothesis 3 PoS can model a reputation system, serving as the foundation for a democratic consensus mechanism independent of proof of resources.
Hypothesis 4 Using reputational signals as the foundation for a consensus mechanism enables a decentralised system to be permissionless, resilient against Sybil attacks, and free.

This thesis employs a mix of methodologies: systematic literature review, survey research, agent-based modelling, and formal logical analysis. This multi-method approach is chosen to investigate the hypotheses comprehensively.

The main result of the thesis is the formalisation of a Sybil-attack vulnerability trilemma, demonstrating that no blockchain protocol can simultaneously achieve permissionlessness, Sybil-attack resistance, and freeness.

The thesis also shows that consensus mechanisms that combine reputation systems with PoS primitives can make a system more democratic. Furthermore, it confirms the energy efficiency of PoS over PoW, thereby emphasising the importance of user awareness of blockchain consensus mechanisms.

1.5 Layout of the Thesis

This thesis is divided into two parts. This is done to separate background and motivation (first part) from the proposal of a novel protocol and its formal evaluation (second part).

The first part provides background on new consensus mechanisms (i.e. those created after the advent of blockchain in 2008) and common criticisms of those mechanisms. In chapter 2, the breadth of consensus mechanisms is shown through a systematic literature review. The reviewed consensus mechanisms are further categorised according to key properties. Chapter 3 describes how popular consensus mechanisms make use of different strategies to prevent Sybil attacks. Using a mathematical model, vastly different energy demands are shown, emphasising the need to focus on less electricity-consumptive mechanisms, such as PoS. Chapter 4 explores user awareness regarding the environmental implications of their selection of blockchain systems with substantial energy consumption. The first part of the thesis closes with an investigation of the impact of energy labelling of blockchain systems. Chapter 5 outlines the potential for influencing consumer preferences towards more energy-efficient blockchain systems.

In the second part, the criticism expressed in the previous part is addressed by proposing a new consensus mechanism. In chapter 6 identity-augmented proof-of-stake (IdAPoS), a modification of PoS is introduced that overlays a reputation system to generate stake. This is done with the goal of creating a novel consensus mechanism that is well suited to self-governance in that it does not require participants to incur significant explicit monetary cost to achieve Sybil attack resistance. Furthermore, in chapter 6, validation of IdAPoS is undertaken using agent-based modelling: a simplified actor model is introduced, consisting of both benign and malicious participants. This experiment shows that a reputation system can be used to deter Sybil attacks, provided that benign participants become aware of an attack sufficiently early. This result is further refined through a formal model in chapter 7, where we show that, in the absence of the simplifying conditions introduced in chapter 6, the conclusions drawn from the agent-based model no longer hold. On the contrary: it is proven that systems cannot be free *and* resistant to Sybil attacks *and* strongly permissionless at the same time. Chapter 8 concludes this thesis by summarising and reflecting on the research carried out, highlighting limitations, and discussing potential for future work.

1.6 Prior Publication

The following chapters present revised excerpts of previously published works. Each chapter, therefore, is presented in such a way as to stand independently.

Chapter 2 is a revised excerpt of an article previously published as Platt, M., & McBurney, P. (2023). Sybil in the haystack: A comprehensive review of blockchain consensus mechanisms in search of strong Sybil attack resistance. *Algorithms*, *16*(1), Article 34. Moritz Platt served as the lead author of this work. Of the authors involved, he made the most significant intellectual contribution to its creation. His contribution included conceptualisation, data curation, formal analysis, investigation, methodology, project administration, validation, visualisation, original draft preparation, as well as review and editing.

Chapter 3 is a revised excerpt of an article previously published as Platt, M., Sedlmeir, J., Platt, D., Xu, J., Tasca, P., Vadgama, N., & Ibañez, J. I. (2021). The energy footprint of blockchain consensus mechanisms beyond proof-of-work. *Companion Proceedings of the 21st Conference on Software Quality, Reliability and Security*. Moritz Platt served as the lead author of this work, and made the most significant intellectual contribution to its creation. His contribution included conceptualisation, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, supervision, validation, visualisation, preparing the original draft, as well as reviewing and editing.

Chapter 4 is a revised excerpt of an article previously published as Platt, M., Ojeka, S., Drăgnoiu, A.-E., Ibelegbu, O. E., Pierangeli, F., Sedlmeir, J., & Wang, Z. (2023). Energy demand unawareness and the popularity of Bitcoin: Evidence from Nigeria. *Oxford Open Energy*, *2*, Article oiad012. Moritz Platt served as the lead author of this work, making the most significant intellectual contribution to its development. His contribution included conceptualisation, data curation, funding acquisition, investigation, methodology, project administration, software, supervision, validation, visualisation, writing the original draft, and review and editing for publication.

Chapter 5 is a revised excerpt of an article previously published as Drăgnoiu, A.-E., Platt, M., Wang, Z., & Zhou, Z. (2023). The more you know: Energy labelling enables more sustainable cryptocurrency investments. *Proceedings of the 43rd International Conference on Distributed Computing Systems Workshops*. Moritz Platt was the lead author of this work, to which he made the most significant intellectual contribution. His contribution included conceptualisation, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, resources, software, supervision, validation, visualisation, writing (original draft preparation, review and editing).

Chapter 6 is an amalgamation of two articles: First; Platt, M., & McBurney, P. (2021a). Self-governing public decentralised systems: Work in progress. T. Groß & L. Viganò (Eds.), *Proceedings of the 10th workshop on socio-technical aspects in security and trust*. Springer. Moritz Platt acted as the lead author of this work, with the most significant intellectual contribution to its creation. His contribution included conceptualisation, investigation, methodology, project administration, validation, visualisation, writing the original draft, and reviewing and editing it for publication. Second; Platt, M., & McBurney, P. (2021b). Sybil attacks on identity-augmented proof-of-stake. *Computer Networks*, *199*, Article 108424. Moritz Platt was the lead author of this work. He played the most significant intellectual role in its creation. His contribution included conceptualisation, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, resources, software, validation, visualisation, preparing the original draft and reviewing and editing it prior to publication.

Chapter 7 is a revised excerpt of an article previously published as Platt, M., Platt, D., & McBurney, P. (2024). Sybil attack vulnerability trilemma. *International Journal of Parallel, Emergent and Distributed Systems*, *39*. Moritz Platt served as the lead author of this work, making the most significant intellectual contribution to its development. His contribution included conceptualisation, formal analysis, investigation, methodology, project administration, software, validation, visualisation, original draft preparation, reviewing and editing.

Blockchain Consensus Mechanisms and Their Criticism

2 Blockchain Consensus Mechanisms: A Review

This chapter initiates the thesis by providing the foundational research context and rationale. To establish a comprehensive understanding of the current landscape, a thorough literature survey is undertaken, focusing on the evolution and challenges of decentralised systems, particularly in relation to Sybil attack resistance. Given their openness, decentralised systems, such as blockchains underpinning cryptocurrencies, have introduced the need for Sybil attack resistance (see subsection 2). As the need for Sybil attack resistance only arises in these truly decentralised systems, research into Sybil attacks is still in its infancy. Many different consensus mechanisms, varying greatly in specification quality, have been proposed after 2008 when Bitcoin (Nakamoto, 2008) set in motion the blockchain era. This environment is particularly conducive to our research: while numerous literature reviews have targeted consensus mechanisms (see subsection 2.7), to the best of our knowledge, no previous work focusing on Sybil attack resistance exists. Given the challenging effects the latest 'crypto winter' (Chohan, 2022) might have on the blockchain industry, academia's role is to research fit-for-purpose approaches on which mature and useful systems can be built. Providing a classification and taxonomy of Sybil attack resistance schemes is key to this.

Nakamoto, S. (2008, November). *Bitcoin: A peer-to-peer electronic cash system.*

2.1 Context

The long and rich history of distributed systems research directly leads up to modern blockchain consensus: contrary to popular belief (Humayun & Belk, 2018; van Flymen, 2020), Nakamoto's consensus mechanism PoW (Nakamoto, 2008) is not an isolated stroke of genius but, like many modern developments in the field of distributed systems, dependent on the rigorous work of previous generations of researchers (Narayanan & Clark, 2017). Nakamoto's contribution is to combine well-studied ideas to enable a new class of *distributed* systems, the *decentralised* systems. This section outlines how researchers contribute to the study of decentralised consensus in creative ways, often combining decades-old findings with modern requirements.

Narayanan, A., & Clark, J. (2017). Bitcoin's academic pedigree. *Queue, 15*(4)

Resilient Distributed Systems

The earliest attempts at remotely controlling computers before the post-war era (Luebbert, 1981) aside, the concept of geographically distributed computers collaborating was introduced in the 1950s. These early computer systems emerged in the military domain and were coordinated by transmitting generalised digital data over standard phone lines (Astrahan & Jacobs, 1983). While issues of reliability were of concern in these early designs, these were met by basic error control techniques like parity checking of input and output data only (Everett *et al.*, 1958). Notably, these early systems assumed that participating computers were fully trusted and secured via military security procedures. Formalisations of computer networks and their failure characteristics were proposed alongside the inception of multiprocess computing in the 1960s (Adair *et al.*, 1966). Back then, Kleinrock (1961) laid the foundation for data networking theory by proposing to research the effects of messaging delays and message

Everett, R. R., Zraket, C. A., & Benington, H. D. (1958). SAGE. *Papers and Discussions Presented at the 1957 Eastern Joint Computer Conference*

Kleinrock, L. (1961, May 31). *Information flow in large communication nets* [PhD Thesis Proposal]. Massachusetts Institute of Technology

Property	Distributed	Verifiable	Blockchain
Sequential	•	•	•
Agreed			•
Ledgered		•	•
Manipulation-Resistant		•	•
Censorship-Resistant			•

Table 2.1: A comparison of distributed system types according to Tai *et al.* (2017, p. 758).

prioritisation in 'large communication nets'. These theoretical foundations led to the practical implementation of formal message protocols for wide-area communication (Marill & Roberts, 1966) and subsequently to the implementation of the Advanced Research Projects Agency Network (Cerf *et al.*, 1972), the predecessor of the modern Internet, that brought with it the notion of various autonomous systems clustered in hierarchical layers (Y. He *et al.*, 2009). These advances, however, focussed mostly on providing stable means of connecting network members in principle and did not govern higher protocol layers.

Byzantine Failures and Malice

Pease *et al.* (1980) describe the problem of complex faults in a distributed system by giving an example of a bad process reporting one value to a given process and a different value to other processes. Such conflicting (and, potentially, malicious) behaviour cannot be controlled via the simple mechanisms outlined before. Instead, more complex approaches, i.e. consensus mechanisms, must be applied. Inconsistent failure modes, later formalised and termed 'byzantine' failures by Lamport *et al.* (1982), have been widely researched and upper bounds for the proportion of byzantine nodes in distributed systems (depending on the consensus mechanism used) are established.

Pease, M., Shostak, R., & Lamport, L. (1980). Reaching agreement in the presence of faults. *Journal of the ACM*, *27*(2)

Lamport, L., Shostak, R., & Pease, M. (1982). The Byzantine generals problem. *ACM Transactions on Programming Languages and Systems*, *4*(3)

From Distributed to Decentralised Systems

As Table 2.1 shows, decentralised systems are different from traditional distributed databases, i.e. geographically displaced, loosely coupled, computing facilities (Davenport, 1979), since these assume a-priori knowledge of participants and their trustworthiness. Furthermore, they fundamentally differ from modern verifiable database systems (Antonopoulos *et al.*, 2021; Y. Zhang, 2020) in that the latter allow cryptographic integrity assurance, but offer no censorship resistance.

Davenport, R. (1979). The state of the art of distributed databases. *Information & Management*, *2*(6)

Decentralised systems are, therefore, a suitable pattern to address entirely permissionless use cases, intended to be open to the world. While, in practicality, many multi-party use cases do not require this degree of decentralisation (see chapter 6), blockchain systems that aspire to be 'unstoppable, uncensorable world computer[s]' (Leising, 2017) do.

Blockchain

The first large-scale implementation of such a system was 'Bitcoin' (Nakamoto, 2008), a decentralised payment system introduced in 2008. It famously put forward the concept of decentrally collecting new transactions into blocks and forming a chain of these blocks via an energy-intensive,

Nakamoto, S. (2008, November). *Bitcoin: A peer-to-peer electronic cash system.*

	Permissioned	Permissionless
Public	*i*	*ii*
Private	*iii*	*iv*

Table 2.2: Four archetypes of blockchain architectures according to Tezel *et al.* (2020, p. 549).

and often criticised (see chapter 3), 'mining' process to make them tamper-proof, thereby coining the term *blockchain*. Beyond Bitcoin, blockchain systems share common attributes[1]: They can be characterised as systems that process transactions sequentially based on majority agreement and, ultimately, persist them in a tamper-resistant form on an append-only ledger (Tai *et al.*, 2017). For the scope of this work, the majority agreement aspect is of primary interest since who constitutes the population for such a majority agreement depends on the type of system.

1: Familiarity with the blockchain concept is assumed, and this definition is provided for completeness only. Introductory monographs provide a more comprehensive foundation from the perspectives of practitioners (Hellwig *et al.*, 2020), financial technologists (Lewis, 2018), and academics (Gayvoronskaya & Meinel, 2021; X. Xu *et al.*, 2019).

Blockchain systems are most commonly categorised along two dimensions: the trust continuum and the anonymity continuum (Tezel *et al.*, 2020). On the trust continuum, systems can be categorised as permissioned or permissionless. A *permissioned* system employs a governance process to determine who can act as a writer, whereas a *permissionless* system allows anyone to potentially assume the role of a writer. On the anonymity continuum, systems can be differentiated by being public or private. A *public* system provides all data in a publicly inspectable form, whereas a *private* one applies explicit or implicit access control mechanisms. These two dimensions can be arbitrarily composed (see Table 2.2). For this chapter, the *writing* permission dimension is most relevant.

While in permissioned contexts (i.e. *i* and *iii*), it can still be necessary to select a leader to orchestrate data replication, trust assumptions are often more relaxed and more akin to those of earlier distributed systems (see subsection 2.1). This is the case because the group of potential leaders in permissioned systems can be curated, making for an inherently trustworthy population of potential leaders. In permissionless contexts (i.e. *ii* and *iv*), however, any member of the public qualifies as a potential leader. This means that entities whose goals do not align with maintaining the stability of a system and adhering to its protocol may make themselves available as leaders. This makes permissionless leader selection the harder problem.

Attacks on Permissionless Systems

Attacks on permissionless systems[2] can be categorised as mining attacks, attacks on network communications/smart contracts, and privacy-related issues.

2: This section follows the taxonomy of attacks established by Y. Chen *et al.* (2022).

Mining Attacks Mining attacks manifest in the block producing phase of a permissionless system protocol. They are only applicable to permissionless systems which apply PoW. Rosenfeld (2011) first discussed 'block withholding', an archetypal mining attack. In this attack, miners following a pooled mining reward strategy in a PoW system may delay the submission of blocks or refrain from submitting them altogether. By delaying the submission of a block, an attacker can receive a disproportionate reward.

Another well-known Mining attack is the '51% attack', first discussed by King and Nadal (2012). In this attack, by contributing more than 50% of the network-wide computing power to mining, an attacker can strongly increase the likelihood of being selected as a miner. An attacker might use this control to overwrite block contents with data of their choosing (e.g. incorrect transactions).

In 'Pool-hopping' (Rosenfeld, 2011) miners participating in pooled mining contribute selectively to the mining pool with the highest expected revenue, thereby increasing their personal expected revenue beyond what is achievable through continuous mining. This negatively influences participants who mine continuously (i.e., those that do not pool-hop).

In 'selfish mining' attacks (Eyal & Sirer, 2014), attackers strategically refrain from announcing found blocks to build an alternative chain in private. By later releasing their privately mined chain, they may overwrite longer sequences of blocks, potentially creating a permanent fork, reverting transactions finalised on the public main chain.

Attacks on Network Communication Beyond the Sybil attack, the main focus of this thesis, further attacks on network communication have been observed.

Distributed denial-of-service attacks are well-known phenomena in internet connected systems (Mirkovic & Reiher, 2004). They are also understood to be prominent attack vectors for permissionless systems (Vasek *et al.*, 2014), where mining pools may be attacked. Where permissionless systems are operated in conjunction with cryptocurrencies, cryptocurrency trading infrastructure, such as centralised exchanges (CEXs) may be attacked. It can be speculated that the motivation behind targeting mining pools or CEXs is monetary.

In 'Eclipse' attacks (Heilman *et al.*, 2016), attackers who control numerous internet protocol (IP) addresses may use these to monopolise connections to network nodes. This behaviour then allows the attacker to conduct further types of attack (e.g. the aforementioned mining attacks) on a victim network node.

Exploits of Smart Contract Vulnerabilities Reentrancy attacks are a common class of smart contract exploits (Al-E'mari & Sanjalawe, 2024). In them, attackers may call a smart contract function before its termination, and before its state is updated, thereby executing the same function repeatedly. Such repeated execution may have unintended consequences that are useful to an attacker.

Privacy Attacks Y. Chen *et al.* (2022) consider de-anonymisation of transactions, i.e., the violation of unlinkability of pseudonyms (Meiklejohn, 2018), another relevant attack type. Commonly, de-anonymisation entails correlating observable transaction data (e.g. public key) with other data sources and possibly network metrics such as propagation times (Biryukov & Tikhomirov, 2019). The result of de-anonymisation, the disclosure of personal data, can be considered a threat to those that use permissionless systems with the goal of being anonymous (Froehlich *et al.*, 2021).

Sybil Identities: A New Threat

Distributed systems theory has provided in-depth analyses of numerous system failure modes and has provided mitigations for many of those. Most of the early work, however, assumed a priori knowledge of system participants (see subsection 2.1). In addition to this, while the abuse of telecommunications infrastructure is documented as early as the 1960s (Lapsley, 2013) and attacks on computer systems occurred frequently from the 1980s onwards (Lukasik, 2011), attacks commonly were carried out by intruders who targeted systems of which they were no members from the outside. Putting that in perspective to the categories outlined before (see Table 2.2), it becomes apparent that research of the pre-blockchain era is most applicable to type *iii* (permissioned/private) systems.

Lapsley, P. (2013). Phreaking out ma bell. *IEEE Spectrum*, *50*(2)

It is not always possible to transfer earlier results to majority-based systems where there is no a-priori knowledge of potential leaders (i.e. type *ii* and *iv* systems) since those constitute a new phenomenon popularised with the emergence of blockchain technology. From an attack perspective, such systems are fundamentally different from prior deployments as, in them, malicious entities can operate on the same level as trustworthy ones for there is no authority to distinguish between those. This means that common approaches to mitigating intrusion risks–such as identity management, user authentication, or encryption (Uzunov *et al.*, 2012)–are not feasible. Furthermore, common strategies to improve fault tolerance, like increasing redundancy, are futile in a scenario where malicious users can present arbitrary identities. The presentation of arbitrary entities with malice–often referred to as 'Sybil Attack' (Douceur, 2002)–threatens large-scale peer-to-peer systems in which trusted authorities are absent. This can be attributed to the fact that, when confronted with conflicting information from multiple peers, decentralised systems need to attempt to resolve those conflicts in a way that maintains a correct system state. Determining which piece of information is to be considered *correct* is the main challenge in such scenarios.

Douceur, J. R. (2002). The Sybil attack. P. Druschel, F. Kaashoek & A. Rowstron (Eds.), *Proceedings of the 1st workshop on peer-to-peer systems*. Springer

Successful Sybil attacks manifest in different ways, depending on the consensus mechanism deployed and the scale of the attack. While successful small-scale attacks might only introduce forks or stall system operation sporadically, others may irrecoverably disable it, forcing users to abandon it or seeking 'hard forks' and subsequent manual removal of the attacking entities (V. Dhillon *et al.*, 2017), a technique frowned upon due to its economical and ethical ambiguity (T. W. Kim & Zetlin-Jones, 2019).

Kim, T. W., & Zetlin-Jones, A. (2019). The ethics of contentious hard forks in blockchain networks with fixed features. *Frontiers in Blockchain*, *2*, Article 9

Consensus Mechanisms and Sybil Resistance Schemes

More formally, this consensus problem can be defined as one in which 'each process proposes some initial value, and processes that do not fail must reach an irrevocable decision on exactly one of the proposed values' (Aguilera, 2010). In the context of blockchain, a *process* can be thought of as any potential writer (see Table 2.2) in a decentralised system, while *values* take the form of transaction proposals. Consensus mechanisms provide algorithms to reach a decision on a canonical system state when presented with ambiguous inputs. Often they do so by dynamically selecting a single entity–or leader–to assume the role of validator and adjudicator of inputs for a limited time. This selection is frequently done pseudorandomly or using voting. The target of such a leader selection process is to select a leader who processes any inputs according to the

Aguilera, M. K. (2010). Stumbling over consensus research: Misunderstandings and issues. B. Charron-Bost, F. Pedone & A. Schiper (Eds.), *Replication*. Springer system protocol while avoiding the selection of adversaries that violate system protocol.

In general, at most m adversaries can be tolerated for systems with $3m + 1$ total participants (Lamport *et al.*, 1982). Therefore, to maintain an operational state, the likelihood of selecting an adversary as a leader in a decentralised system needs to be minimised. This is the purpose of Sybil resistance schemes, many of which will be analysed in this chapter. Academic works discussing Sybil resistance schemes come in many forms, regularly as a key component of a consensus mechanism, other times as a standalone proposal. In our work, we focus on extracting any information relevant to Sybil attack resistance from the papers analysed, regardless of which context it is discussed in.

Lamport, L., Shostak, R., & Pease, M. (1982). The Byzantine generals problem. *ACM Transactions on Programming Languages and Systems*, *4*(3)

2.2 Motivation

Making sound architectural decisions for decentralised systems is challenging: a variety of different potential designs exist and, apart from usual functional and non-functional requirements, the choice of consensus mechanism shapes decentralised systems in hard-to-revert ways. The choice of consensus mechanism dictates whether a system requires a central admissions process or can be entirely open to the public. Due to the underlying trust assumptions, it furthermore influences the performance of a system, the hardware demands on participants, and the cost of deployment and operation.

Particularly the choice for or against a mechanism with strong Sybil attack resistance properties is decisive: due to being rooted in the blockchain movement, many of the existing consensus mechanisms claim Sybil attack resistance. However, as our analysis will show, many schemes only deliver on that claim under certain, limited, conditions. It is, therefore, necessary to firmly establish a framework that established Sybil attack resistance as one of the key elements of decentralised systems design. This will help the field to focus research efforts on useful areas in which decentralised technology can be applied, rather than reinventing the wheel or making costly design mistakes.

2.3 Organisation

The remainder of this chapter is structured as follows: In section 2.6 we describe the search strategy applied along with the search systems used to gain the broadest possible exposure to relevant works. In section 2.7, we present our results, initially analysing existing secondary literature on consensus mechanisms (see subsection 2.7). Subsequently, we analyse the primary literature, i.e., the original works discovered. We group works into mechanisms for democratic processes (see subsection 2.7), mechanisms for the education sector (see subsection 2.7), mechanisms for the energy sector (see subsection 2.7), general purpose mechanisms (see subsection 2.7), mechanisms for healthcare (see subsection 2.7), mechanisms that seek to improve performance (see subsection 2.7), mechanisms for internet of things (IoT) (see subsection 2.7), mechanisms for media and entertainment (see subsection 2.7), mechanisms for supply chain management (see subsection 2.7), mechanisms for the telecommunications sector (see subsection 2.7), proof of useful work (PoUW) schemes (see subsection 2.7), and mechanisms for vehicles (see subsection 2.7). We then conclude with a comment on the current state of literature (see section 2.8).

2.4 Scope

For a comprehensive review, we apply a broad definition of academic works, including research papers, technical reports, pre-prints, and theses. We consider any work that describes a consensus mechanism, a leader selection approach, or a Sybil attack resistance scheme well enough to understand its Sybil attack resistance properties. This includes many works that describe technologies that do not have Sybil attack resistance as a design goal (including many Byzantine fault tolerance (BFT) schemes). These are included for completeness.

2.5 Contribution

This work offers a comprehensive review of the blockchain consensus mechanism landscape. While many similar efforts have been undertaken (see subsection 2.7), this manuscript can be differentiated from those by the method of categorising relevant works: it taxonomises individual contributions in this landscape using a novel composition of key mechanism attributes (Sybil attack resistance class, reward scheme, and leader selection methodology). This approach allows us to provide an objective and comprehensive overview of the field that clearly describes which Sybil attack resistance class is prevalent. Grouping by industry, furthermore, allows the reader to determine which verticals have seen research interest. Lastly, by analysing the full breadth of the literature, this work shows which technological building blocks are commonly applied within different Sybil attack resistance classes (e.g. PoW in strongly Sybil attack resistant systems). These analyses provide the groundwork for researchers wishing to extend state of the art in consensus mechanism research.

2.6 Method

The goal of this work is to gain an understanding of the full breadth of available leader selection strategies. Based on preliminary screening, consensus mechanisms in the context of blockchain are proposed in various types of academic literature. While many algorithms are published in dedicated papers that are easy to discover based on their title alone, others are discussed as part of wider research works, such as technical papers introducing new system constructs.

Thus, to also include manuscripts in which consensus mechanisms are not the central concern, queries need to be broad. The selection of academic search systems to retrieve manuscripts relevant to the research question is informed by the findings of Gusenbauer and Haddaway (2020) who categorise academic search systems. All systems that provide the following functions are considered:

- The system focuses on Computer Science or has a multidisciplinary focus
- The system supports Boolean operators (at least `AND` and `OR`)

Listing 2.1: Basic search string used in manuscript retrieval.

```
(
  Blockchain OR
  "Distributed Ledger" OR
  "Crypto Currency" OR
  Cryptocurrency OR
  Cryptocurrencies
) AND (
  Consensus OR
  "Proof-of-" OR
  Membership
)
```

Operator	Search System	Results
ACM	Guide to Computing Literature	519
Bielefeld University Library	Bielefeld Academic Search Engine	5,890
IEEE	Xplore	1,846
Elsevier	ScienceDirect	3,186
Elsevier	Scopus	4,084
Springer Nature	SpringerLink	3,519
Clarivate Analytics	Web of Science	2,755

Table 2.3: Search results returned from targeted search systems.

- The system supports parentheses
- The system allows for bulk download of 50 results or more

Seven academic search systems (see Table 2.3) provide this functionality (Gusenbauer & Haddaway, 2020), including the highly relevant offerings by the Association for Computing Machinery (ACM) and the Institute of Electrical and Electronics Engineers (IEEE). For each of the academic search systems, a broad query is formulated. While individual query syntax varies, queries are modelled after Listing 2.1.

Since this review targets leader selection strategies in the context of blockchain, the term *blockchain*, or the closely related terms *Distributed Ledger* and *Cryptocurrency* are required. Similarly, since leader selection is central to consensus mechanisms, the term *Consensus*, or the related term *Membership* is required. Furthermore, the commonly used construct *Proof-of-** is admissible.

Search Results The total number of results returned (21,799)[3] raises confidence in the search queries being sufficiently broad. Since the queries do not impose any limitations on the manuscript type, a broad range of scientific works, both peer-reviewed and not peer-reviewed are returned (see section 2.4). For this review, all result types, including preprints, are considered. These results form the input to the following multistep manuscript screening process (see Figure 2.1).

3: The queries were executed in the week of 26 April 2021. The number of results reflects the manuscripts indexed at this time.

Duplicate Resolution ‘EPPI-Reviewer’[4], a software suite for research synthesis, is used to manage sources and screen manuscripts. The step in the review process is to apply the built-in ‘Manage Duplicates’ functionality to correlate duplicates automatically. While this functionality was found to be less effective than other de-duplication software (Jubb *et al.*, 2020), a significant number of duplicates (7,861) is discovered. Additional instances, however, remain undetected and will be treated manually after this automated step.

4: Software developed by the Social Science Research Unit in the Department of Social Science at University College London. Available at https://eppi.ioe.ac.uk/.

Abstract Screening Following duplicate resolution, titles and abstracts of results are screened for relevancy. Here, the following questions are answered:

- Is the result published in 2008 or after?
- Is the result written in the English language?
- Does the result describe a primary article (e.g. an original research article) or a secondary article (e.g. a review)?
- Does the article concern the field of computer science?

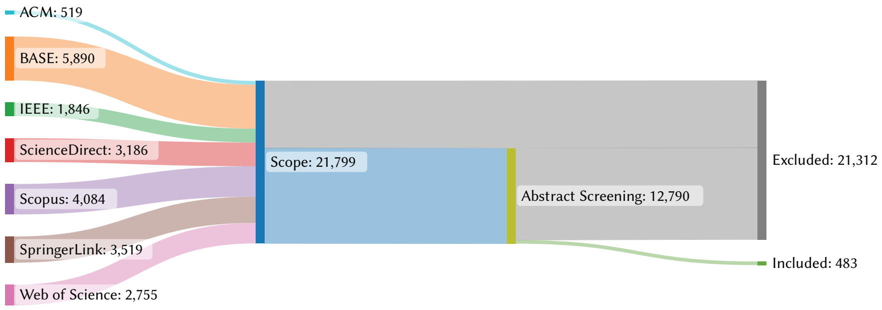

Figure 2.1: A total of 21,799 records–many of which were duplicates–were obtained from scientific search engines. During the manuscript screening process, 12,790 scientific manuscripts were initially analysed. After de-duplication and analysis of the abstracts and manuscript contents, 483 were found relevant.

- Does the article concern decentralised computing?
- Does the article describe a consensus mechanism?

Should either of the questions be answered with *no*, the manuscript is automatically excluded. Should all questions be answered with *yes* or *unclear*, the manuscript will be included for further analysis.

The abstract screening process eliminated many manuscripts (9,009). However, the remaining candidate list still contained numerous duplicates which were removed manually based on similarities of title, author list, or digital object identifier during abstract screening.

During the manual screening, the full texts of the documents were obtained online. Documents were only considered if digital access was granted to them via King's College London library subscriptions. Manuscripts that would only have been available in print or via interlibrary loan were not obtained. Still, the vast majority of full texts were obtainable and only a very small number could not be accessed. The full texts were then briefly examined to understand whether they were indeed relevant and, only if found so, were fully analysed. They were then categorised as either primary literature (i.e. works that proposed an individual mechanism) or secondary literature (i.e. surveys and taxonomies).

2.7 Results

We organise the results by common sectors to gain a better understanding of the needs of particular industry verticals. This also helps us to further analyse which system properties occur together, e.g., we find that many use cases in IoT (see subsection 2.7) have no or limited Sybil attack resistance due to the limited computational capabilities of IoT devices.

Aspects of Interest We are particularly interested in some key elements of Sybil attack resistance schemes. First, what level of Sybil attack resistance they provide when deployed. We assume that the mechanism in question is deployed to a *large-scale* system, as many Sybil attack resistance effects only materialise with high usage volume, as can be seen when considering the likelihood of 51 % attacks on PoW systems with low popularity (Sayeed & Marco-Gisbert, 2019). For Sybil attack resistance, we introduce three categories: Strong Sybil attack resistance; meaning a mechanism that, when applied to a sizeable system, could not be attacked,

Sayeed, S., & Marco-Gisbert, H. (2019). Assessing blockchain consensus and security mechanisms against the 51% attack. *Applied Sciences*, *9*(9), Article 1788 even by an irrational attacker, unless they incur a tremendous cost. Limited Sybil attack resistance; meaning an attacker might be successful with an attack if they are ready to incur significant costs and orchestrate an attack over a long time. No Sybil attack resistance, meaning an attacker could easily overwhelm a public/permissionless system using the mechanism in question. The latter category is not to be seen as a criticism of the mechanism: it mostly applies to environments in which the mechanism designers do not have to consider Sybil attacks.

Table 2.4: Three categories of Sybil attack resistance.

Category	Cost	Coordination effort	Time expenditure
Strong Sybil attack resistance	Excessive	High	Extreme
Limited Sybil attack resistance	High	High	High
No Sybil attack resistance	Negligible	Negligible	Negligible

Second, whether an incentive scheme is part of the proposed mechanism. Here, we analyse whether participants receive rewards for adherence to the protocol and/or punishment for deviating from it. Fourth, the application of random/pseudorandom numbers during the execution of the mechanism. Fifth, whether the mechanism is somehow linked to the physical world, e.g. by evaluating messages generated by users having access to central processing units (CPUs) with trusted execution environments (TEEs). Sixth, whether a reputation system is applied in the context of the mechanism. Such a system may be used to inform leader selection based on reputation signals from other participants or from central sources. Seventh, whether the mechanism is intended to be applied in permissioned or permissionless settings. Here we are particularly interested in outlier mechanisms with strong Sybil attack resistance that are applied in permissioned settings or mechanisms with weak Sybil attack resistance being applied in permissionless settings. Last, we outline the leader selection method, foremost by analysing whether the mechanism relies on (pseudo-)randomness, an election, or a pre-formed committee. Many mechanisms combine multiple factors. In such cases, we aim to call out the dominant factor.

Secondary Literature

To gain an overview of the available works and to understand which mechanisms are deemed most relevant by the research community, the secondary literature was analysed (see Table 2.5). Here, we find that most researchers align on few major mechanisms: in the 62 pieces of secondary literature analysed, only PoW, PoS, practical Byzantine fault tolerance (PBFT), delegated proof-of-stake (DPoS), and proof of elapsed time (PoET) are mentioned by more than half of the works. This foreshadows the analysis of the primary literature where we find several works that have received hardly any attention so far. These most commonly discussed mechanisms reflect the breadth of the entire field well as they include permissionless mechanisms (e.g. PoW), permissioned mechanisms (e.g. PBFT), and mechanisms that make use of physical world linking (e.g. PoET by using TEE).

Table 2.5: Secondary literature analysed during manuscript screening.

	PoW	PoS	PBFT	DPoS	PoET	Proof-of-Space	Proof-of-Burn	Proof-of-Activity	Proof-of-Importance	Raft	Proof-of-Authority	dBFT	Proof-of-Luck
S. Aggarwal and Kumar, 2021	●	●	●	●	●	●	●	●	●	○	○	○	○
A. Aggarwal *et al.*, 2021	●	●	●	●	●	●	●	○	○	○	●	○	○
Ahmad *et al.*, 2021	●	●	●	○	●	○	○	○	○	○	○	○	○
Ali Syed *et al.*, 2019	●	●	●	○	●	●	○	○	○	○	○	○	○
Alsaqqa and Almajali, 2020	●	○	●	○	○	●	●	●	●	○	●	○	○
Alsunaidi and Alhaidari, 2019	●	●	●	●	○	○	○	●	●	●	○	○	●
Andoni *et al.*, 2019	●	●	●	●	●	●	●	●	○	○	●	○	○
Andrey and Petr, 2019	●	●	○	●	○	●	●	●	●	○	○	○	○
Antal *et al.*, 2021	●	●	○	○	○	○	●	●	○	●	●	○	○
Azbeg *et al.*, 2021	●	●	●	●	●	●	●	○	○	○	●	○	○
Bach *et al.*, 2018	●	○	○	●	○	○	○	○	●	○	○	●	○
Bamakan *et al.*, 2020	●	●	●	●	●	●	●	●	●	○	○	●	○
G. Bashar *et al.*, 2019	●	●	○	○	●	●	●	●	●	○	●	○	●
Belotti *et al.*, 2019	●	●	●	●	●	○	○	○	●	○	○	○	○
Bhutta *et al.*, 2021	●	●	●	●	●	●	●	○	●	○	○	●	●
Dabbagh *et al.*, 2021	●	●	●	○	○	○	○	●	○	●	○	○	○
Dhaiouir and Assar, 2020	●	●	●	●	○	○	○	○	○	○	○	○	○
Ferdous *et al.*, 2021	●	●	○	○	○	○	●	○	●	○	○	○	○
Ferrag *et al.*, 2020	●	●	●	●	●	●	●	●	●	○	●	●	○
Fournier and Petrillo, 2020	●	○	○	○	○	○	○	○	○	○	○	○	●
X. Fu *et al.*, 2021	●	●	●	●	○	○	●	●	○	○	○	●	●
Y. Hao *et al.*, 2018	●	○	●	○	○	○	○	○	○	○	○	○	○
Hazari and Mahmoud, 2019a	●	●	●	○	○	○	●	○	●	○	○	○	○
Q. He, Guan *et al.*, 2018	●	●	○	●	●	●	●	●	●	○	○	○	●
Hellwig *et al.*, 2020	●	●	●	●	●	●	○	○	○	○	●	○	○
Honnavalli *et al.*, 2020	○	○	●	○	○	○	○	○	○	○	●	○	○
Indhuja and Venkatesulu, 2021	●	●	●	●	●	●	○	●	●	○	○	●	○
Ismail and Materwala, 2019	●	●	●	●	●	●	●	●	●	●	●	●	○
Jayabalan and N, 2021	●	●	●	○	●	○	○	○	○	●	○	○	○
Jennath and Asharaf, 2020	●	●	○	○	●	●	●	●	●	○	●	○	○
Katarya and Vats, 2021	●	●	●	●	○	○	○	○	○	●	○	○	○
Kaur *et al.*, 2021	●	●	●	●	●	○	○	○	○	○	●	○	○
Khamar and Patel, 2021	●	●	●	●	●	○	○	●	●	●	○	●	●
Lao *et al.*, 2021	●	●	●	○	○	●	●	○	●	○	●	○	○
Lashkari and Musilek, 2021	●	●	●	●	●	●	●	●	○	●	●	●	●
Lasisi and Hsu, 2019	○	○	●	○	○	○	○	○	○	○	○	○	○
Lepore *et al.*, 2020	●	●	○	○	○	○	○	○	○	○	○	○	○
MacKenzie *et al.*, 2018	●	●	●	○	○	○	○	○	○	○	○	○	●
Maple and Jackson, 2019	●	●	●	○	●	○	○	○	○	○	○	○	○
Masood and Faridi, 2018	●	●	○	○	●	●	●	○	○	○	○	○	○
Masood and Faridi, 2019	○	○	●	○	○	○	○	○	○	●	○	○	○
Mingxiao *et al.*, 2017	●	●	●	●	○	○	○	○	○	●	○	○	○

Continued on next page

Monrat *et al.*, 2019	●	●	●	●	○	○	○	○	○	○	○	○	○
Naz and Lee, 2020	●	●	●	●	●	○	○	○	○	○	○	○	○
Nguyen and Kim, 2018	●	●	●	●	●	●	●	●	○	●	○	○	●
Nijsse and Litchfield, 2020	●	●	●	●	●	○	○	○	○	○	○	○	○
Pahlajani *et al.*, 2019	●	●	○	○	●	●	○	○	○	○	○	○	●
Panda *et al.*, 2019	●	●	○	○	○	○	○	○	○	●	○	○	○
Perez *et al.*, 2019	●	●	○	○	○	○	○	○	●	○	○	○	○
Praveen *et al.*, 2021	●	●	●	●	●	●	●	●	●	●	○	●	○
Ramkumar *et al.*, 2020	●	●	●	○	●	●	●	●	○	○	○	●	○
Sharma and Jain, 2019	●	●	●	●	●	●	●	●	●	○	○	○	○
Srivastav *et al.*, 2020	●	●	●	●	○	●	●	○	○	○	○	●	○
Verma *et al.*, 2021	●	●	●	○	●	●	●	●	●	●	○	○	●
Q. Wang *et al.*, 2020	●	●	●	●	○	○	○	○	○	○	○	○	○
Wazid *et al.*, 2020	●	●	●	●	●	●	●	●	●	○	○	●	○
Xiao *et al.*, 2019	●	●	●	○	●	○	○	○	○	○	○	○	○
Xiao *et al.*, 2020	●	●	●	●	●	○	○	○	○	○	●	○	○
H. Yin *et al.*, 2020	●	●	●	○	●	●	●	○	○	●	○	○	○
Yousuf *et al.*, 2021	●	●	○	○	○	○	○	○	○	○	○	○	○
S. Zhang and Lee, 2020	●	●	●	●	○	○	○	○	○	○	○	○	○
W. Zhao *et al.*, 2019	●	●	○	○	●	●	○	○	○	○	○	○	○

●: Included, ○: Not included

Mechanisms for Democratic Processes

Only five mechanisms were categorised as targeting the facilitation of democratic processes (see Table 2.6). A key responsibility for such systems is to orchestrate voting on ledger: to support this, an underlying consensus mechanism should make block producer selection subject to a free election (A. Dhillon *et al.*, 2021), a prerequisite that is met in the Proof-of-Credibility mechanism proposed by Abuidris *et al.* (2021). The majority of the mechanisms target permissionless systems, thereby contributing to the debate about the suitability of consensus mechanisms for public democratic blockchain systems (Tozzi, 2019). Some mechanisms approach Sybil attack resistance through personal encounters in the physical world (Borge *et al.*, 2017; Brenzikofer, 2019), while others apply PoS-like mechanics to credibility scores (Abuidris *et al.*, 2021; Poupko & Talmon, 2020).

Dhillon, A., Kotsialou, G., McBurney, P., & Riley, L. (2021). Voting over a distributed ledger: An interdisciplinary perspective. *Foundations and Trends in Microeconomics*, *12*(3)

Table 2.6: Consensus mechanisms in *democracy*.

		Inc					Perm		
	SA	**+**	**-**	**Ra**	**Ph**	**Re**	**Pn**	**Pl**	**Sel**
Democratic Byzantine Fault Tolerance (Crain *et al.*, 2018)	○	○	○	○	○	○	●	○	None
Consensus for e-Democracy (Poupko & Talmon, 2020)	◐	○	○	○	○	○	○	●	
Proof-of-Credibility (Abuidris *et al.*, 2021)	◐	○	○	●	○	●	○	●	Elec.
Encointer (Brenzikofer, 2019)	◐	○	○	●	●	○	○	●	Rand.
Proof-of-Personhood (Borge *et al.*, 2017)	◐	●	○	●	●	○	○	●	Rand.

Continued on next page

Table 2.6: Consensus mechanisms in *democracy*.

		Inc					Perm		
	SA	**+**	**-**	**Ra**	**Ph**	**Re**	**Pn**	**Pl**	**Sel**

SA: Sybil attack resistance class, **Inc +**: Reward scheme, **Inc -**: Punishment scheme,
Ra: Randomisation, **Ph**: Physical world linking, **Re**: Reputation system,
Perm Pn: Permissioned, **Perm Pl**: Permissionless, **Sel Rand.**: Random leader selection,
Sel Comm.: Committee-based leader selection, **Sel Elec.**: Leader selection via election

● Strong Sybil attack resistance, ◐ Limited Sybil attack resistance, ○ No Sybil attack resistance

● Applicable, ○ Not applicable

Mechanisms for Education

Steiu (2020) identified 'digitalization and decentralization of educational certifications' and 'lifelong learning' as key drivers for the deployment of blockchain technology in an educational context. Additionally, Hsu *et al.* (2022) define the main challenges of smart education environments as 'trust, privacy, and transparency'. Consequently, some researchers suggested the application of blockchain technology to these environments. While, with only three mechanisms, the education category is sparse, some commonalities can be made out: as Table 2.7 shows, none of the proposed Sybil attack resistance schemes explicitly assume a permissionless system. The 'Group-Based Consensus for Educational Systems' (B. Wang *et al.*, 2019), for example, assumes a hierarchical system of educational stakeholders.

Steiu, M.-F. (2020). Blockchain in education: Opportunities, applications, and challenges. *First Monday*, *25*(9)

Table 2.7: Consensus mechanisms in *education*.

		Inc					Perm		
	SA	**+**	**-**	**Ra**	**Ph**	**Re**	**Pn**	**Pl**	**Sel**
Group-Based Consensus for Educational Systems (B. Wang *et al.*, 2019)	○	○	○	○	○	●	●	○	Elec.
Proof-of-Reputation (Qin *et al.*, 2018)	◐	●	○	●	○	●	○	○	
Improved DPoS (X. Liu, 2021)		○	○	○	○	○	○	○	

SA: Sybil attack resistance class, **Inc +**: Reward scheme, **Inc -**: Punishment scheme,
Ra: Randomisation, **Ph**: Physical world linking, **Re**: Reputation system,
Perm Pn: Permissioned, **Perm Pl**: Permissionless, **Sel Rand.**: Random leader selection,
Sel Comm.: Committee-based leader selection, **Sel Elec.**: Leader selection via election

● Strong Sybil attack resistance, ◐ Limited Sybil attack resistance, ○ No Sybil attack resistance

● Applicable, ○ Not applicable

Mechanisms for Energy

The energy sector is seeing a considerable amount of blockchain-related research (see Table 2.8). This is not surprising since, as Brilliantova and Thurner (2019) attest, relevant requirements are arising in the energy sector as a result of the movement towards peer-to-peer (P2P) energy trading. As Andoni *et al.* (2019) asserted, blockchain technology can empower consumers and small electricity generators. While challenges, such as regulatory uncertainty and environmental questions remain (N. Wang *et al.*, 2019), decentralised technology was found to be a good fit for blockchain technology, specifically to combat the existing security issues of smart grid systems (Hasan *et al.*, 2022). In regard to Sybil attack resistance it is worth mentioning that, apart from two (X. Fu *et al.*, 2020; B.

Andoni, M., Robu, V., Flynn, D., Abram, S., Geach, D., Jenkins, D., McCallum, P., & Peacock, A. (2019). Blockchain technology in the energy sector: A systematic review of challenges and opportunities. *Renewable and Sustainable Energy Reviews*, *100*

Hu *et al.*, 2021), all reviewed mechanisms expose limited or no Sybil attack resistance. This can, in most cases, be attributed to the fact that energy trading needs to rely on central entities, like distribution system operators, to provide reliable meter readings (Montakhabi *et al.*, 2021) both for the consumption and production of electricity. Therefore, permissioned systems (W. Cai *et al.*, 2020; Chowdhury *et al.*, 2022; Guan *et al.*, 2021; J. Guo *et al.*, 2021; M. Hu *et al.*, 2020; Huh & Kim, 2019; L. Jiang *et al.*, 2019; Q. Liu *et al.*, 2020; Moniruzzaman *et al.*, 2019; Olivares-Rojas *et al.*, 2020; L. Wang *et al.*, 2020; Yahaya *et al.*, 2020; F. Zhao *et al.*, 2020) are dominating the energy space.

Table 2.8: Consensus mechanisms in *energy*.

		Inc					**Perm**		
	SA	**+**	**-**	**Ra**	**Ph**	**Re**	**Pn**	**Pl**	**Sel**
Communicate Proof-of-Credit (X. Fu *et al.*, 2020)	●	●	○	●	○	●	○	●	Rand.
Decentralised Consensus Decision-Making (B. Hu *et al.*, 2021)	●	○	○	●	○	○	○	●	Rand.
Consensus Resource Slicing Model (M. Hu *et al.*, 2020)	○	○	○	●	○	○	●	○	Rand.
Credit-Based PoW (J. Guo *et al.*, 2021)	○	●	○	●	○	○	●	○	Rand.
Dynamic-Reputation Practical Byzantine Fault Tolerance (W. Cai *et al.*, 2020)	○	○	○	○	○	●	●	○	Elec.
Enhanced Proof-of-Work (Keshk *et al.*, 2020)	○	○	○	○	○	○	○	○	
Fast, Secure and Distributed Consensus Mechanism for Energy Trading Among Vehicles using Hashgraph (Bansal & Bhatia, 2020)	○	○	○	●	○	●	○	○	None
Hyper Delegation Proof-of-Randomness (Huh & Kim, 2019)	○	○	○	○	○	○	●	○	
Improved Proof of Work (Q. Liu *et al.*, 2020)	○	○	○	●	○	○	●	○	Rand.
Lightweight DPoS for Energy Transmitters (L. Jiang *et al.*, 2019)	○	●	●	●	○	○	●	○	
Proof-of-Credit-Threshold (L. Wang *et al.*, 2020)	○	○	○	○	○	○	●	○	
Proof-of-Energy-Generation (Moniruzzaman *et al.*, 2019)	○	●	○	○	○	○	●	○	
Proof-of-Cooperation (R. Khalid *et al.*, 2021)	○	○	○	●	○	●	○	○	Elec.
Proof-of-Efficiency (Olivares-Rojas *et al.*, 2020)	○	○	○	○	○	●	●	○	
Proof-of-Generation (F. Zhao *et al.*, 2020)	○	○	○	●	○	○	●	○	Rand.
Proof-of-Work based on Reputation (Yahaya *et al.*, 2020)	○	○	○	●	○	○	●	○	Comm.
Credibility Consensus (Guan *et al.*, 2021)	◐	●	○	○	○	○	●	○	Elec.
Credit-Based Concurrent Block Building Consensus (N. Liu *et al.*, 2020)	◐	●	○	●	○	●	○	○	Elec.
Lightweight Credibility-Based Equity Proof Consensus (X. Lu *et al.*, 2019)	◐	●	○	●	○	○	○	○	Rand.
Cross-Layer Trust-Based Consensus (Chowdhury *et al.*, 2022)		●	○	●	○	●	●	○	Rand.
Proof-of-Benefit (C. Liu *et al.*, 2019a, 2019b, 2021; H. Liu *et al.*, 2018)		○	○	○	○	●	○	●	

SA: Sybil attack resistance class, **Inc +**: Reward scheme, **Inc -**: Punishment scheme, **Ra**: Randomisation, **Ph**: Physical world linking, **Re**: Reputation system, **Perm Pn**: Permissioned, **Perm Pl**: Permissionless, **Sel Rand.**: Random leader selection, **Sel Comm.**: Committee-based leader selection, **Sel Elec.**: Leader selection via election

● Strong Sybil attack resistance, ◐ Limited Sybil attack resistance, ○ No Sybil attack resistance

● Applicable, ○ Not applicable

General Purpose Mechanisms

This catch-all category captures the majority (57 %) of all algorithms analysed. It contains all algorithms that are not industry-specific, as well as some targeted at exotic domains, such as astronautics (Ling *et al.*, 2020). In this category, many schemes with strong Sybil attack resistance are contained, including those found in subsection 2.7 to be particularly popular in the field (Bentov *et al.*, 2014; Castro & Liskov, 1999; Dziembowski *et al.*, 2015; Intel, 2018; Larimer, 2017; Milutinovic *et al.*, 2016; NEM, 2018; Ongaro & Ousterhout, 2014; QuantumMechanic, 2011; Stewart, 2012). As seen in Table 2.9, mechanisms with strong Sybil attack resistance are commonly targeted towards permissionless systems and often apply random leader selection strategies. Mechanisms without Sybil attack resistance, unsurprisingly, target permissioned systems and make use of election-based leader selection. In some cases, these mechanisms follow deterministic leader selection methods that are not captured by our categorisation. Schemes with limited Sybil attack resistance make heavy use of physical world linking to achieve Sybil attack resistance: some by using CPUs (Ahmed-Rengers & Kostiainen, 2018) or TEEs (Andreina *et al.*, 2022; G. D. Bashar *et al.*, 2020; Carlyle *et al.*, 2021; X. Chen *et al.*, 2018; Intel, 2018; Martin *et al.*, 2018; Milutinovic *et al.*, 2016), others by relying on IP addresses (Auvolat *et al.*, 2021; Cabellos Aparicio, 2018; X. Cong *et al.*, 2021), mobile phones (Martinez *et al.*, 2020), or physical proximity (Amoretti *et al.*, 2018; Chatzopoulos *et al.*, 2016, 2020; Z. Jiang *et al.*, 2020; Pournaras, 2020). Incentive schemes are common in mechanisms with strong Sybil attack resistance, likely because these are expected in permissionless systems.

Ling, X., Gao, Z., Le, Y., You, L., Wang, J., Ding, Z., & Gao, X. (2020). Satellite-aided consensus protocol for scalable blockchains. *Sensors*, *20*(19), Article 5616

Table 2.9: *General* consensus mechanisms.

		Inc					Perm		
	SA	**+**	**-**	**Ra**	**Ph**	**Re**	**Pn**	**Pl**	**Sel**
Bitcoin-NG (Das, 2021)	●	●	○	●	○	○	○	●	Rand.
Elliptic curve discrete logarithm problem-based PoW (Meneghetti *et al.*, 2020)	●	●	○	●	○	○	○	●	Rand.
PoS Based on Credit Rewards and Punishments (L. Li *et al.*, 2020)	●	○	○	○	○	●	○	●	Elec.
PoS Based on Verifiable Random Functions (Gilad *et al.*, 2017)	●	●	○	●	○	○	○	●	Rand.
PoS for Bitcoin Sidechains (Longo *et al.*, 2020)	●	●	●	●	○	○	○	●	Rand.
PoS with Behavior Score and Trust Rating (Y. Cheng *et al.*, 2019)	●	●	○	○	○	●	○	●	Elec.
PoS with Waiting-Time First-Price Auctions (Deuber *et al.*, 2020)	●	●	○	●	○	○	○	●	Rand.
PoS with Weighted Voting (S. Leonardos *et al.*, 2019, 2020)	●	●	○	●	○	●	○	●	Rand.
PoW Based on Power Analysis of Low-End Microcontrollers (H. Kim *et al.*, 2020)	●	●	○	●	○	○	○	●	Rand.
PoW on quadratic multivariate equations (J. Chen *et al.*, 2021a, 2021b)	●	○	○	●	○	○	○	●	Rand.
PoW with Early Stage PoS (L. Chen *et al.*, 2018)	●	●	○	●	○	○	○	●	Rand.
PoW with Personalised Difficulty Adjustment (Chou *et al.*, 2018)	●	●	○	●	○	○	○	●	Rand.
PoW with Quantum-Resistant Hash Collision (D. Aggarwal *et al.*, 2018)	●	●	○	○	○	○	○	●	Rand.

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verifiable random functionss for PoW (R. Han *et al.*, 2020)	●	○	○	●	○	○	○	●	Rand.
Albatross (Berrang *et al.*, 2019)	●	●	●	●	○	○	○	●	Rand.
Alt-PoW (Sharkey & Tewari, 2019)	●	●	○	●	○	○	○	●	Rand.
Attack-Tolerant PoW (Kitakami & Matsuoka, 2018)	●	●	○	●	○	○	○	●	Rand.
B4SDC (G. Liu *et al.*, 2022)	●	●	○	●	○	○	○	●	Rand.
BeaconBlocks (Hartl *et al.*, 2019)	●	○	○	●	○	○	○	○	Rand.
Block Maturity Level (Memon *et al.*, 2018)	●	●	○	●	○	○	○	●	Rand.
Bobtail (Bissias & Levine, 2017)	●	●	○	●	○	○	○	●	Rand.
Chains of Activity (Bentov *et al.*, 2016)	●	●	○	●	○	○	○	●	Rand.
Client-Assisted Consensus (J. Liu & Ren, 2022)	●	○	○	●	○	○	○	○	
CloudPoS (Tosh *et al.*, 2018)	●	○	○	○	○	○	○	●	Elec.
Composite Framework Leveraging Proof-of-Stake and Proof-of-Work (Baudlet *et al.*, 2020)	●	●	○	●	○	●	○	●	Rand.
Conflux (C. Li *et al.*, 2020)	●	●	○	●	○	○	○	●	Rand.
Crux (P. Li *et al.*, 2018)	●	○	○	●	○	○	○	●	Elec.
Cumulative Proof-of-Work (Boyen *et al.*, 2018)	●	●	○	●	○	○	○	●	Rand.
Delegated Proof of Stake with Downgrade (F. Yang *et al.*, 2019)	●	○	○	●	○	●	○	●	Rand.
Delegated Proof-of-Reputation (Do *et al.*, 2019)	●	●	○	○	○	●	○	●	Elec.
Delegated Proof-of-Stake (Larimer, 2017)	●	●	○	●	○	○	○	●	Elec.
Deterministic Proof-of-Work (Z. Cheng *et al.*, 2018)	●	○	○	○	○	○	○	●	Elec.
Equihash (Biryukov & Khovratovich, 2017)	●	●	○	●	○	○	●	●	Rand.
Error-Correction Code Based Proof-of-Work (Jung & Lee, 2020)	●	●	○	○	○	○	○	●	Rand.
Estimable PoW (Hsueh & Chin, 2017)	●	●	○	●	○	○	○	●	Rand.
extended PoS (Saad *et al.*, 2021)	●	○	○	●	○	○	○	●	Rand.
Fair Proof-of-Work System With Computing Power Rating (Nakahara & Inaba, 2018)	●	●	○	●	○	●	○	●	Rand.
Fair selection protocol for committee-based permissionless blockchains (Y. Liu, Liu, Zhang & Yu, 2020)	●	○	○	○	○	○	●	●	Rand.
Fantômette (Azouvi *et al.*, 2018a)	●	●	●	●	○	○	○	●	Rand.
Filtered Proof-of-Work (Ashik *et al.*, 2020)	●	○	○	●	○	○	○	●	Rand.
Goshawk (Wan *et al.*, 2019)	●	●	○	●	○	○	○	●	Rand.
Greedy Observed Largest Forest (J. Zhao *et al.*, 2020)	●	○	○	●	○	○	○	●	Rand.
HashCore (Georghiades *et al.*, 2019)	●	○	○	●	○	○	○	●	Rand.
Hybrid PoW/PoS (Kolokotronis *et al.*, 2019)	●	○	○	●	○	●	○	○	Rand.
Hybrid PoW/PoS (Harvilla & Du, 2019)	●	●	○	●	○	○	○	●	Rand.
Hybrid Consensus with flexible Proof-of-Activity (Z. Liu *et al.*, 2019)	●	●	○	●	○	○	○	●	Rand.
Improved DPoS with K-Means (W. Liu *et al.*, 2021)	●	●	○	●	○	●	○	○	Elec.
Interactive Proof-of-Stake (Chepurnoy, 2016)	●	●	○	●	○	○	○	●	Rand.
Itsuku (Coelho *et al.*, 2017)	●	○	○	●	○	○	○	●	Rand.
MaGPoS (Mckinnon, 2019)	●	●	●	○	○	○	○	●	
Multi-tokens Proof of Stake (Pang, 2020)	●	●	○	●	○	○	○	●	Rand.
Multiple Winners Proof of Work (Y. Xu & Huang, 2019)	●	●	○	●	○	○	○	●	Rand.
NeuCoin (Davarpanah *et al.*, 2015)	●	●	●	●	○	○	○	●	Rand.
Open Representative Voting (Morais *et al.*, 2020)	●	○	○	○	○	○	○	●	Elec.
Optical Proof-of-Work (Dubrovsky *et al.*, 2019)	●	●	○	●	○	○	○	●	Rand.
Ouroboros Crypsinous (Kerber *et al.*, 2019)	●	○	○	○	○	○	○	●	Elec.
Ouroboros Genesis (Badertscher *et al.*, 2018)	●	○	○	○	○	○	○	●	Rand.

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Ouroboros Praos (David *et al.*, 2018)	●	○	○	●	○	○	○	●	Rand.
Parallel Proof-of-Work (Hazari & Mahmoud, 2019b)	●	●	○	●	○	○	○	●	Rand.
Penalty by Consensus in PoW (Adewumi & Liwicki, 2020)	●	●	●	○	○	○	○	●	Rand.
Pixel (Drijvers *et al.*, 2020)	●	○	○	●	○	○	○	●	Rand.
PoolCoin (Kaci & Rachedi, 2020)	●	●	○	●	○	●	○	●	Rand.
Prism (Bagaria *et al.*, 2019)	●	●	○	●	○	○	○	●	Rand.
Private Proof-of-Effort (Alberini *et al.*, 2015)	●	○	○	●	○	○	○	●	Rand.
Proof of Adjourn (Sayeed & Marco-Gisbert, 2020)	●	○	○	●	○	○	○	●	Rand.
Proof of Contribution (Xue *et al.*, 2018)	●	●	○	●	○	●	○	●	Rand.
Proof of Experience (Masseport, Darties *et al.*, 2020)	●	●	○	●	○	○	○	●	Rand.
Proof of Segmented Work (Monem *et al.*, 2020)	●	●	○	●	○	○	○	○	Rand.
Proof-of-Accumulated-Work (Amar & Zilpa, 2019)	●	○	○	●	○	○	○	●	Rand.
Proof-of-Activity (Bentov *et al.*, 2014)	●	●	○	●	○	○	○	●	Rand.
Proof-of-Behavior (L.-e. Wang *et al.*, 2021)	●	○	○	●	○	●	○	●	
Proof-of-Burn (Stewart, 2012)	●	○	○	○	○	○	○	●	
Proof-of-Credit (X. Han *et al.*, 2019)	●	○	●	●	○	●	○	○	Rand.
Proof-of-Discrete Logarithm (K. Huang *et al.*, 2020)	●	●	○	●	○	○	○	●	Rand.
Proof-of-Equivalence (Chatzopoulos *et al.*, 2020)	●	○	○	○	○	●	○	●	Rand.
Proof-of-Human-Work (Blocki & Zhou, 2016)	●	●	○	●	○	○	○	●	Rand.
Proof-of-Importance (NEM, 2018)	●	●	○	○	○	○	○	●	
Proof-of-Interaction (Abegg *et al.*, 2021)	●	●	○	●	○	○	○	●	Rand.
Proof-of-Lottery (S.-b. Lee *et al.*, 2020)	●	●	○	●	○	○	○	●	Rand.
Proof-of-Participation (Nandwani *et al.*, 2019)	●	●	○	●	○	○	○	●	Rand.
Proof-of-Prestige (Krol *et al.*, 2019)	●	○	○	●	○	●	○	●	Rand.
Proof-of-Replicated-Storage (Damgård *et al.*, 2019)	●	○	○	○	○	○	○	●	
Proof-of-Reproducibility (Al-Mamun *et al.*, 2018)	●	○	○	○	○	●	○	●	Rand.
Proof-of-Reputation with Nakamoto Fallback (Kleinrock *et al.*, 2020)	●	○	○	○	○	●	○	●	Elec.
Proof-of-Space (Dziembowski *et al.*, 2015)	●	○	○	○	○	○	○	●	
Proof-of-Stake (QuantumMechanic, 2011)	●	●	○	○	○	○	○	●	
Proof-of-Stake for Bitcoin Subchains (Bartoletti *et al.*, 2017)	●	●	○	●	○	○	○	●	Rand.
Proof-of-Strategy (X. Cai *et al.*, 2020)	●	●	○	●	○	●	○	●	Elec.
Proof-of-Work (Castro & Liskov, 1999; Nakamoto, 2008)	●	●	○	●	○	○	○	●	Rand.
Proof-of-Work Applied to the Clique Problem (Bag *et al.*, 2016)	●	○	○	●	○	○	○	●	Rand.
Proof-of-Work Based on Analog Hamiltonian (Kalinin & Berloff, 2018)	●	○	○	●	○	○	○	●	Rand.
Proof-of-Work on the Inflation Propensity of Collatz Orbits (Bocart, 2018)	●	○	○	○	○	○	○	●	Rand.
Reputation Based Hybrid Consensus (Y. Sun *et al.*, 2020)	●	●	○	●	○	○	○	○	Rand.
Robust Proof-of-Stake (A. Li *et al.*, 2020)	●	○	○	○	○	○	○	○	Rand.
Roll-DPoS (X. Fan & Chai, 2018)	●	●	○	●	○	○	○	●	Rand.
Rollerchain (Chepurnoy *et al.*, 2016)	●	●	○	●	○	○	○	●	Rand.
Satellite-Aided Consensus (Ling *et al.*, 2020)	●	○	○	○	○	○	○	●	
Service-Zone-Based Hierarchical Consensus (Kwak *et al.*, 2020)	●	○	○	●	○	○	○	●	Rand.
SklCoin (Jaroucheh *et al.*, 2020)	●	○	○	●	○	○	○	●	Rand.
Solida (I. Abraham *et al.*, 2018)	●	●	○	○	○	○	○	●	Rand.

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Thinkey (S. Chen *et al.*, 2019)	●	●	○	●	○	○	○	●	Rand.
Time-Memory-Data Trade-Off (Mihaljevic, 2020)	●	●	○	●	○	○	○	●	Rand.
Proof-of-Trusted-Execution-Environment-Stake (Andreina *et al.*, 2022)	●	○	○	●	●	○	○	●	Elec.
BFT with Satellite Chains (W. Li *et al.*, 2017)	○	○	○	○	○	○	●	○	
PBFT with Node Quality Control (X. Bao, 2020)	○	○	○	○	○	●	●	○	Elec.
PoW with Integer Prime Factorisation (Janjanam *et al.*, 2019)	○	●	○	●	○	○	○	●	Rand.
Adaptive Wide-Area Replication (Berger *et al.*, 2022)	○	○	○	●	○	○	●	○	Elec.
AdRaft (W. Fu *et al.*, 2021)	○	○	○	●	○	○	●	○	Elec.
Alzahrani and Bulusu's Decentralised Consensus Protocol Utilizing Game Theory and Randomness (Alzahrani & Bulusu, 2018)	○	○	○	●	○	○	●	○	Elec.
Amoeba Paxos (Tan & Golab, 2021)	○	○	○	○	○	○	●	○	Elec.
Assigned-Majority-Validation (Jill, 2018)	○	○	○	○	○	○	●	○	
Auction-Based Consensus (Ai *et al.*, 2020)	○	○	○	○	○	○	●	○	Elec.
Authorised Proof of Stake (Van Toan *et al.*, 2018)	○	○	○	●	○	○	●	○	Rand.
BLIC (Bose *et al.*, 2018)	○	○	○	●	○	○	●	○	Rand.
Blockchain for the Common Good (Jain & Simha, 2018)	○	●	○	○	○	○	●	○	Elec.
Blockchain-Based Federated Learning Framework with Committee Consensus (Y. Li, Chen *et al.*, 2021)	○	○	○	○	○	○	●	○	Elec.
Byzantine Set Union Consensus (Dold, 2019)	○	○	○	●	○	○	●	○	Elec.
Casanova (Butt *et al.*, 2018)	○	○	○	○	○	○	○	○	None
Caucus (Azouvi *et al.*, 2018b)	○	○	○	●	○	○	○	●	Rand.
CloudPoS (with Cloud service provider Invovlement) (Tosh *et al.*, 2018)	○	○	○	○	○	○	●	○	Elec.
Coinami (Ileri *et al.*, 2016)	○	○	○	●	○	○	●	○	Rand.
Committee-Based Byzantine Consensus (Meng *et al.*, 2018)	○	○	○	●	○	○	●	○	Elec.
Consensus Through Herding (Hubert Chan *et al.*, 2019)	○	○	○	○	○	○	●	○	
Credence-Based Consensus (G. He *et al.*, 2020, 2021)	○	○	○	●	○	●	●	○	Elec.
Cross-Application Permissioned Blockchain (Amiri *et al.*, 2019a)	○	○	○	○	○	○	●	○	
Dynamic PBFT (X. Hao *et al.*, 2018)	○	○	○	○	○	○	●	○	Rand.
DagGrid (W. Chen *et al.*, 2021)	○	○	○	●	○	●	●	○	Rand.
Delegate Consensus Algorithm (K. Lei *et al.*, 2020)	○	○	○	●	○	○	○	○	Rand.
Delegated Adaptive Byzantine Fault Tolerance (Deng, 2019)	○	○	○	○	○	○	●	○	Elec.
Delegated Proof of Economic Value (Deng, 2019)	○	○	○	○	○	●	●	○	
Dependability-Rank-based Consensus (J. Chen *et al.*, 2018)	○	○	○	●	○	●	●	○	Rand.
DEXON (T.-Y. Chen *et al.*, 2018)	○	○	○	●	○	○	●	○	Rand.
DFINITY (Hanke *et al.*, 2018)	○	○	○	●	○	○	○	●	Rand.
Dynamic Hierarchical Byzantine Fault-Tolerant Consensus Based on Credit (F. Li *et al.*, 2020)	○	○	○	○	○	●	●	○	
Egalitarian Practical Byzantine Fault Tolerance (L. He & Hou, 2019)	○	○	○	○	○	○	●	○	
Electronic Identification, Authentication and Trust Services Validating Indy-Plenum (A. Abraham *et al.*, 2018)	○	○	○	○	○	○	●	○	Comm.
Elpis (Garg *et al.*, 2019)	○	○	○	●	○	○	●	○	Elec.
Extensible-PBFT (Y. Li *et al.*, 2019)	○	○	○	●	○	○	●	○	Elec.

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Fast leader-based, randomised Byzantine Agreement (Locher, 2020)	○	○	○	●	○	○	●	○	Rand.
FastBFT (J. Liu *et al.*, 2019)	○	○	○	○	○	○	●	○	
Geo-Scale Byzantine Fault Tolerance (S. Gupta *et al.*, 2020)	○	○	○	○	○	○	●	○	
Graph Learning BFT (Oh *et al.*, 2020)	○	○	○	○	○	○	●	○	
Group-Based Optimised Practical Byzantine Fault Tolerance (Z. Bao *et al.*, 2021)	○	○	○	○	○	●	●	○	Elec.
HotStuff (M. Yin *et al.*, 2019)	○	○	○	○	○	○	●	○	
Hybrid Byzantine Agreement (Kuo *et al.*, 2020)	○	○	○	●	○	○	●	○	Rand.
Identifiable Practical Byzantine Fault Tolerance (L. Liang *et al.*, 2020)	○	○	○	●	○	○	●	○	Elec.
Istanbul BFT Consensus (Moniz, 2020)	○	○	○	○	○	○	●	○	
Leader-Stable Fast Byzantine Fault Tolerance (Y. Liu, Liu, Li *et al.*, 2020)	○	○	○	○	○	○	○	○	
LFT2 (Lim *et al.*, 2020)	○	○	○	○	○	○	○	○	Elec.
Lisk-BFT (Hackfeld, 2019)	○	○	○	○	○	○	○	○	Elec.
Majority vOting Cellular Automata (Z. Wang, 2018)	○	○	○	○	○	○	●	○	
Mchain Consensus (B. Zhao *et al.*, 2018)	○	○	○	●	○	○	●	○	Rand.
Mobile Crowdsourcing Chain (W. Feng & Yan, 2019)	○	●	○	●	○	○	○	○	Elec.
Multi-Block BFT (S. Kim *et al.*, 2020)	○	○	○	○	○	○	●	○	
Multi-Round Concession Negotiation (Y. Jiang *et al.*, 2021)	○	●	●	○	○	●	○	○	Elec.
Multi-Supervised Permissioned Blockchain (Z. Bao *et al.*, 2019)	○	○	○	○	○	○	●	○	Comm.
Multisignature-BFT (C.-W. Chen *et al.*, 2018)	○	○	○	●	○	○	●	○	Elec.
Musch (Jalalzai & Busch, 2018)	○	○	○	○	○	○	●	○	
Open Business Environment BFT (W. Wu & Gao, 2020)	○	○	○	○	○	●	○	○	
PeerBFT (J. Ma *et al.*, 2020)	○	○	○	○	○	○	●	○	
Personal Archive Service System (Z. Chen *et al.*, 2020)	○	○	○	○	○	○	●	○	Comm.
POA-PBFT (J. Zhang *et al.*, 2021)	○	○	○	○	○	○	●	○	Comm.
Practical Byzantine Fault Tolerance (Castro & Liskov, 1999)	○	○	○	○	○	○	●	○	
Practical Layered Consensus Mechanism (S. He *et al.*, 2019)	○	○	○	○	○	○	○	○	
Proof of Rest (Shen *et al.*, 2020)	○	○	○	●	○	○	●	○	Rand.
Proof of Training Quality (Y. Lu *et al.*, 2020)	○	○	○	○	○	○	●	○	
Proof of Usage (Masseport, Lartigau *et al.*, 2020)	○	●	○	●	○	○	●	○	Rand.
Proof-of-Accuracy (Kudin *et al.*, 2019)	○	○	○	●	○	○	○	○	Rand.
Proof-of-Achievement (Komiya & Nakajima, 2019b)	○	●	○	○	○	○	●	○	
Proof-of-Activity (Belfer *et al.*, 2020)	○	○	○	●	○	○	●	○	Rand.
Proof-of-Atomicity (E. Lee & Yoon, 2019; E. Lee *et al.*, 2020)	○	○	○	○	○	○	●	○	Comm.
Proof-of-Authority (Wood, 2015)	○	○	○	○	○	○	●	○	
Proof-of-Business (Q. Hu *et al.*, 2020)	○	●	○	●	○	●	●	○	Elec.
Proof-of-Communication (Y. Chen *et al.*, 2019)	○	○	○	○	○	●	○	○	Elec.
Proof-of-Contribution (Komiya & Nakajima, 2019a)	○	●	○	○	○	●	●	○	
Proof-of-Lucky-Id (Ogawa *et al.*, 2018)	○	○	○	●	○	○	●	●	Rand.
Proof-of-Majority (J.-T. Kim *et al.*, 2018)	○	○	○	●	○	○	●	○	Rand.
Proof-of-Participation-and-Fees (X. Fu *et al.*, 2018)	○	○	○	○	○	●	○	●	Elec.
Proof-of-Points (Z. Lin *et al.*, 2020)	○	●	○	●	○	○	○	●	Elec.
Proof-of-Reputation (F. Gai *et al.*, 2018)	○	○	○	○	○	●	●	○	Elec.

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Proof-of-Review (Matsuyama, 2021)	○	●	○	○	○	○	○	○	
Proof-of-Sovereignty (Hegadekatti & S G, 2016)	○	○	○	○	○	○	●	○	
Proof-of-Stack (Barhanpure *et al.*, 2019)	○	○	○	●	○	●	●	○	Rand.
Proof-of-Trust (Zou *et al.*, 2019)	○	○	○	●	○	●	●	●	Comm.
Proof-of-Validation (Hassanzadeh-Nazarabadi *et al.*, 2021)	○	●	○	●	○	○	○	●	Rand.
Proof-of-Verifying (Kang *et al.*, 2020)	○	●	●	●	○	○	●	○	Rand.
Proof-of-Vote (K. Li *et al.*, 2020)	○	○	○	●	○	○	●	○	Comm.
Proof-of-Win (H. Gupta & Janakiram, 2019)	○	○	○	●	○	○	●	○	Rand.
Proteus (Jalalzai *et al.*, 2019a, 2019b)	○	○	○	○	○	○	●	○	
Raft (Ongaro & Ousterhout, 2014)	○	○	○	○	○	○	●	○	Elec.
Randition (Moh *et al.*, 2020)	○	○	○	●	○	○	●	○	Rand.
Random Leader Selection based on Credit Value (G. He *et al.*, 2018)	○	○	○	●	○	●	●	○	Elec.
Randomization to PoW (Sumathy Kingslin, 2019)	○	●	○	●	○	○	○	●	Rand.
Regulated Bitcoin (Ahuja *et al.*, 2021)	○	●	○	●	○	○	●	○	Rand.
Reputaion-Based BFT (Dingyu *et al.*, 2020)	○	○	○	●	○	●	●	○	Elec.
Reputation-based Byzantine Fault Tolerance (K. Lei *et al.*, 2018)	○	○	○	●	○	●	○	○	Rand.
Rift (F. A. Khan *et al.*, 2019)	○	○	○	○	○	○	○	○	Elec.
Robust Byzantine Agreement (Kuo *et al.*, 2020)	○	○	○	●	○	○	●	○	Rand.
Rock-Scissors-Paper (D.-H. Kim *et al.*, 2019)	○	●	●	○	○	○	●	○	
Rotating Multiple Random Sampling (Y. Fan *et al.*, 2020)	○	○	○	●	○	○	●	○	Rand.
Saguaro (Amiri *et al.*, 2021)	○	○	○	○	○	○	●	○	
Scalable Byzantine Fault Tolerance (Golan Gueta *et al.*, 2019)	○	○	○	●	○	○	●	○	Rand.
Scalable Hierarchical Byzantine Fault Tolerance (Y. Li, Qiao & Lv, 2021)	○	○	○	○	○	○	●	○	Elec.
Scalable Network-Coded PBFT (Choi *et al.*, 2019)	○	○	○	○	○	○	●	○	Elec.
Scalable Practical Byzantine Fault Tolerance with Short-Lived Signature Schemes (X. Fan, 2018)	○	○	○	●	○	○	●	○	Comm.
Score Voting-based BFT Consensus (C. Li *et al.*, 2021)	○	○	○	○	○	●	●	○	Elec.
Security-Aware Genetic Algorithm based Practical Byzantine fault Tolerance (Kashyap *et al.*, 2019)	○	○	○	○	○	○	●	○	
Self-Stabilizing Byzantine Consensus (Binun *et al.*, 2019)	○	○	○	●	○	○	●	○	Rand.
separate-proof-of-deep-learning (X. Luo *et al.*, 2021)	○	○	○	○	○	○	○	○	
Sharding Permissioned Blockchains Over Network Clusters (Amiri *et al.*, 2019b)	○	○	○	○	○	○	●	○	
Streamlet (B. Y. Chan & Shi, 2020)	○	○	○	○	○	○	●	○	Elec.
Torneo (Lucas & Paez, 2019)	○	●	○	●	○	○	●	○	Rand.
Two-Tier Voting System Architecture (M. H. Khalid *et al.*, 2020)	○	○	○	○	○	○	●	●	
uMine (Kopp *et al.*, 2018)	○	○	○	○	○	○	●	○	
Unitary Interchain Network (H. Guo, Zheng *et al.*, 2018)	○	○	○	○	○	○	○	○	
Weak Centralised Consensus Mechanism with Incentive Effects (Mu *et al.*, 2019)	○	●	●	●	○	○	○	○	Rand.
Weight of Authentication Byzantine Fault Tolerance (G. Heo *et al.*, 2020)	○	○	○	○	○	○	○	●	
What, Where, How much (Mazurok *et al.*, 2019)	○	●	○	○	○	○	●	○	

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Efficient, General, and Scalable Consensus (X. Chen *et al.*, 2018)	○	○	○	●	●	○	●	○	Rand.
PoS with Robust Round Robin (PoW Bootstrap) (Ahmed-Rengers & Kostiainen, 2018)	◐	●	○	○	○	○	○	●	Comm.
PoUW as a Problem-Solving Market (Dotan & Tochner, 2020)	◐	●	○	●	○	○	○	●	Rand.
Blockchain Reputation-Based Consensus (de Oliveira *et al.*, 2020)	◐	●	○	●	○	●	○	●	Elec.
Circle of Trust (Ravindran, 2019)	◐	●	○	●	○	●	●	●	Elec.
Consensus of Trust (Lv *et al.*, 2020)	◐	○	○	●	○	●	○	○	Rand.
Consensus-based Oracle Protocol for the Secure Trade of Digital Goods (M. Muller *et al.*, 2020)	◐	●	○	●	○	○	○	○	Rand.
Credit-Based Verifier Selection with Double Consensus (D. Liang *et al.*, 2018)	◐	○	○	○	○	●	○	○	
Decision-Theoretic Online Learning Consensus (Bugday *et al.*, 2019)	◐	○	○	●	○	●	○	●	Rand.
Delegated Proof-of-Reputation (Z. Hu *et al.*, 2020)	◐	●	○	●	○	●	●	○	Elec.
Delegated Proof-of-Reputation (An *et al.*, 2020)	◐	○	○	●	○	●	○	●	Elec.
Green-PoW (Lasla *et al.*, 2022)	◐	●	○	●	○	○	○	●	Rand.
Guru (Biryukov *et al.*, 2017)	◐	○	○	●	○	●	○	●	Rand.
Identity-Augmented Proof-of-Stake (Platt & McBurney, 2021b)	◐	○	○	○	○	●	○	●	Rand.
Permissionless Proof-of-Reputation-X (Bou Abdo *et al.*, 2021)	◐	○	○	●	○	●	○	●	Rand.
Proof-of-Balance (Ehrlich & Guzova, 2019)	◐	○	○	○	○	○	○	●	
Proof-of-Bid (W. K. Chan *et al.*, 2020)	◐	●	○	●	○	●	○	●	Rand.
Proof-of-Human-Engagement (Q. Chen *et al.*, 2019)	◐	●	○	●	○	○	○	●	Rand.
Proof-of-Networking (Ghiro *et al.*, 2018)	◐	●	○	●	○	○	○	○	
Proof-of-Notarised-Work (Abubakar *et al.*, 2020)	◐	●	○	●	○	○	○	●	Rand.
Proof-of-Review (D. Khan *et al.*, 2021)	◐	○	○	●	○	●	○	●	Elec.
Proof-of-Spending (C. Liu, 2018)	◐	●	○	●	○	○	○	●	Rand.
Proof-of-Stake with Time Staking (Burmaka *et al.*, 2021)	◐	○	○	●	○	○	○	●	Rand.
Proof-of-Trust (Bahri & Girdzijauskas, 2018, 2019)	◐	○	○	●	○	●	○	●	Rand.
Proof-of-Unique-Human (Hajialikhani & Jahanara, 2018)	◐	○	○	○	○	●	○	●	
Reputation Consensus (Biryukov & Feher, 2020)	◐	○	○	●	○	●	○	●	Rand.
Reverse Hash Chain (D. Kim & Lee, 2021)	◐	○	○	●	○	○	○	○	Rand.
Semada Proof-of-Reputation (Calcaterra & Kaal, 2020)	◐	●	○	●	○	●	○	●	Rand.
Software Guard Extension-enabled decentralised intrusion detection framework (G. Liu *et al.*, 2021)	◐	●	○	○	○	○	○	○	Rand.
Sybil Tolerant Equality Protocol (H. Shin *et al.*, 2020)	◐	●	●	●	○	●	○	●	
token age based consensus (S. Sun *et al.*, 2019)	◐	●	○	○	○	●	○	○	
Trust Consensus Protocol (Shala *et al.*, 2019)	◐	○	○	●	○	●	○	○	Elec.
Twice Verifications and Consensuses of Blockchain (An, Liang *et al.*, 2019; An, Yang *et al.*, 2019)	◐	○	○	○	○	●	●	●	Elec.
PoS with Robust Round Robin (Intel Intel Software Guard Extensions Variant) (Ahmed-Rengers & Kostiainen, 2018)	◐	●	○	○	●	○	○	●	Comm.
Basalt (Auvolat *et al.*, 2021)	◐	○	○	●	●	○	○	●	Rand.
DTNB (X. Cong *et al.*, 2021)	◐	○	○	●	●	○	○	●	Rand.
LocalCoin (Chatzopoulos *et al.*, 2016)	◐	●	○	●	●	○	○	●	Rand.

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Proof of Block Inclusion (Carlyle *et al.*, 2021)	◐	●	○	○	●	○	○	●	Rand.
Proof of Elapsed Time (Intel, 2018)	◐	○	○	●	●	○	○	●	Rand.
Proof of Luck (Milutinovic *et al.*, 2016)	◐	○	○	●	●	○	○	○	Rand.
Proof of Social Contact (Martinez *et al.*, 2020)	◐	○	○	○	●	○	○	●	
Proof of witness presence (Pournaras, 2020)	◐	○	○	○	●	○	○	○	
Proof-of-Context (Chatzopoulos *et al.*, 2020)	◐	○	○	○	●	●	○	●	Rand.
Proof-of-Location (Amoretti *et al.*, 2018)	◐	○	○	●	●	●	○	●	Rand.
Proof-of-Queue (G. D. Bashar *et al.*, 2020)	◐	○	○	●	●	○	●	○	Rand.
Rationality-proof consensus (Martin *et al.*, 2018)	◐	●	●	○	●	○	○	●	
Staked IP-Address Selection (Cabellos Aparicio, 2018)	◐	○	○	●	●	○	○	●	Rand.
Sybil-Proof Wireless Network Coordinate Based Byzantine Consensus (Z. Jiang *et al.*, 2020)	◐	○	○	○	●	○	○	●	Elec.
DPoS with Quantum Entanglement (Y.-L. Gao *et al.*, 2020)		○	○	○	○	○	○	○	
Delegated Byzantine Fault Tolerance (Neo, 2022)		○	○	○	○	○	○	○	Elec.
Delegation Based Scalable Byzantine False [*sic*] Tolerance Consensus (Y. Liu, Ai *et al.*, 2020)		○	○	○	○	○	○	●	
Fast Probabilistic Consensus with Weighted Votes (Müller *et al.*, 2021)		○	○	●	○	○	●	●	Rand.
Green Mining (Jacquet & Mans, 2019, 2020)		○	○	●	○	○	○	●	Rand.
Permissioned Trusted Trading Network Consensus Algorithm (Min *et al.*, 2016)		○	○	●	○	○	●	○	Rand.
Proof of Kernel Work (Lundbæk *et al.*, 2018)		○	○	○	○	●	○	○	Rand.
Proof-by-Approval (Akhilesh *et al.*, 2020)		○	○	○	○	○	●	○	Comm.
Proof-of-Behaviour (Grollemund *et al.*, 2021)		●	○	●	○	●	●	●	
Proof-of-Belief (Abramowicz, 2016)		●	○	○	○	○	○	●	
Proof-of-Phone (J. M. Kim *et al.*, 2019)		○	○	○	○	○	○	●	
Proof-of-Probability (S. Kim & Kim, 2018)		○	○	●	○	○	○	●	Rand.
Proof-of-Work-or-Knowledge (Baldimtsi *et al.*, 2016)		○	○	●	○	○	○	●	Rand.
Secure and Scalable Hybrid Consensus (Y. Liu, Liu *et al.*, 2022)		○	○	○	○	○	○	●	

SA: Sybil attack resistance class, **Inc +**: Reward scheme, **Inc -**: Punishment scheme, **Ra**: Randomisation, **Ph**: Physical world linking, **Re**: Reputation system, **Perm Pn**: Permissioned, **Perm Pl**: Permissionless, **Sel Rand.**: Random leader selection, **Sel Comm.**: Committee-based leader selection, **Sel Elec.**: Leader selection via election

● Strong Sybil attack resistance, ◐ Limited Sybil attack resistance, ○ No Sybil attack resistance

● Applicable, ○ Not applicable

Mechanisms for Healthcare

Blockchain algorithms in healthcare are proposed to improve patient centricity of electronic healthcare systems (Junaid *et al.*, 2022; Roman-Belmonte *et al.*, 2018), the effectiveness of legal medicine (Roman-Belmonte *et al.*, 2018), the credibility and interoperability between healthcare stakeholders (Krichen *et al.*, 2022), as well as medical supply chain management (Shah *et al.*, 2022). Mackey *et al.* (2019) attested that legal barriers constitute a major obstruction to blockchain adoption in the healthcare domain, as record sharing (predominantly of electronic health records and personal health records Hasselgren *et al.*, 2020) is a highly regulated activity (T.-F. Lee *et al.*, 2021) with disjoint approaches throughout the healthcare field (Hashim *et al.*, 2022). Furthermore, as Platt, Hasselgren *et al.* (2021) show, blockchain technology may offer limited benefits

Mackey, T. K., Kuo, T.-T., Gummadi, B., Clauson, K. A., Church, G., Grishin, D., Obbad, K., Barkovich, R., & Palombini, M. (2019). 'Fit-for-purpose?' challenges and opportunities for applications of blockchain technology in the future of healthcare. *BMC Medicine*, *17*(1), Article 68 over traditional centralised systems in the context of patient data gathering activities, like algorithmic contact tracing. Consequently, as seen in Table 2.10, only a single mechanism targets permissionless systems (A. Kumar *et al.*, 2020) with others targeting permissioned or unspecified deployments (K. Fan *et al.*, 2018; N. Kumar *et al.*, 2020; B. Li, Chenli, Xu, Jung & Shi, 2019; B. Li, Chenli, Xu, Shi & Jung, 2019; Malamas *et al.*, 2019; Mythili & Venkataraman, 2021; Pohl *et al.*, 2019; J. Yang *et al.*, 2019).

Table 2.10: Consensus mechanisms in *healthcare.*

		Inc					**Perm**		
	SA	**+**	**-**	**Ra**	**Ph**	**Re**	**Pn**	**Pl**	**Sel**
PoW Applied to Biomedical Image Segmentation (B. Li, Chenli, Xu, Jung & Shi, 2019)	○	○	○	●	○	○	●	○	Rand.
Deep Learning Based Consensus (B. Li, Chenli, Xu, Shi & Jung, 2019)	○	○	○	●	○	○	○	○	Rand.
Lightweight Proof-of-Game (A. Kumar *et al.*, 2020)	○	○	○	●	○	○	●	●	Rand.
MedBlock (K. Fan *et al.*, 2018)	○	○	○	○	○	○	●	○	Comm.
Proof of Policy (Pohl *et al.*, 2019)	○	○	○	○	○	○	●	○	
Proof of Policy (Mythili & Venkataraman, 2021)	○	○	○	○	○	○	●	○	Comm.
Proof-of-Familiarity (J. Yang *et al.*, 2019)	○	○	○	○	○	●	●	○	
proof-of-medical-stake (Malamas *et al.*, 2019)	○	○	○	○	○	●	●	○	
Proof of Artificial Intelligence (N. Kumar *et al.*, 2020)	◐	○	○	●	○	●	○	○	Rand.

SA: Sybil attack resistance class, **Inc +**: Reward scheme, **Inc -**: Punishment scheme,
Ra: Randomisation, **Ph**: Physical world linking, **Re**: Reputation system,
Perm Pn: Permissioned, **Perm Pl**: Permissionless, **Sel Rand.**: Random leader selection,
Sel Comm.: Committee-based leader selection, **Sel Elec.**: Leader selection via election

● Strong Sybil attack resistance, ◐ Limited Sybil attack resistance, ○ No Sybil attack resistance

● Applicable, ○ Not applicable

Mechanisms for High Performance

As Oyinloye *et al.* (2021) summarised, 'throughput, scalability, security, energy consumption, and finality' can be considered the main attributes of the overall performance of a blockchain system. This is despite an ongoing debate on the appropriateness of applying traditional performance metrics for centralised systems to decentralised systems (Merrad *et al.*, 2022). Consequently, most papers analysed have the goal of putting forward designs that improve on traditional metrics such as throughput and latency. Of the papers in this category (see Table 2.11), the vast majority provide no Sybil attack resistance by building on BFT principles. For these papers, similar to early work (Lamport *et al.*, 1982; Pease *et al.*, 1980), commonly a maximum threshold of byzantine processes (e.g. 1/3) is assumed. Most papers perform benchmarking against similar mechanisms that do not provide Sybil attack resistance.

Oyinloye, D. P., Teh, J. S., Jamil, N., & Alawida, M. (2021). Blockchain consensus: An overview of alternative protocols. *Symmetry*, *13*(8), Article 1363

Table 2.11: Consensus mechanisms with 'high performance'.

		Inc					**Perm**		
	SA	**+**	**-**	**Ra**	**Ph**	**Re**	**Pn**	**Pl**	**Sel**
PoS using Fair and Dynamic Sharding Management (D. R. Lee *et al.*, 2019)	●	●	○	●	○	○	○	●	Rand.
Bicomp (Jiao *et al.*, 2018)	●	●	○	●	○	○	○	●	Rand.

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Hybrid PoW/PoS (K. D. Gupta *et al.*, 2019)	●	●	○	●	○	○	○	●	Rand.
Scalable BFT consensus mechanism through aggregated signature gossip (Long & Wei, 2019)	●	○	○	●	○	○	●	●	
BFT Consensus on FPGA (Bravo *et al.*, 2020)	○	○	○	●	○	○	●	○	Elec.
BFT Consensus with smart network interface controller Offloading (Bravo *et al.*, 2020)	○	○	○	●	○	○	●	○	Elec.
PBFT with support vector machine-Based Trust Evaluation (N. Du *et al.*, 2020)	○	○	○	○	○	●	●	○	Elec.
Autonomous and Controllable High-performance Consensus (Chengfu, 2020)	○	○	○	○	○	○	○	○	
BEAT (Duan *et al.*, 2018)	○	○	○	●	○	○	●	○	Elec.
BlockDAG (K. Gai *et al.*, 2020)	○	○	○	○	○	○	●	○	Elec.
Checkpoint Consensus (K. Cong *et al.*, 2018)	○	○	○	●	○	○	●	○	Rand.
Concordia (Santiago *et al.*, 2021)	○	○	○	●	○	○	●	○	Rand.
Consensus based on the Mortgage Model (T. Lin *et al.*, 2020)	○	○	○	●	○	○	●	○	Elec.
Consensus for Mobile Devices using Online Brokers (Kiamari *et al.*, 2020)	○	○	○	○	○	○	●	○	
FAST (N. Khan, 2018)	○	○	○	○	○	○	●	○	None
Fast Probabilistic Consensus within Byzantine Infrastructures (Popov & Buchanan, 2021)	○	○	○	●	○	○	●	○	None
Fast, Dynamic and Robust Byzantine Fault Tolerance (A. Song *et al.*, 2019)	○	○	○	●	○	○	●	○	Rand.
FastPay (Baudet *et al.*, 2020)	○	○	○	○	○	○	●	○	None
FireLedger (Buchnik & Friedman, 2020)	○	○	○	○	○	○	●	○	Elec.
FRChain Consensus (Chander *et al.*, 2019)	○	○	○	●	○	○	●	○	Elec.
Gosig (P. Li *et al.*, 2020)	○	○	○	●	○	○	●	○	Rand.
High Performance and Scalable Byzantine Fault Tolerance (Y. Jiang & Lian, 2019a)	○	○	○	○	○	○	●	○	Rand.
HPBC (Y. Jiang & Ding, 2018)	○	○	○	●	○	○	●	○	Elec.
Improved Practical Byzantine Consensus (Z. Gao & Yang, 2020)	○	○	○	○	○	○	●	○	Elec.
MinBFT-Based Consensus (T.-L. Huang & Huang, 2020)	○	○	○	○	○	○	●	○	
Mixed Byzantine Fault Tolerance (M. Du, Chen & Ma, 2020; M. Du *et al.*, 2021)	○	○	○	○	○	○	●	○	Elec.
Proof-of-Execution (S. Gupta *et al.*, 2019)	○	○	○	○	○	○	●	○	Elec.
Proof-of-Scalable-Traceability (Al-Mamun & Zhao, 2020b)	○	○	○	○	○	○	●	○	Comm.
Proof-of-Vote (K. Li *et al.*, 2017)	○	○	○	○	○	○	●	○	Elec.
Raft with Network Stability Evaluation (D. Kim *et al.*, 2021)	○	○	○	○	○	○	●	○	Elec.
Random-Checkers Proof-of-Stake (Chomsiri & Kongsup, 2018)	○	●	●	○	○	○	○	●	Elec.
SACZyzzyva (Gunn *et al.*, 2019)	○	○	○	●	○	○	●	○	Rand.
Scalable Dynamic Multi-Agent Byzantine Fault-Tolerance (L. Feng *et al.*, 2018)	○	○	○	○	○	○	●	○	Elec.
Scalable Efficient Byzantine Fault Tolerance (Y. Jiang & Lian, 2019b)	○	○	○	○	○	○	●	○	Elec.
Scored PBFT (Y. Chen *et al.*, 2020)	○	○	○	●	○	●	●	○	Elec.
Stream of Distributed Secrets for Quantum-safe Blockchain (Dolev & Wang, 2020)	○	○	○	●	○	○	○	○	Rand.

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T-PBFT (S. Gao *et al.*, 2019)	○	○	○	○	○	●	●	○	Elec.
Blinkchain (Basescu *et al.*, 2017)	◐	○	○	●	○	○	○	●	Rand.
High-Performance Blockchain Enhanced Consensus (Al-Mamun & Zhao, 2020a)	◐	○	○	●	○	○	○	●	Rand.
Multi Party Computation DPoS (Y. Luo *et al.*, 2018, 2019)	◐	●	○	●	○	○	○	○	Rand.
Self-Referencing Directed Acyclic Graph and Voting-Based PBFT Consensus (Cao *et al.*, 2019)		○	○	●	○	●	○	●	Elec.

SA: Sybil attack resistance class, **Inc +**: Reward scheme, **Inc -**: Punishment scheme,
Ra: Randomisation, **Ph**: Physical world linking, **Re**: Reputation system,
Perm Pn: Permissioned, **Perm Pl**: Permissionless, **Sel Rand.**: Random leader selection,
Sel Comm.: Committee-based leader selection, **Sel Elec.**: Leader selection via election

● Strong Sybil attack resistance, ◐ Limited Sybil attack resistance, ○ No Sybil attack resistance

● Applicable, ○ Not applicable

Mechanisms for IoT

Many papers apply blockchain to IoT (see Table 2.12), often with the goal of improving the security of IoT stacks (Sadhu *et al.*, 2022) or to enable immutable traceability (M. Lei *et al.*, 2022). In common IoT deployments, there is a large disparity of computational capabilities between IoT devices and edge devices (Gu *et al.*, 2022). This disparity is also one of the drivers of hierarchicality in these use cases (Platt, Bandara *et al.*, 2021). This favours algorithms with limited or no Sybil attack resistance that offload computation to edge nodes: typically these edge nodes are operated centrally, by trusted entities.

Table 2.12: Consensus mechanisms for the 'internet of things'.

		Inc					Perm		
	SA	**+**	**-**	**Ra**	**Ph**	**Re**	**Pn**	**Pl**	**Sel**
PoW with Mining Tokens (Y. Gupta *et al.*, 2018)	●	●	○	●	○	○	○	○	Rand.
Collaborative Proof-of-Work (Y. Liu, Wang *et al.*, 2020)	●	●	●	●	○	○	○	●	Rand.
Credit-Based Consensus Mechanism (J. Huang, Kong, Chen, Cheng *et al.*, 2019; J. Huang, Kong, Chen, Wu *et al.*, 2019)	●	○	○	●	○	●	○	●	Rand.
Hybrid PoW/PoS (Y. Huang *et al.*, 2020)	●	●	○	●	○	○	○	●	Rand.
Proof-of-Trading (X. Lin *et al.*, 2019)	●	●	○	○	○	○	○	○	Rand.
Random Proof of Work (X.-F. Du *et al.*, 2020)	●	●	○	●	○	○	●	●	Rand.
Three-Dimensional Greedy Heaviest-Observed Sub-Tree Consensus (M. Du, Wang *et al.*, 2020)	●	●	○	●	○	○	○	●	Rand.
BFT Consensus Based on Dynamic Permission Adjustment (D. Liao *et al.*, 2021)	○	○	○	○	○	●	●	○	Elec.
IoT Adaptive Dynamic Consensus (X. Hu *et al.*, 2020)	○	○	○	○	○	○	●	○	
Adaptive Proof-of-Work (Chuang *et al.*, 2020)	○	○	○	●	○	○	●	○	Rand.
Collaborative Trust Based Delegated Proof-of-Stake (J. Chen *et al.*, 2020)	○	○	○	●	○	○	●	○	Rand.
Consensus with Elected Leader (Melnik *et al.*, 2020)	○	○	○	●	○	○	●	○	Elec.
Context-based consensus (Lunardi *et al.*, 2020)	○	○	○	○	○	○	○	○	Elec.
Double-Layer PBFT (W. Li *et al.*, 2021)	○	○	○	○	○	○	●	○	Elec.
Dynamic Blind Voting (Cho *et al.*, 2017)	○	○	○	●	○	○	○	○	Rand.
Improved PBFT (K. Fan, Wang, Ren *et al.*, 2019)	○	○	○	○	○	○	●	○	Comm.

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Lightweight Blockchain based Cybersecurity (Abdulkader *et al.*, 2019)	○	○	○	○	○	○	●	○	Comm.
Lightweight Consensus for IoT (Moudoud *et al.*, 2019)	○	○	○	○	○	○	●	○	
Luckyminer (C.-W. Huang & Chen, 2019)	○	●	○	●	○	○	○	●	Rand.
Multi-Chain Proof of Rapid Authentication (Alkhodair *et al.*, 2020)	○	○	○	○	○	○	●	○	Comm.
Optimised PBFT Consensus with Speaker (S. Guo *et al.*, 2020)	○	○	○	●	○	○	●	○	Rand.
Proof-of-Presence (Mamun & Khan, 2020)	○	○	○	○	○	○	●	○	Rand.
Proof of Random Count in Hashes (Hossain *et al.*, 2020)	○	○	○	●	○	○	●	○	Rand.
Proof of Reputation (El Majdoubi *et al.*, 2020)	○	○	○	○	○	○	●	○	
Proof-of-Balance (S. Liao *et al.*, 2020)	○	○	○	○	○	○	●	○	Elec.
Proof-of-Block-and-Trade (Biswas *et al.*, 2020)	○	○	○	○	○	○	●	○	Rand.
Proof-of-Common-Interest (Doku & Rawat, 2020; Doku *et al.*, 2020)	○	○	○	○	○	○	●	○	
Proof-of-Popularity (G. Li *et al.*, 2020)	○	○	○	●	○	●	●	○	Rand.
Proof-of-Stability (K. Fan, Sun *et al.*, 2019)	○	○	○	○	○	○	●	○	Comm.
Proof-of-Validity (Falcone *et al.*, 2019)	○	○	○	○	○	○	●	○	
ReBFT (W. Lin, Yin *et al.*, 2020)	○	○	○	○	○	○	●	○	
SCBFT (Hou *et al.*, 2020)	○	○	○	○	○	○	●	○	
Time-Dependent Consensus (S. N. Mohanty *et al.*, 2020)	○	○	○	●	○	○	○	○	Rand.
Tree-Chain (Dorri & Jurdak, 2020)	○	○	○	●	○	○	●	○	Rand.
Two-Layer-Consensus Architecture for IoT (Bai *et al.*, 2019)	○	○	○	●	○	○	●	○	Rand.
Weighted Majority Consensus Algorithm (Alhejazi & Mohammad, 2022)	○	○	○	●	○	●	●	○	Rand.
DPoS for Network Intrusion Detection (Jinhua *et al.*, 2019)	◐	○	○	○	○	●	○	○	Elec.
PBFT Based on Reputation Value (S. Guo *et al.*, 2022)	◐	○	○	○	○	●	●	○	Elec.
Consensus Algorithm for Mobile-Edge Computing (Asheralieva & Niyato, 2020)	◐	○	○	●	○	●	●	●	Elec.
Credit Reinforce Byzantine Fault Tolerance (P. Chen *et al.*, 2021)	◐	○	○	●	○	○	○	●	Elec.
Distributed Time-Based Consensus (Dorri *et al.*, 2019)	◐	●	○	●	○	○	○	●	Rand.
Edge Computing Blockchain Security Consensus Model (Xuan *et al.*, 2021)	◐	○	○	●	○	●	○	○	
Geographic-PBFT (Lao *et al.*, 2020)	◐	○	○	○	○	○	○	●	Elec.
Honesty-Based Distributed Proof-of-Work (S. Alrubei *et al.*, 2020)	◐	○	○	●	○	●	●	●	Elec.
Honesty-Based Distributed Proof-of-Authority via Scalable Work (S. M. Alrubei *et al.*, 2022)	◐	○	○	●	○	●	●	●	Elec.
PF-BVM (Baniata & Kertesz, 2020)	◐	○	○	●	○	●	●	●	
Predictive Proof of Metrics (Bhamidipati *et al.*, 2019)	◐	●	○	●	○	●	●	●	Elec.
Proof of Elapsed Work and Luck (Raghav *et al.*, 2020)	◐	○	○	●	○	○	○	●	Rand.
Proof-of-Honesty (Makhdoom *et al.*, 2020a, 2020b)	◐	○	○	○	○	●	○	○	Rand.
Proof-of-Negotiation/Proof-of-Trust Negotiation (J. Feng *et al.*, 2019, 2020)	◐	●	○	●	○	●	○	○	Rand.
Register, Deposit, Vote (Solat, 2018)	◐	○	○	○	○	○	○	○	None
Sybil Resistant IoT Trust Model (Asiri & Miri, 2018)	◐	○	○	●	○	●	●	○	Rand.
Synergistic Multiple Proof (Y. Liu *et al.*, 2019)	◐	○	○	○	○	●	○	●	
Delegated Proof-of-Proximity (Ledwaba *et al.*, 2020)	◐	○	○	○	●	○	○	○	

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	SA	Inc +	Inc -	Ra	Ph	Re	Perm Pn	Perm Pl	Sel
Proof-of-Authentication (Maitra *et al.*, 2020; Puthal & Mohanty, 2019; Puthal *et al.*, 2019)	◐	○	○	●	●	○	○	○	Elec.
Proof-of-Physical Unclonable Function (Asif *et al.*, 2020)	◐	○	○	●	●	○	○	●	Rand.
Hybrid Consensus (J. Chen, 2018)		○	○	●	○	○	●	●	Rand.
Proof-of-Reputation-X (E. K. Wang *et al.*, 2020)		○	○	○	○	●	○	○	
Proof-of-Work Using Maximization-Factorization Statistics (G. Kumar *et al.*, 2019)		○	○	●	○	○	○	○	Rand.
Proof of PUF-Enabled Authentication (S. P. Mohanty *et al.*, 2020)		○	○	○	●	○	○	○	

SA: Sybil attack resistance class, **Inc +**: Reward scheme, **Inc -**: Punishment scheme,
Ra: Randomisation, **Ph**: Physical world linking, **Re**: Reputation system,
Perm Pn: Permissioned, **Perm Pl**: Permissionless, **Sel Rand.**: Random leader selection,
Sel Comm.: Committee-based leader selection, **Sel Elec.**: Leader selection via election

● Strong Sybil attack resistance, ◐ Limited Sybil attack resistance, ○ No Sybil attack resistance

● Applicable, ○ Not applicable

Mechanisms for Media and Entertainment

Few mechanisms have been proposed for the media and entertainment domain (see Table 2.13). While non-fungible tokens have gained considerable popularity in the arts (Ippolito, 2022; Sinnott & Zhou, 2024; Valeonti *et al.*, 2021) as well as media and entertainment (Colicev, 2023; Zaucha & Agur, 2022) and have since generated sales of over 940 million USD (H. Bao & Roubaud, 2021), according to our analysis, no mechanisms specifically for this phenomenon have been developed. Instead, we see a mechanism tailored to gaming which requires centralised infrastructures for score assessment (Yuen *et al.*, 2019) and a mechanism to counteract false and misleading information presented as news (Q. Chen *et al.*, 2020). No mechanisms with strong Sybil attack resistance were found in this category.

Yuen, H. Y., Wu, F., Cai, W., Chan, H. C., Yan, Q., & Leung, V. C. (2019). Proof-of-play. *Proceedings of the 2019 International Symposium on Blockchain and Secure Critical Infrastructure*

Table 2.13: Consensus mechanisms in *media*.

		Inc					Perm		
	SA	+	-	Ra	Ph	Re	Pn	Pl	Sel
Proof-of-Play (Yuen *et al.*, 2019)	○	○	○	●	○	○	●	○	Rand.
Credibility Score (Q. Chen *et al.*, 2020)	◐	○	○	○	○	●	○	●	Elec.
Proof-of-Contribution (H. Song *et al.*, 2021)	◐	●	○	○	○	●	○	○	

SA: Sybil attack resistance class, **Inc +**: Reward scheme, **Inc -**: Punishment scheme,
Ra: Randomisation, **Ph**: Physical world linking, **Re**: Reputation system,
Perm Pn: Permissioned, **Perm Pl**: Permissionless, **Sel Rand.**: Random leader selection,
Sel Comm.: Committee-based leader selection, **Sel Elec.**: Leader selection via election

● Strong Sybil attack resistance, ◐ Limited Sybil attack resistance, ○ No Sybil attack resistance

● Applicable, ○ Not applicable

Mechanisms for Supply Chain

The supply chain sector can be characterised as a highly relevant domain for blockchain due to its promise of delivering information transparency and immutability (J.-S. Kim & Shin, 2019) as well as its potential to bring about disintermediation (Dutta *et al.*, 2020). Supply chain management and IoT (see subsection 2.7) often go hand in hand to improve traceability and interoperability and make frequent use of sensor data originating from IoT devices (Rejeb *et al.*, 2019). Therefore, it is not surprising that, in terms of Sybil attack resistance, many properties are shared with the IoT domain: The mechanisms proposed (see Table 2.14) only deliver limited Sybil attack resistance (G. Kumar *et al.*, 2020; H. Li *et al.*, 2021; X. Li *et al.*, 2020; T. Mao *et al.*, 2021) or no Sybil attack resistance (Z. Fang *et al.*, 2020; D. Mao *et al.*, 2018, 2019). This can be attributed to the trustworthy provenance problem: supply chain efforts require trustworthy intermediaries to prevent physical tampering (e.g. the manipulation of sensors or interference with physical products). Therefore, the mechanisms in this category commonly rely on permissioned systems which can cater to hierarchical architectures (Platt, Bandara *et al.*, 2021).

Kim, J.-S., & Shin, N. (2019). The impact of blockchain technology application on supply chain partnership and performance. *Sustainability*, *11*(21), Article 6181

Table 2.14: Consensus mechanisms in *supply chain.*

		Inc					Perm		
	SA	**+**	**-**	**Ra**	**Ph**	**Re**	**Pn**	**Pl**	**Sel**
Consensus Mechanism for Marine Data Management System (Z. Fang *et al.*, 2020)	○	○	○	○	○	○	●	○	Elec.
Improved Practical Byzantine Fault Tolerance (D. Mao *et al.*, 2018, 2019)	○	○	○	○	○	○	●	○	Comm.
Multi-Center Practical Byzantine Fault Tolerance (X. Li *et al.*, 2020)	○	○	○	○	○	○	●	○	
Proof of Accomplishment (G. Kumar *et al.*, 2020)	○	○	○	○	○	●	●	○	
C-dBFT (T. Mao *et al.*, 2021)	◐	○	○	○	○	○	●	○	
Group-Based PoW (H. Li *et al.*, 2021)	◐	●	○	●	○	○	○	○	Rand.
Proof-of-Location (Y. Fu *et al.*, 2020)		○	○	●	○	○	○	●	None

SA: Sybil attack resistance class, **Inc +**: Reward scheme, **Inc -**: Punishment scheme,
Ra: Randomisation, **Ph**: Physical world linking, **Re**: Reputation system,
Perm Pn: Permissioned, **Perm Pl**: Permissionless, **Sel Rand.**: Random leader selection,
Sel Comm.: Committee-based leader selection, **Sel Elec.**: Leader selection via election

● Strong Sybil attack resistance, ◐ Limited Sybil attack resistance, ○ No Sybil attack resistance

● Applicable, ○ Not applicable

Mechanisms for Telecommunications

The application of blockchain in telecommunications is nascent with only two dedicated mechanisms found (see Table 2.15). Darmwal (2017) explained this by showing that the telecom domain is bound by numerous historic standards of interoperability and predicts that, therefore, adoption will be slow and should focus on lighthouse projects led by established consortia. This prediction is reflected in the fact that mechanisms in this vertical expose only limited Sybil attack resistance (Lwin *et al.*, 2020) or no Sybil attack resistance (W. Lin, Xu *et al.*, 2020) at all, hinting at the application of the proposed mechanisms in permissioned contexts.

Darmwal, R. (2017). Blockchain in telecom sector: An analysis of potential use cases. *Telecom Business Review, 10*(1)

Table 2.15: Consensus mechanisms in *telecommunications.*

		Inc					Perm		
	SA	**+**	**-**	**Ra**	**Ph**	**Re**	**Pn**	**Pl**	**Sel**
Proof-of-Majority (W. Lin, Xu *et al.*, 2020)	○	○	○	○	○	○	●	○	
Delegated Proof-of-Trust (Lwin *et al.*, 2020)	◐	○	○	●	○	○	○	○	

SA: Sybil attack resistance class, **Inc +**: Reward scheme, **Inc -**: Punishment scheme,
Ra: Randomisation, **Ph**: Physical world linking, **Re**: Reputation system,
Perm Pn: Permissioned, **Perm Pl**: Permissionless, **Sel Rand.**: Random leader selection,
Sel Comm.: Committee-based leader selection, **Sel Elec.**: Leader selection via election

● Strong Sybil attack resistance, ◐ Limited Sybil attack resistance, ○ No Sybil attack resistance

● Applicable, ○ Not applicable

Mechanisms Undertaking Useful Work

Useful work schemes are ubiquitous (Ball *et al.*, 2018), created with the goal of deriving useful outputs from PoW. As most mechanisms implement mechanics similar to conventional PoW, their Sybil attack resistance properties are strong (see Table 2.16). It can be differentiated between fixed PoUW schemes, in which a well-defined computational problem exists and marketplace mechanisms in which problems can be proposed

Ball, M., Rosen, A., Sabin, M., & Vasudevan, P. N. (2018). Proofs of work from worst-case assumptions. H. Shacham & A. Boldyreva (Eds.), *Proceedings of the 38th annual international cryptology conference.* Springer dynamically. The former are often well-specified and provide appropriate difficulty adjustment mechanisms, whereas the latter sometimes lack such considerations and, therefore have unclear security properties. Marketplace mechanisms can typically only provide limited Sybil attack resistance due to the risk of collusion between sellers and buyers and the challenge of objectively establishing the best solution. This is particularly apparent in AI-based marketplace mechanisms that require the objective evaluation of a potentially large number of proposed machine learning (ML) models.

Table 2.16: Consensus mechanisms with 'proof of useful work'.

		Inc					**Perm**		
	SA	**+**	**-**	**Ra**	**Ph**	**Re**	**Pn**	**Pl**	**Sel**
PoUW for ML Training (Lihu *et al.*, 2020)	●	○	○	●	○	○	○	●	Rand.
PoW Applied to High-Dimension, Non-Linear Optimisation Problems (W. Li, 2018)	●	●	○	●	○	○	○	●	Rand.
PoW Based On nilcatenation problem-Solving (Géraud *et al.*, 2018)	●	○	○	●	○	○	○	●	Rand.
PoW Based on Random Multivariate Quadratic Equations (Ding, 2019)	●	○	○	●	○	○	○	●	Rand.
PoW on Elements in a Cyclic Group (Hastings *et al.*, 2019)	●	○	○	○	○	○	○	●	Rand.
Calibration of Public Key Cryptographic Systems via Proof-of-Work (Boyd & Carr, 2018)	●	●	○	●	○	○	○	●	Rand.
Coin.AI (Baldominos & Saez, 2019)	●	●	○	●	○	○	○	●	Rand.
Conquering Generals (Loe & Quaglia, 2018)	●	●	○	●	○	○	○	●	Rand.
Difficulty-based Incentives for Problem Solving (Philippopoulos *et al.*, 2020)	●	●	○	●	○	○	○	●	Rand.
Hybrid Mining (Chatterjee *et al.*, 2019)	●	●	○	●	○	○	○	●	Rand.
Image-based Proof-of-Work (Billings, 2017)	●	●	○	●	○	○	○	●	Rand.
Proof of Catalytic Space (Pietrzak, 2018)	●	○	○	●	○	○	○	●	
Proof of Deep Learning with Hyperparameter Optimization (Mittal & Aggarwal, 2020)	●	○	○	●	○	○	○	●	Rand.
Proof of Evolution (Bizzaro *et al.*, 2020)	●	●	○	○	○	○	○	●	Rand.
Proof-of-Deep-Learning (Chenli *et al.*, 2019, 2020)	●	●	○	●	○	○	○	●	Rand.
Proof-of-Exercise (Shoker, 2017)	●	●	○	●	○	○	○	●	Rand.
Proof-of-Learning (Bravo-Marquez *et al.*, 2019)	●	●	○	●	○	○	○	●	Rand.
Proof-of-Learning (Qiu *et al.*, 2021)	●	●	○	●	○	○	○	●	Rand.
Proof-of-WorkStore (Arslan & Goker, 2019)	●	○	○	●	○	○	○	●	Rand.
Reciprocally Useful Work (Mauri *et al.*, 2020)	●	●	○	●	○	○	○	●	Rand.
Proof of Federated Learning (Qu *et al.*, 2021)	○	○	○	○	○	○	○	○	
Proof-of-Accuracy (Z. Li *et al.*, 2021)	○	○	○	○	○	○	●	○	
Proof-of-Learning (Lan *et al.*, 2020)	○	●	○	●	○	○	○	○	Rand.
BlockML (Merlina, 2019)	◐	●	○	●	○	○	○	●	Rand.
Proof-of-Search (Shibata, 2019)	◐	●	○	●	○	○	○	○	Rand.
Susreum (Badreddin *et al.*, 2019)	◐	○	○	○	○	○	○	●	Elec.
VBFL (H. Chen *et al.*, 2021)	◐	●	○	●	○	●	○	○	Rand.

SA: Sybil attack resistance class, **Inc +**: Reward scheme, **Inc -**: Punishment scheme,
Ra: Randomisation, **Ph**: Physical world linking, **Re**: Reputation system,
Perm Pn: Permissioned, **Perm Pl**: Permissionless, **Sel Rand.**: Random leader selection,
Sel Comm.: Committee-based leader selection, **Sel Elec.**: Leader selection via election

● Strong Sybil attack resistance, ◐ Limited Sybil attack resistance, ○ No Sybil attack resistance

● Applicable, ○ Not applicable

Mechanisms for the Vehicle Domain

A surprisingly rich domain is the domain of vehicles which shares many benefits and challenges with the domain of IoT. Blockchain applications in the vehicle domain predominantly target data exchange between vehicles, resource sharing (e.g., collaborative use of computing resources), parking coordination, and traffic management (Alladi *et al.*, 2022). As shown in Table 2.17, schemes with strong Sybil attack resistance (Q. Han *et al.*, 2020; Javaid & Sikdar, 2021a, 2021b; Javaid *et al.*, 2020; X. Liu *et al.*, 2020) are outnumbered by schemes with limited or no Sybil attack resistance. In line with X. Wang *et al.*, 2020, who attest that blockchain in vehicular security is in its infancy, we find that some mechanisms proposed don't make use of the full potential of a decentralised paradigm by being overly hierarchical and by relying on centralised actors.

Wang, X., Xu, C., Zhou, Z., Yang, S., & Sun, L. (2020). A survey of blockchain-based cybersecurity for vehicular networks. *Proceedings of the 2020 Conference on International Wireless Communications and Mobile Computing*

Table 2.17: Consensus mechanisms for the vehicle domain.

		Inc					Perm		
	SA	**+**	**-**	**Ra**	**Ph**	**Re**	**Pn**	**Pl**	**Sel**
Consensus Mechanism for Blockchains on internet of vehicles (IoV) (Q. Han *et al.*, 2020)	●	○	○	●	○	○	○	○	Elec.
dynamic Proof-of-Work (Javaid & Sikdar, 2021a, 2021b; Javaid *et al.*, 2020)	●	●	○	●	○	○	○	●	Rand.
Mixed Consensus Algorithm Based on PoW and PBFT (X. Liu *et al.*, 2020)	●	○	○	●	○	●	○	○	Rand.
Consensus Program for Charging Piles (Q. He, Xu *et al.*, 2018)	○	●	○	●	○	●	●	○	Elec.
Enhanced DPoS (Kang *et al.*, 2019)	○	○	○	○	○	●	●	○	
Improved Byzantine Consensus for IoV (W. Hu *et al.*, 2019)	○	○	○	●	○	○	●	○	Rand.
Multipoint-Relay-Driven Consensus (Jian *et al.*, 2021)	○	○	○	●	○	○	●	○	Rand.
Practical Byzantine Fault Tolerance with Forwarding (Ferrag & Maglaras, 2019)	○	○	○	○	○	○	●	○	None
Proof of Event and Location (Nurfatih *et al.*, 2020)	○	○	○	○	○	○	●	○	
Proof-of-Communication (Ghribi *et al.*, 2020)	○	○	○	●	○	○	○	○	Elec.
Proof-of-Driving (Kudva *et al.*, 2021)	○	○	○	●	○	●	○	○	Rand.
Proof-of-Event with Dynamic Federation (H. Guo, Meamari & Shen, 2018; H. Guo *et al.*, 2020)	○	○	○	●	○	○	●	○	Rand.
Proof-of-Matching (Bathen *et al.*, 2020)	○	●	○	○	○	○	●	○	Comm.
Proof-of-Nonce (Ahn & Lee, 2020)	○	○	○	●	○	○	●	○	Rand.
Proof-of-Reputation (Chai *et al.*, 2019)	○	○	○	○	○	●	●	○	None
Proof-of-Utility (Y. Dai *et al.*, 2019, 2020)	○	○	○	●	○	○	●	○	Elec.
Proof-of-Vehicular-Services BFT (Islam *et al.*, 2020)	○	○	○	○	○	●	●	○	Elec.
Reputation-based Miner Node Selection (Maskey *et al.*, 2021)	○	○	○	○	○	○	●	○	
Secured Event-Information Sharing (Dwivedi *et al.*, 2020)	○	○	○	○	○	○	●		Comm.
Semi-Autonomous Distributed Blockchain-Based Framework for unmanned aerial vehicles (Ge *et al.*, 2020)	○	○	○	○	○	○	●	○	Elec.
Time-Oriented Proof of Work (J. Liu *et al.*, 2020)	○	○	○	○	○	○	○	○	
Voting-based Consensus Protocol for vehicular ad-hoc network (Ayaz, Sheng, Tian & Leung, 2021; Ayaz *et al.*, 2020)	○	○	○	●	○	●	●	○	Comm.

Continued on next page

Proof-of-Work-at-Proximity (Fujihara, 2020)	◐	●	○	●	○	○	○	○	Rand.
Reputation-Based DPoS (X. Ma *et al.*, 2019)	◐	○	○	○	○	●	○	○	
Proof-of-Quality-Factor (Ayaz, Sheng, Tian & Guan, 2021)	◐	●	●	●	●	○	●	○	Elec.

SA: Sybil attack resistance class, **Inc +**: Reward scheme, **Inc -**: Punishment scheme,
Ra: Randomisation, **Ph**: Physical world linking, **Re**: Reputation system,
Perm Pn: Permissioned, **Perm Pl**: Permissionless, **Sel Rand.**: Random leader selection,
Sel Comm.: Committee-based leader selection, **Sel Elec.**: Leader selection via election

● Strong Sybil attack resistance, ◐ Limited Sybil attack resistance, ○ No Sybil attack resistance

● Applicable, ○ Not applicable

2.8 Conclusion

The bulk of research interest in the field is focused on a small set of popular mechanisms (see subsection 2.7) with Proof-of-Work (PoW), Proof-of-Stake (PoS), and practical Byzantine fault tolerance as the most popular mechanisms. The literature review has shown that, beyond these, many more, seemingly disparate, consensus mechanisms exist (see section 2.7). However, upon closer examination, it can be concluded that there are strong similarities between them.

Among the systems with *strong Sybil attack resistance*, we find mostly PoW-like mechanisms that utilise some form of scarce resource constraint (CPU, memory, or otherwise) and PoS-like systems that rely on staking of resources (e.g., cryptocurrencies, stablecoins, reputation tokens, etc.). We also find combinations of the two, e.g., where a PoW bootstrapping phase is combined with a PoS execution phase. Sometimes these mechanisms are combined with reputation systems, but the latter are rarely central to their Sybil attack resistance properties.

Systems with *limited Sybil attack resistance* are among the most interesting. These often make creative use of reputation systems (e.g. systems in which the right of a user to mine blocks correlates with their peer-to-peer rating). Physical world linking, i.e. the use of information emanating from objects in the physical world, is also most prevalent in this category: we find CPUs, trusted execution environments, and mobile phones to be common objects used. Furthermore, some mechanisms in this category use properties of well-recognised Internet standards such as the domain name system or public IP addresses to determine uniqueness. This category, regrettably, is also one that contains mechanisms with overstated Sybil attack resistance properties and, regularly, poor specification.

Systems *without Sybil attack resistance* are mostly contained for completeness. Apart from rare instances where authors claim Sybil attack resistance, most works in this category target permissioned systems (that do not require mechanisms with Sybil attack resistance). Some authors explicitly exclude Sybil attack resistance considerations from their proposals and suggest applying alternative schemes to achieve it.

State of the Literature

Sybil attack resistance is a hard-to-achieve property that comes with significant trade-offs in terms of complexity, performance, and cost. Many mechanisms that aspire to provide strong Sybil attack resistance fall short of this goal. The choice of consensus mechanisms is overwhelming, and system designers have frequently opted for designing new mechanisms without a strong need for it. In many ways, this is symptomatic for the wider subject area of distributed ledger technology where, despite some signs of the field maturing, *solutions looking for problems* are still prevalent.

In his 2017 manuscript 'Blockchains and Consensus Protocols: Snake Oil Warning', Cachin (2017) alerts of the risks arising from consensus mechanism being developed without 'agreement on trust assumptions, security models, formal reasoning methods, and protocol goals'. We want to reiterate and enforce this warning: many protocols that suggest Sybil attack resistance do not deliver on this implication. Overall, significant quality issues in many of the protocols proposed are apparent. Many of the papers analysed lack formalisation, leaving essential aspects as an exercise to the reader, thereby exposing implementors and users to security risks. Some papers, while peer-reviewed and, in some cases, highly cited, misunderstand fundamental aspects of decentralised computing. This is particularly evident in reputation-based consensus methods that often address Sybil attack resistance through mitigation schemes that, under scrutiny, do not hold.

Cachin, C. (2017). Blockchains and consensus protocols: Snake oil warning. *Proceedings of the 13th European Dependable Computing Conference*

However, it is not all doom and gloom: recent years have seen some high-quality contributions that have further enhanced the performance of consensus mechanisms, significantly improved their Sybil attack resistance, or produced innovative approaches to saving energy. It is also to be welcomed that authors analyse the use of consensus mechanisms in industry-specific settings to drive appropriate solutions forward. It can therefore be speculated that the works that will stand the test of time will be those that apply rigorous modelling, formal security analysis, and comprehensive threat modelling.

Gaps

It is a challenging undertaking to compare the Sybil attack resistance properties of consensus mechanisms: there is no canonical approach to specifying consensus mechanisms for blockchain making it hard to formally scrutinise them. Furthermore, there is no common understanding of what constitutes *good* Sybil attack resistance, let alone recognised ways of asserting the Sybil attack resistance qualities of a proposed mechanism. Mechanism designers, in many cases, lack consideration for Sybil attacks in their work. Especially for domain-specific consensus mechanisms, it should be established what the permissioning approach and, therefore, the threat model is.

Research Directions

Given the challenges discussed, four dominant directions for consensus mechanism research are conceivable. First, research should focus on introducing more rigorous approaches to the field by adapting existing formal methodologies to blockchain environments. Second, research should be undertaken to formally identify suitable canonical consensus mechanisms for common standard use cases. Third, research should enable incremental innovation of such canonical consensus mechanisms, thereby creating 'best-in-class' mechanisms for various use cases. Fourth, realistic benchmarks should be developed that allow for the objective comparison of non-functional aspects of blockchain systems (e.g. throughput performance or time-to-finality).

The Energy Footprint of Blockchain Consensus 3

This chapter follows the preceding literature survey (chapter 2), which established the dominance of PoW and PoS in permissionless systems. Consequently, this chapter undertakes an analysis of the energy consumption properties of both algorithms. This analysis aims to determine whether either of these consensus algorithms can serve as a baseline for democratic self-governance while highlighting their downsides.

In decentralised distributed ledger technology (DLT) systems, consensus mechanisms fulfil multiple purposes surrounding the proposal, validation, propagation and finalisation of data (Xiao *et al.*, 2020). Voting is used extensively in common consensus mechanisms to resolve ambiguity arising from actors that spread incorrect or conflicting information (Lamport *et al.*, 1982). Sybil attacks, which pose a critical problem for DLT systems, occur when an attacker creates an artificially large number of bogus identities (Douceur, 2002) to skew the results of majority decisions on the admission and order of transactions. In *permissioned* networks, gatekeeping strategies can be applied that limit access to a network to previously vetted actors (see chapter 6), thereby preventing such attacks.

Xiao, Y., Zhang, N., Lou, W., & Hou, Y. T. (2020). A survey of distributed consensus protocols for blockchain networks. *IEEE Communications Surveys & Tutorials*, *22*(2)

However, for *permissionless* networks, in which participants can partake in consensus without any control, more complex mechanisms need to be applied to combat Sybil attacks. These commonly entail aligning entitlement to participate in consensus proportionally with the possession or expenditure of resources that can be digitally verified (Xiao *et al.*, 2020). Proof-of-Work (PoW) is an example of a Sybil attack resistance scheme that has been used in most early cryptocurrencies such as Bitcoin (Nakamoto, 2008). To counteract Sybil attacks, PoW uses cryptographic puzzles of configurable difficulty with efficient verification so that it becomes computationally expensive for attackers to interfere with consensus (Back, 1997). However, the electricity consumption of PoW-based cryptocurrencies is connected to their respective market capitalisations, leading to extreme energy demand for popular implementations (Sedlmeir *et al.*, 2020b). For instance, the electricity demand of Bitcoin is now in the same range as that of entire industrialised nations (de Vries, 2018) and has been positioned as a dangerous contributor to global warming, producing up to 22.9 Mt CO_2 (Stoll *et al.*, 2019). Against this backdrop, many alternatives to PoW have been proposed that do not rely on extensive computational effort (Ismail & Materwala, 2019). Among these is Proof-of-Stake (PoS), in which participants with larger holdings of a cryptocurrency have a greater influence in transaction validation. While PoS is generally understood as being more energy-efficient than PoW, the exact electricity consumption characteristics of PoS-based systems, and the influence that network throughput has on them, are not widely understood.

Back, A. (1997, March). *A partial hash collision based postage scheme.*

de Vries, A. (2018). Bitcoin's growing energy problem. *Joule*, *2*(5)

Two main approaches to quantifying the electricity consumption of a DLT system have been used in the past. One is to measure the consumption of a representative participant node and then extrapolate from this measurement. An alternative approach is to develop a mathematical model that includes the core metrics of a DLT system to calculate its energy footprint. Extensive research efforts have cumulated in best practices for determining the energy consumption of DLT systems (N. Lei *et al.*, 2021). So far, most work has focused on PoW blockchains[1], and some research has investigated individual non-PoW systems. However, there is

1: For the purpose of this manuscript, the term 'blockchain' refers to any type of DLT, even if it does not make use of the 'block' concept, first described by Nakamoto (2008).

no widely agreed-upon methodology for comparing energy consumption against traditional financial infrastructures (Platt *et al.*, 2025).

In this chapter we propose a simple electricity consumption model, applicable to a broad range of DLT systems that use PoS for Sybil attack resistance. Specifically, this model considers the number of validator nodes, their electricity consumption, and the network throughput, based on which the energy consumption per transaction is estimated. We present the results of applying this model to six PoS-based systems. Our results suggest that, while negligible compared to PoW, the electricity consumption of PoS systems can still vary significantly.

The next section reviews related work in both experimental and mathematical models. We then briefly describe the relevant architectural features of selected PoS systems. In the following section, we introduce our model in detail and describe how the underlying data were obtained. Finally, we apply the model to the systems selected, present the comparative results, and discuss the limitations.

3.1 Related Work

We conducted an informal literature review[2] using the 'Bielefeld Academic Search Engine', a popular academic search system meeting all the necessary quality requirements for a systematic literature review (Gusenbauer & Haddaway, 2020). We thereby obtained 413 results of prior studies analysing the energy demand of different DLT systems, with a significant focus on PoW blockchains in general, and specifically on Bitcoin. Commonly, models take one of the following two forms.

2: This was done using the query `("Blockchain" OR "DLT" OR "Distributed Ledger") AND ("Energy Consumption" OR "Energy Demand" OR "Electricity Demand" OR "Carbon Footprint") year:[2008 TO *]`.

Experimental Models

The first form revolves around conducting experiments using mining hardware and measuring its actual electricity consumption, as performed by Igumenov *et al.* (2019) with different configurations of computational resources. This approach has been used to derive consumption characteristics for different usage scenarios. The 'BCTMark' [*sic*] framework (Saingre *et al.*, 2020), for instance, allows for the deployment of an entire experiment stack, including the DLT system under test. Using load generators, a realistic network workload can be created. The effects on the electricity consumption of this setup under varying loads can subsequently be measured via energy sensors connected to the testbed. An experimental study on the electricity consumption of the permissioned, non-PoW XRP ledger demonstrates that customising validator hardware can yield reductions in energy demand (Roma & Hasan, 2020). Metrics reported for common cryptocurrencies have been combined with testbed experiments to model the electricity consumption behaviours of various consensus algorithms (Cole & Cheng, 2018).

Igumenov, A., Filatovas, E., & Paulavičius, R. (2019, November). Experimental investigation of energy consumption for cryptocurrency mining. J. Bernatavičienė (Ed.), *Proceedings of the 11th international workshop on data analysis methods for software systems.* Vilnius University Press

Mathematical Models

An alternative method is to quantify assumptions about the environment in which a DLT system operates. Often, such models use a 'top-down' approach that relies on publicly observable factors – such as the hash rate in the case of Bitcoin – and associates them with common mining hardware or even seeks to determine the hardware used via surveys (N. Lei *et al.*, 2021). The papers of Gallersdörfer *et al.* (2020), Küfeoglu and Özkuran (2019), and Zade *et al.* (2019) are examples of this hash rate-based approach. Sedlmeir *et al.* (2020b) undertake a basic comparison of different DLT architectures with the conclusion that the electricity consumption differs significantly depending on the design chosen. A further study by the same authors (Sedlmeir *et al.*, 2020a) refines previous models for measuring the power consumption of Bitcoin, such as the one by Vranken (2017), and emphasises that the driving forces behind power consumption are the Bitcoin price and the availability of cheap electricity. Eshani *et al.* (2021) use a linear regression model to predict Ethereum's electricity consumption based on the observed hash rate and difficulty level; however, the use of simplistic interpolation techniques alone is likely not an appropriate method for PoW blockchains (N. Lei *et al.*, 2021). Powell *et al.* (2021) derive a mathematical model for the electricity consumption of the PoS-based Polkadot blockchain by extrapolating from the power demand of a single validator machine.

Gallersdörfer, U., Klaaßen, L., & Stoll, C. (2020). Energy consumption of cryptocurrencies beyond Bitcoin. *Joule*, *4*(9)

Powell, L. M., Hendon, M., Mangle, A., & Wimmer, H. (2021). Awareness of blockchain usage, structure, & generation of platform's energy consumption: Working towards a greener blockchain. *Issues In Information Systems*, *22*(1)

3.2 Systems Reviewed

This paper compares six DLT systems, both permissioned and permissionless, with high market capitalisation that use PoS as part of the consensus mechanism. To reiterate: PoS systems are conceptualised in such a way that the eligibility to actively participate in their consensus mechanism is proportional to the amount of cryptocurrency a participant holds. This means that the resource 'computing power', as used in PoW, is replaced by the resource 'capital' (Sedlmeir *et al.*, 2020b). While the systems compared use different PoS-based consensus mechanisms that have different participation requirements, such as a minimum balance or a commitment to staking capital for a pre-defined period, they share important commonalities. A commonality particularly relevant to electricity consumption is the need for dedicated servers, validator nodes, to validate sets of transactions and attach generation proofs. This activity is essential, irrespective of whether transactions are organised as chains of blocks (Xiao *et al.*, 2020), or as directed acyclic graphs (Živić *et al.*, 2020). It is this validation and proof generation aspect that attracts stern criticism in the context of PoW since it is responsible for its problematic electricity consumption characteristics (Xiao *et al.*, 2020).

Sedlmeir, J., Buhl, H. U., Fridgen, G., & Keller, R. (2020b). The energy consumption of blockchain technology: Beyond myth. *Business & Information Systems Engineering*, *62*(6)

The selection process for validation and proof generation in PoS constitutes a pseudo-random procedure in which a higher stake yields a higher probability of being selected (Deirmentzoglou *et al.*, 2019). PoS can be applied in both permissioned (Baird *et al.*, 2021) and permissionless (Oyinloye *et al.*, 2021) settings (see Table 3.1). Participants in *permissionless* networks can contribute validator nodes by following the respective admission procedures, such as broadcasting a key registration transaction

Baird, L., Harmon, M., & Madsen, P. (2021, August 15). *Hedera: A governing council & public hashgraph network: The trust layer of the internet* (2.1).

Platform	Permissioned	Permissionless
Ethereum 2		•
Algorand		•
Cardano		•
Polkadot		•
Tezos		•
Hedera	•	

Table 3.1: Comparison of the analysed DLT systems in node permission setting.

in the case of Algorand (Gilad *et al.*, 2017). The procedures often involve staking cryptocurrency and emitting messages on the ledger to indicate the willingness to act as validator (S. Aggarwal & Kumar, 2021). In *permissioned* systems, such as Hedera, participation in the pseudo-random selection procedure is, however, limited to participants previously determined via an off-ledger process (Baird, 2016). Once a node is recognised as a validator, it may participate in validation and proof generation. Depending on the particulars of the consensus mechanism, selection may require additional protocol steps, such as election onto a committee (Gilad *et al.*, 2017). Once selected, validator nodes will verify the transactions received and generate proofs of validation, most commonly in the form of cryptographic signatures.

Gilad, Y., Hemo, R., Micali, S., Vlachos, G., & Zeldovich, N. (2017). Algorand. *Proceedings of the 26th Symposium on Operating Systems Principles*

In summary, the commonalities of the protocols analysed can be described as follows: in all protocols, participants can act as validators, thereby qualifying to perform transaction validation and proof generation. In permissioned networks, the set of participants that can act as validators is limited. In permissionless networks, there are no such limitations. To act as a validator, a participant needs to be operating a computer that can send and receive data across the Internet. This computer must be able to perform the computations required to establish the correctness of proposed transactions and to make other calculations mandated by the consensus protocol. Operating such a validator node is an opt-in process, which means that participants can choose whether to run a validator node. Validator nodes need to remain in an active state as, due to pseudo-random selection, periods of activity often cannot be predicted in advance. Focusing on these similarities allows us to devise a method that is applicable to all six analysed systems.

3.3 Method

Our model differs from previous work (see section 3.1) in that we also consider electricity consumption *per transaction*, as opposed to only the overall energy footprint of an entire DLT system. Nevertheless, existing models can be combined with additional data arising from the scientific literature, reports, and public ledger information to form a baseline that can be used to avoid time-consuming experimental validation. Powell *et al.* (2021) define an elementary mathematical model for the electricity consumption of the Polkadot blockchain that can be generalised as:

Powell, L. M., Hendon, M., Mangle, A., & Wimmer, H. (2021). Awareness of blockchain usage, structure, & generation of platform's energy consumption: Working towards a greener blockchain. *Issues In Information Systems, 22*(1)

$$p_t = p \cdot n_{\text{val}}, \tag{3.1}$$

where p_t is the overall average power the DLT system consumes, p is the average power consumed by a validator node, and n_{val} is the number of validator nodes. As this model constitutes the baseline of our work, the results it yields for the Polkadot blockchain are comparable to ours. Due to the comparatively low computational effort associated with PoS and the intentionally relatively low throughput of permissionless blockchains to avoid centralisation because of computing, bandwidth, or storage constraints (Buterin, 2021), it can be assumed that validating nodes run on similar types of commodity server hardware, irrespective of the network load.

Buterin, V. (2021, April). *Why sharding is great: Demystifying the technical properties.*

Under this assumption, the overall energy need of such a protocol is solely contingent on the number and hardware configuration of validator nodes. In the context of this chapter, we only consider the energy footprint of the consensus mechanism itself. We therefore only consider *validators*[3],

3: Nodes fulfilling this role go by various names, e.g., 'participation nodes' for Algorand, or 'bakers' for Tezos.

Platform	# Validators	TPS Cont. (tx/s)	TPS Max. (tx/s)
Ethereum 2	2,649*	15.4*	3,000
Algorand	1,126	9.8	1,000
Cardano	8,874	0.4	257
Polkadot	297	0.1	1,000
Tezos	399	1.7	40
Hedera	21	48.2	10,000

* Ethereum 1 Mainnet measurements used as approximation

Table 3.2: The current number of validators, contemporary throughput, and the upper bound of throughput postulated.

i.e., nodes that actively participate in a network's consensus mechanism by submitting and verifying the proofs necessary for Sybil attack resistance (Xiao *et al.*, 2020). The overall number of nodes, including other *full nodes* that replicate the transaction history without participating in consensus, is likely higher for all systems analysed. A key model parameter, therefore, is the number of validator machines running concurrently (n_{val}). This number can be established reliably, since it is stored on-chain as a key aspect of any PoS-based protocol. Table 3.2 shows the number of validators currently operating on each of the networks considered.

Electricity Consumption per Transaction

To arrive at an electricity consumption per transaction metric (c_{tx}), the number of transactions per unit of time needs to be considered. The actual numbers are dynamic and fluctuate over time. The contemporary network throughput (*Cont.*) is defined as the actual throughput recently experienced by a system. As a key metric, this can be derived from approximate timestamps that are associated with transactions on public ledgers. The maximum postulated sustainable system throughput (*Max.*) of the different protocols is derived from casual sources. Note that these postulated figures are probably optimistic, that is, not necessarily reliable, as they originate not from controlled experiments, but are anecdotal or come from promotional materials. However, we consider these estimates acceptable as they have no direct influence on the electricity consumption per transaction for a fixed contemporary network throughput. They merely dictate the domain of the consumption function $f_{c_{\text{tx}}}(l)$ that calculates the consumption per transaction depending on the overall system throughput l (measured in tx/s). Treating the average power consumed by a validator node (p, measured in W) as a constant means that an inverse relationship between consumption per transaction (c_{tx}) and system throughput (l) can be established within the bounds of $(0, l_{\max}]$:

$$f_{c_{\text{tx}}}(l) = \frac{n_{\text{val}} \cdot p}{l}. \tag{3.2}$$

Modelling c_{tx}

Equation (3.2) depends on two variables: n_{val} and l. We will now present a model for c_{tx} that depends on one variable, namely l, only. Data from the Cardano blockchain (Platt, 2021a) suggest that the number of validators n_{val} and the number of transactions per second l are positively correlated. Namely, Pearson's correlation coefficient[4] for n_{val} and l for 375 data points from 29 July 2020 to 7 August 2021 is 0.80. The correlation coefficient for n_{val} delayed by 28 days and l (not delayed) for the same data is 0.87. This is plausible for the following reason: as the total number of users

4: The correlation coefficient takes values in $[-1, 1]$ and a value of ± 1 would imply that n_{val} is an affine function in l.

in a permissionless system increases, of the new users, a share becomes validators and another non-disjoint share executes transactions, meaning that n_{val} and l are positively correlated. For permissioned systems, it is still conceivable that the number of validators and throughput are positively correlated because, as new partner organisations are invited to run validator nodes, these partners may decide to use the system for their own applications, thereby increasing the number of transactions. We also observe that in the case of Hedera, the number of validators and the throughput are positively correlated: the number of validator nodes has been continuously increasing, and throughput, while fluctuating from month to month, has increased year to year. Furthermore, it can be observed for the Algorand and Hedera systems that n_{val} and l have increased from July to August 2021. On the Polkadot blockchain, n_{val} has remained constant from February to July 2021. An exception is the Tezos blockchain for which n_{val} has decreased while l has increased from February to August 2021. This trend has so far held true throughout the lifetime of the Tezos blockchain. We note that, in this case, an affine function is not appropriate for modelling the dependence of n_{val} on l because n_{val} would become negative for large values of l. We will still compute the affine best approximation of n_{val} in terms of l for the Tezos blockchain, as it is an approximation of the first Taylor polynomial of n_{val}, and therefore a local model for n_{val}.

For simplicity, we assume that the correlation is perfect, i.e., $n_{\text{val}} = \kappa + \lambda \cdot l$ for some $\kappa, \lambda \in \mathbb{R}, \lambda > 0$, and using (3.2) we obtain

$$f_{c_{\text{tx}}}(l) = \frac{(\kappa + \lambda l) \cdot p}{l}. \tag{3.3}$$

Because we could not obtain high-resolution historic data for Algorand, Polkadot, Tezos, and Hedera, we will compute κ, λ later based on two data points. For Cardano, we use linear regression implemented as ordinary least squares regression to compute κ, λ that have the maximum likelihood of modelling $f_{c_{\text{tx}}}(l)$ under the assumption that $f_{c_{\text{tx}}}(l)$ is an affine function with Gaussian noise. The resulting values for κ, λ can be found in Table 3.3.

Platform	κ	λ
Algorand	102.8	103.9
Cardano	3,803.4	8,877.6
Polkadot	297	0
Tezos	440.7	−24.6
Hedera	7.6	0.3

Table 3.3: Estimates for κ, λ used in Equation 3.3 to model the number of validators depending on the number of transactions per second.

Hardware Considerations

In contrast to energy-intensive PoW systems, in PoS, the computational effort relating to the participation in the consensus protocol can almost be considered independent of extraneous factors like cryptocurrency capitalisation. Numerous factors influence the overall electricity consumption of a server, with CPU activity, hard disk drive operations, and cooling being the most significant (Jaiantilal *et al.*, 2010). The consensus-related energy demand in PoS is generally constant, meaning it occurs irrespective of system load (Jaiantilal *et al.*, 2010). Energy demand relating to CPU time and input/output operations is, however, highly load-dependent (Z. Zhou

Jaiantilal, A., Jiang, Y., & Mishra, S. (2010). Modeling CPU energy consumption for energy efficient scheduling. *Proceedings of the 1st Workshop on Green Computing* *et al.*, 2019). Therefore, a realistic electricity consumption estimate for a validator node needs to factor in both the minimum hardware requirement (i.e., how many CPU cores or how much memory is required) and the utilisation of that hardware.

Since it is nearly impossible to determine which type of hardware is used by validators, we use an approximation derived from industry recommendations. Dramatically different hardware recommendations are put forward for permissionless systems and permissioned systems. The permissionless systems analysed in this chapter, all traditional blockchains with comparatively large numbers of validators running full nodes that verify every transaction (Buterin, 2021), demand comparatively low-powered hardware. Hedera, the only permissioned system analysed here, constitutes a high-tps (transactions per second) system. Such systems are characterised by few nodes maintaining consensus (Buterin, 2021). The maximum network performance is determined by the lowest-performing validator node (Hunjan, 2020). Therefore, to achieve the postulated maximum throughput values, the network operator demands highly performant server hardware. We assumed that similar high-tps systems would have energy requirements in the same range. This explains the difference in the electricity consumption per validator node between Hedera and the other traditional blockchain systems.

Hunjan, S. (2020, July 7). *Node requirements.*

Table 3.4: Conceivable upper and lower bounds for the power demand of a validator machine.

Configuration	Hardware Type	Exemplar	Demand (W)
Minimum	Small single-board computer	Raspberry Pi 4	5.5
Medium	General purpose server	Dell PowerEdge R730	168.1
Maximum	High-performance server	HPE ProLiant ML350 Gen10	328

To capture the uncertainty regarding appropriate hardware and expected hardware utilisation in the model, three different validator configurations are considered (see Table 3.4): a single-board computer, a general-purpose rackmount server for midsize and large enterprises, and a high-performance server. For all configurations, hardware utilisation based on typical workloads is assumed. For traditional blockchains, we assume a power demand in the minimum to medium range (5.5 to 168.1 W). For high-tps systems, the medium to maximum range (168.1 to 328 W) is assumed.

3.4 Results

Table 3.5 illustrates the application of the models described in Equation 3.2 and Equation 3.3 to estimate the electricity consumption of the protocols considered under contemporary throughput, i.e., based on recent throughput measurements (see section 3.3). To facilitate a broad overview, we also provide the global system-wide consumption of each DLT system according to the model. Furthermore, the table presents two estimates for electricity consumption per DLT system: an *optimistic* estimate assuming validator nodes are operated on the lower bound of the system range and a *pessimistic* estimate that assumes validators utilise hardware on the higher bound (see Table 3.4). As, at the time of writing, the merging of the Ethereum Mainnet with the Beacon Chain was outstanding, no contemporary throughput figures for Ethereum 2 were established. Instead, the current throughput of Ethereum Mainnet (15.4 tx/s) is presented.

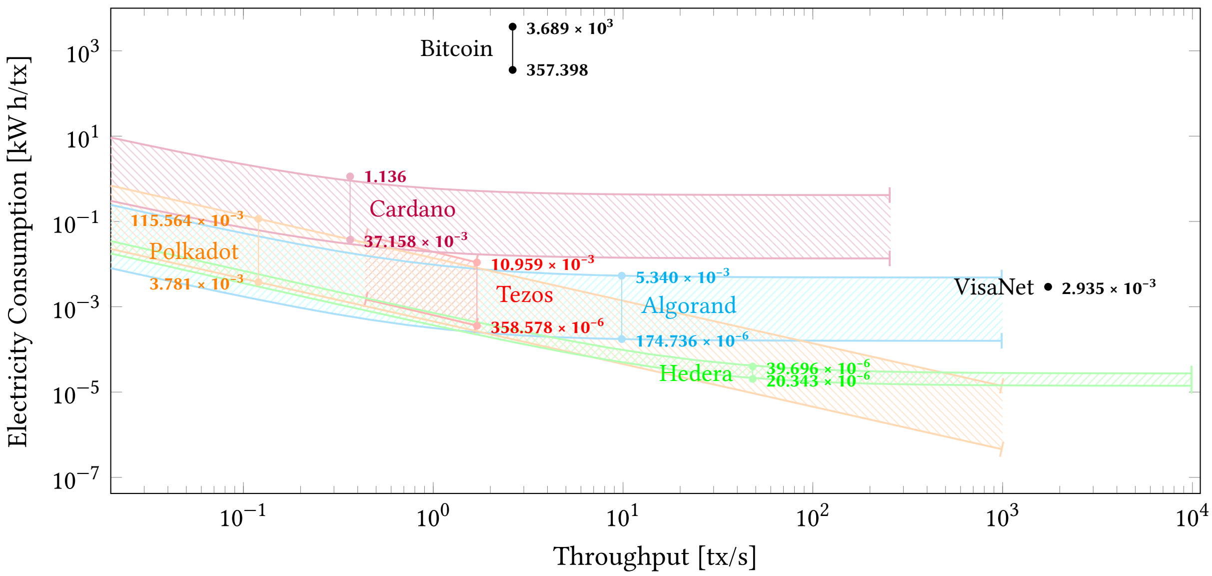

Figure 3.1: The electricity consumption per transaction is close to inversely correlated with throughput. For each system, the lower mark indicates the energy consumption under an optimistic validator hardware assumption while the upper mark indicates a pessimistic model. The consumption figures for Bitcoin and VisaNet are plotted for comparison. Ethereum 2 is not plotted as Ethereum has not yet merged with the Beacon Chain. For Algorand, Hedera Hashgraph, Polkadot and Tezos, contemporary throughput and validator count data points lie exactly on the graph because the interpolation was computed by requiring the graph to pass through them; for Cardano, the graph is a regression obtained from many data points, so the contemporary data points do not lie exactly on the graph.

All estimates are based on the validator counts established earlier (see section 3.3). The plot of the model function shown in Figure 3.1 visualises the inverse relationship described earlier within the boundaries of the postulated throughput values (see Table 3.2). It also provides a projection of electricity consumption as a function of system load, based on the model presented earlier which predicts the number of validators as a function of system load. This projection is equally illustrated within the boundaries of the postulated throughput values, except in the case of the Tezos blockchain, for which no global model could be derived.

Based on this data, we can compare the electricity consumption per transaction on two related systems: first, the PoW cryptocurrency Bitcoin, and second, the VisaNet payment network (see Figure 3.1). It becomes evident that the consumption of Bitcoin – overall and per transaction – is at least three orders of magnitude higher than that of the highest consuming PoS system even under favourable assumptions. While the difference between PoS systems and VisaNet is less pronounced, it is evident that most of the former undercut the electricity consumption of VisaNet in many configurations.

Pronounced differences between PoS-based systems are equally evident from the results. We observe a low energy demand per transaction in active permissioned DLT systems that are characterised by high throughput and comparatively small numbers of validators with a corresponding low degree of decentralisation. Less active permissionless systems show a higher energy demand per transaction due to comparatively low throughput and a high number of validators. This illustrates that not only for PoW (Carter, 2021), but also for PoS blockchains, 'electricity consumption per transaction' should not be the only metric considered for assessing sustainability. In particular, when utility is not approximately proportional to throughput, total electricity consumption may be a more appropriate key figure.

Carter, N. (2021, May 5). *How much energy does Bitcoin actually consume?* Harvard Business Review.

Table 3.5: Global power consumption ranges (i.e., the network-wide consumption of the DLT systems under consideration and of VisaNet) and ranges of the electricity consumed per transaction for contemporary throughput.

Platform	Global (kW)			Per transaction (kW h/tx)		
Ethereum 2[*]	14.6	–	445.3	0.00026	–	0.00803
Algorand	6.2	–	189.3	0.00017	–	0.00534
Cardano	48.8	–	1,491.7	0.03716	–	1.13562
Polkadot	1.6	–	49.9	0.00378	–	0.11556
Tezos	2.2	–	67.1	0.00036	–	0.01096
Hedera	3.5	–	6.9	0.00002	–	0.00004
Bitcoin	3,373,287.7	–	34,817,351.6	360.39300	–	3,691.40700
VisaNet			22,387.1			0.00358

[*] Ethereum 1 Mainnet measurements used as approximation

3.5 Discussion

Interpretations

These results can primarily be understood as a clear confirmation of the common opinion that the electricity consumption of PoW systems, especially Bitcoin, is excessive. Therefore, they can be interpreted as a strong argument for the modernisation of PoW-based systems towards PoS. Ethereum is taking a commendable lead in this respect with the development of Ethereum 2. Furthermore, the results indicate that the electricity consumption of different non-PoW blockchains is surprisingly divergent. In absolute terms, however, the consumption rates of PoS-based systems are moderate and thus also much closer to the figures for traditional, centralised payment systems such as VisaNet. The main reason why our model yields considerable divergence between PoS systems is the difference in the number of validators and throughput. Specifically, in permissioned systems, electricity consumption can be controlled through the ability to limit the number of validators on a network, so the permissioned network analysed in this chapter is characterised by low energy consumption.

This result should not be misinterpreted as an argument for increased centralisation or for permissioned networks over permissionless ones. This becomes obvious when considering a permissioned DLT system in extremis: such a system would consist of only a single validator node and would thus be effectively centralised. This hypothetical scenario shows that, if a permissioned paradigm is applied, close attention should be paid to system entry barriers enforced through gatekeeping capabilities. If not, there is a risk of centralisation, which may offer minuscule advantages in terms of electricity consumption but will negate the functional advantages of a decentralised paradigm. The fact that the selection of suitable validator hardware is central to electricity consumption is also of practical relevance. Information regarding adequate hardware for validators is often inconsistent. Therefore, standardised recommendations should be put forward to help operators of validator nodes in selecting the most energy-efficient hardware configurations.

This chapter is only a first step towards quantifying the electricity consumption of PoS systems. Its results reveal a need for comprehensive models that account for the entire blockchain ecosystem, contrasting significantly with traditional financial systems (Platt *et al.*, 2024). However, despite its limitations, it gives impetus to designers of decentralised

Platt, M., Platt, D., & McBurney, P. (2024). Sybil attack vulnerability trilemma. *International Journal of Parallel, Emergent and Distributed Systems, 39* systems by revealing the dependency between validator number, load, and hardware configuration. Our model can thus be used to determine the carbon footprint of a particular use case. It can, furthermore, prompt operators of validator nodes to carefully select suitable hardware.

Limitations

So far, we have used broad consumption ranges to model the electricity consumption of individual validator nodes. While we are confident that the actual electricity consumption is in fact within these ranges, the underlying characteristics of different PoS protocols that might impact energy consumption, such as the accounting model, have been ignored. Second, while assuming that the electricity consumption of a validator node is independent of system throughput is well justified for the permissionless systems analysed (Buterin, 2021), permissioned systems that are designed to support high throughput may not warrant such an assumption. We have accounted for this by assuming more powerful hardware for high-tps systems, but more work is needed to better understand permissioned blockchains' electricity consumption characteristics. Moreover, the impact of different workloads on electricity consumption should be considered; for example, simple payment transactions may have lower computational requirements when compared to more complex smart contract invocations, but we have not so far distinguished between transaction types.

While our model suggests that PoS systems can remain energy-efficient while scaling up to VisaNet throughput levels, there is no hard evidence in support of this argument, as, to our knowledge, no DLT-based system has experienced a sustained volume of this magnitude to date on the base level. On the other hand, we ignored the possibility of achieving effectively higher throughput than the specified maximum through layer 2 solutions, such as the Lightning network or via optimistic rollups and zero-knowledge-rollups that are receiving increasing attention.

Finally, although there are reasons to support its plausibility, the assumption that an affine function can be used to express the number of validators in terms of throughput is questionable. While we assume that it is applicable to Hedera, this might not be a justifiable assumption for other permissioned settings. The applicability of this model to other permissioned systems should therefore be more formally analysed, for example, by collecting more data points and comparing these with the model outputs.

3.6 Conclusion

The increasing popularity of DLT systems since the invention of Bitcoin, and with it the energy-intensive PoW consensus mechanism, has produced a variety of alternative approaches. PoS is a particularly popular method that is commonly assumed to be more energy efficient than PoW. In this chapter, we tested this hypothesis using a mathematical consumption model that predicts expected electricity consumption per transaction as a function of network load. Applying this model to six different PoS-based DLT systems supports the hypothesis and suggests that their electricity consumption per transaction is indeed at least three orders of magnitude lower than that of Bitcoin. Furthermore, we discover significant differences among the analysed PoS-based systems themselves. Here, a permissioned system was found to consume significantly less energy per transaction than a permissionless system. This difference can be attributed to the gatekeeping capabilities offered by permissioned systems.

These results can be understood as an urgent call for the modernisation of PoW systems and a shift towards PoS, as well as a recommendation to practitioners to consider energy-saving hardware which aligns with minimal supported configurations. They are also intended to provide a basis for the future comparative study of the energy friendliness of PoS systems and to facilitate the development of more rigorous consumption models. Given the enormous challenges posed by climate change, avoiding unnecessary electricity consumption needs to be a high priority. Our work shows that PoS-based systems can contribute to this and could even undercut the energy needs of traditional central payment systems, raising hopes that DLT can contribute positively towards combatting climate change.

Energy Demand Unawareness of Blockchain Users 4

This chapter follows the mathematical modelling of PoW and PoS energy consumption in chapter 3. It extends that work by investigating whether users are aware of the significant energy consumption associated with PoW to gauge their awareness of the technical aspects of the systems they use. Understanding user sentiment is crucial for assessing the acceptability of novel consensus mechanisms for self-governance lateron.

Bitcoin is arguably the most popular cryptocurrency (Mikhaylov, 2020) and has a major impact on the wider crypto-asset ecosystem (Ante, 2022). It has experienced enormous success since its invention in 2008, as evidenced by its peak market capitalisation exceeding 1 trillion United States dollars (USD) (de Best, 2021) and its growing user base (Park *et al.*, 2019). Yet, this original 'pure digital asset' (F. Fang *et al.*, 2022) has been criticised outright for reproducing societal inequality (Walsh, 2021) and for its enormous electricity demand caused by the underlying PoW consensus mechanism (Badea & Mungiu-Pupazan, 2021; de Vries, 2018; Gehlot & Dhall, 2022; Truby, 2018). While Bitcoin's exact energy footprint cannot be established with certainty, and estimates vary considerably depending on the method of measurement applied (Gallersdörfer *et al.*, 2020; N. Lei *et al.*, 2021), the cryptocurrency is commonly considered a substantial contributor to global warming. As such, it was found to produce up to 65.4 Mt CO_2 annually: the equivalent of the total emissions of Greece (de Vries, 2022). There are now numerous cryptocurrencies that incorporate more sustainable consensus mechanisms (Miraz *et al.*, 2021), including the second-largest cryptocurrency by market capitalisation, Ethereum, that has recently transitioned to PoS (de Vries, 2022). Still, many cryptocurrencies, first and foremost Bitcoin, continue to apply PoW. Consequently, a critical examination of this technology remains pressing beyond the ongoing debate on cryptocurrencies as instruments of payment in the context of criminal activity (Kethineni & Cao, 2019; Treiblmaier & Gorbunov, 2022).

Ante, L. (2022). The non-fungible token (NFT) market and its relationship with Bitcoin and Ethereum. *FinTech*, *1*(3)

Miraz, M. H., Excell, P. S., & Sobayel, K. (2021). Evaluation of green alternatives for blockchain proof-of-work (PoW) approach. *Annals of Emerging Technologies in Computing*, *5*(4)

Although many experts, including many in the wider Bitcoin community, continue to emphasise that the security of the tried and tested PoW mechanism is unrivalled (Brown-Cohen *et al.*, 2019; Houy, 2014; Shifferaw & Lemma, 2021), this view is not shared universally (Kiayias *et al.*, 2017; Rieger *et al.*, 2022; Saleh, 2020). The strengths and weaknesses of PoW and PoS from the perspective of economic security and decentralisation remain the subject of debates (Nair & Dorai, 2021). Some publications address misconceptions concerning the electricity consumption characteristics of blockchain applications in general (N. Lei *et al.*, 2021) and PoW cryptocurrencies in particular (Sedlmeir *et al.*, 2020b). However, we have no knowledge of academic works that directly assess the awareness of electricity consumption of cryptocurrency users.

Nair, P. R., & Dorai, D. R. (2021). Evaluation of performance and security of proof of work and proof of stake using blockchain. *Proceedings of the 3rd International Conference on Intelligent Communication Technologies and Virtual Mobile Networks*

Research Objectives

This research seeks to bridge the gap between cryptocurrency understanding and environmental consciousness within the Nigerian context. As the global conversation around Bitcoin's environmental impact intensifies and with the rising prominence of cryptocurrency transactions in Nigeria, the insights from this chapter can potentially shape policies, steer educational initiatives, and guide stakeholder decisions.

Research Gap and Context

Despite studies on the sustainability of payment systems by regulatory bodies (Agur *et al.*, 2022, 2023) and various attempts to regulate cryptocurrencies (Ioannou, 2020; Truby *et al.*, 2022), legislators still lack an understanding of how users perceive policies targeting cryptocurrencies and whether they have the knowledge needed to understand the motivation behind these policies. Due to the decentralised nature of cryptocurrencies and the corresponding challenges in banning PoW-based cryptocurrencies, measures that consider users' perspectives and actions are likely to be the only way to bring about long-term improvements in energy use. Considering that the cryptocurrency market offers products with dramatically different carbon footprints, there is a clear gap in research on potential mechanisms to encourage users to explore more sustainable alternatives.

This chapter aims to address this gap through field research in Nigeria. Nigeria's social, cultural, and economic realities are particularly conducive to such research: as a result of a recession and rising inflation, the Nigerian economy is experiencing stress (Yusuf *et al.*, 2022). Coupled with the deficiencies of an outdated and costly traditional banking sector (Osuagwu *et al.*, 2018), this implies that many Nigerians regularly and routinely use cryptocurrencies as a means of payment (Lawal, 2021) despite the rejective position of the government (Bakare, 2021b). This application stands in stark contrast to industrial countries where cryptocurrencies are pursued primarily as a speculative form of investment (Auer & Tercero-Lucas, 2021; Baek & Elbeck, 2014; Bindseil & Schaaf, 2024; Glaser *et al.*, 2014; Steinmetz *et al.*, 2021). Furthermore, the Nigerian public is aware that, due to their location, they could be severely affected by the effects of climate change (Abah, 2014; Haider, 2019; Mustapha *et al.*, 2013). The combination of the extensive use of cryptocurrencies and the high awareness of the effects of climate change makes Nigeria a suitable setting for our work.

Osuagwu, E. S., Isola, W. A., & Nwaogwugwu, I. C. (2018). Measuring technical efficiency and productivity change in the Nigerian banking sector: A comparison of non-parametric and parametric techniques. *African Development Review*, *30*(4)

Hypotheses

We collect and quantitatively analyse data from 158 cryptocurrency users in Nigeria, focusing on hypotheses in three areas: awareness, actionability, and responsibility.

Awareness

The starting hypothesis of this work is that most Nigerian Bitcoin users are unaware of its high energy demand. This hypothesis is motivated by broader research on the technological awareness of Bitcoin users in similar markets (Ku-Mahamud *et al.*, 2019). It is furthermore influenced by earlier findings that showed that cryptocurrency users did not take sustainability into account when selecting a cryptocurrency to use or mine (Shehhi *et al.*, 2014).

Shehhi, A. A., Oudah, M., & Aung, Z. (2014). Investigating factors behind choosing a cryptocurrency. *Proceedings of the 2014 International Conference on Industrial Engineering and Engineering Management*

Actionability

It can be further hypothesised that users who misestimate electricity consumption see less need to counteract it. This is conceivable since a correct understanding of the causes of global warming was found to be a key determinant of behavioural intentions to act against it (Bord *et al.*, 2000). Furthermore, it can be speculated that those users who can accurately estimate electricity consumption possess sufficient expertise to contemplate countermeasures.

Bord, R. J., O'Connor, R. E., & Fisher, A. (2000). In what sense does the public need to understand global climate change? *Public Understanding of Science*, *9*(3)

Responsibility

A further hypothesis is that participants who see a clear need for action counteracting the electricity consumption of Bitcoin feel that nongovernmental actors are responsible as distrust in government has been found to be linked with cryptocurrency adoption (Bratspies, 2018).

Outline

In section 4.1 of this chapter, we introduce the fundamentals of blockchain technology, covering technical concepts like consensus mechanisms and their application to cryptocurrencies like Bitcoin. In this context, we also describe the drivers of electricity consumption in PoW cryptocurrencies. Subsequently, we present critical insights into cryptocurrency adoption in Nigeria and explain the situation in China as a case study. Next, in section 4.2, we give an overview of related work with a focus on research on cryptocurrency user attitudes. The remainder of this chapter features a description of the questionnaire-based online survey conducted (section 4.3), followed by a discussion of the results obtained (section 4.4). We conclude with policy considerations (section 4.6).

4.1 Background

Blockchain technology, a kind of DLT, goes back to the work of Nakamoto (2008) and forms the foundation of the most common cryptocurrencies (Garriga *et al.*, 2020), including Bitcoin. A *blockchain* is commonly characterised as a linear, append-only collection of data elements ('blocks'), all of which are linked to construct a tamper-evident chain using hash-pointers (Butijn *et al.*, 2020; M. Zhang *et al.*, 2020). The data in blocks can be arbitrary (Gregoriadis *et al.*, 2022) but, in the case of blockchains with native cryptocurrencies, consist mainly of transfer instructions between accounts (J. Wu *et al.*, 2021). Although, in theory, transfers can occur between arbitrary addresses, real-world systems typically constitute small-world networks (Serena *et al.*, 2021). Different entities hold replicas of chains and synchronise them by means of a consensus mechanism that facilitates decentralised agreement on which data elements to append next (Tai *et al.*, 2017). Often, blockchains that expose a native cryptocurrency provide incentives to those users who participate in consensus (Mukhopadhyay *et al.*, 2016). In the context of PoW, the incentives are called 'mining rewards'. Stakeholders who aim to transfer amounts of cryptocurrency offer transaction proposals to block producers, who, in turn, select transactions to maximise their reward in terms of fees (Pontiveros *et al.*, 2018).

Garriga, M., Palma, S. D., Arias, M., Renzis, A., Pareschi, R., & Tamburri, D. A. (2020). Blockchain and cryptocurrencies: A classification and comparison of architecture drivers. *Concurrency and Computation: Practice and Experience*, *33*(8), Article e5992

Tai, S., Eberhardt, J., & Klems, M. (2017). Not ACID, not BASE, but SALT - a transaction processing perspective on blockchains. *Proceedings of the 7th Conference on Cloud Computing and Services Science*

Blockchain technology constitutes the foundation for most existing cryptocurrencies. This is because blockchain technology is well suited to record account balances in decentral systems that allow anyone to participate, yet do not require a distinguished trusted authority (Ehrenberg & King, 2019).

Ehrenberg, A. J., & King, J. L. (2019). Blockchain in context. *Information Systems Frontiers*, *22*(1)

Cryptocurrency Mining

As noted, blockchain technology constitutes decentralised ledger technology that synchronises across distributed replicas. To provide synchronisation that is not dependent on the availability and honesty of a distinguished entity, a wide variety of consensus mechanisms can be applied (see chapter 2). These typically combine economic incentives and cryptographic protocols to achieve a system state in which all honest nodes come to agreement under the assumption of an honest majority of nodes (Sedlmeir *et al.*, 2020b; N. Wang *et al.*, 2019). Initial research on consensus mechanisms dates back to the 1980s with the work of Lamport (1998) on 'Paxos' and Castro and Liskov (1999) on PBFT as key contributions.

Castro, M., & Liskov, B. (1999). Practical byzantine fault tolerance. *Proceedings of the 3rd Symposium on Operating Systems Design and Implementation*

Early work focused on *closed* systems in which the number and identity of participating parties are determined in advance. For example, in an aeroplane that requires a particularly high reliability of sensor information, it may be of interest that a coherent overall picture of the system state can be formed even if some components behave unpredictably, for instance, in the presence of cosmic radiation. In such closed systems, the problem of consensus can be solved efficiently by majority voting combined with appropriate communication protocols.

In contrast, *open* systems, such as many cryptocurrencies, do not have predetermined groups of users. Consequently, they do not conform to the principle of 'one participant, one vote' (Sedlmeir *et al.*, 2020b). In such systems, an entity that intends to control the system could skew majority votes by registering many bogus accounts, a technique known as 'Sybil attack' (Douceur, 2002). Most commonly, preventing such attacks is done by linearly tying the weight of a vote to a scarce resource provided by the participants that is verifiable digitally, and by encouraging the provision of this resource through economic incentives (Sedlmeir *et al.*, 2020b). The earliest and, arguably, the simplest approach to satisfy this requirement is to utilise computational power as a scarce resource, as first proposed by Dwork and Naor (1992) and later applied by Nakamoto (2008) in the context of the consensus mechanism for the first cryptocurrency Bitcoin. This approach, commonly termed Proof-of-Work (PoW), ultimately ties a voting weight to hardware and energy and, thus, to capital. More precisely, miners compete by solving cryptographically hard puzzles through trial-and-error (Back, 2002). Whoever solves a puzzle can submit its solution along with the transactions collected as a new block, which will be accepted by other honest nodes. Next, all honest nodes aim to find a subsequent block, including a corresponding solution to a puzzle that is linked to the previous block.

Sedlmeir, J., Buhl, H. U., Fridgen, G., & Keller, R. (2020b). The energy consumption of blockchain technology: Beyond myth. *Business & Information Systems Engineering*, *62*(6)

Electricity Demand

As a consequence, the electricity demand of a PoW cryptocurrency can be determined via a simple approximation: assuming participating miners are rational, they will only provide computational power if their expected revenue (i.e. rewards for finding new blocks and the fees of the transactions included in it) exceeds the cost that they incur for buying, maintaining, and operating hardware. At the time of writing, Bitcoin releases a reward of 6.25 Bitcoin (BTC) for creating a block; and, on average, producing a block takes 10 minutes (de Vries, 2021). Cumulative transaction fees have consistently been one to two orders of magnitude lower per block than mining rewards in many cryptocurrencies, including Bitcoin (Sedlmeir *et al.*, 2020b) and, therefore, can be ignored for rough estimates. Following this line of reasoning, a simple worst-case model for electricity consumption can be established by assuming electricity costs are the only costs for miners and a lower bound on electricity prices is 0.05 USD per kW h (de Vries, 2021; Sedlmeir *et al.*, 2020b). The accuracy of this model can be improved by considering that the share of electricity costs in mining is only around 40 % (de Vries, 2018). In general, the decline in hash rate following price shocks on the revenue side (e.g., a sharp decrease of the Bitcoin price and halving events) and on the cost side (e.g., a sharp increase in electricity prices at the end of the rainy season in China) suggest that the upper bound is relatively accurate (Sedlmeir *et al.*, 2020a; Stinner, 2022). On the other hand, a lower bound for electricity consumption can be derived by observing the complexity of the solved puzzles and the distribution of the energy efficiency of the mining hardware deployed. A variation of this method is also applied by the Cambridge Bitcoin Electricity Consumption Index (CBECI) Bitcoin network power demand model[1], a widely recognised consumption model (Kohli *et al.*, 2023; Maiti *et al.*, 2023; D. Zhang *et al.*, 2023), that forms the basis of the consumption figures applied in the survey (see subsection 4.3). As of mid-July 2022, Bitcoin's annual electricity consumption is within the theoretical limits of 40 to 138 TW h, with an estimate of 84 TW h according to the CBECI model.

de Vries, A. (2021). Bitcoin boom: What rising prices mean for the network's energy consumption. *Joule*, *5*(3)

1: See https://ccaf.io/cbeci/index.

This number appears enormous on its own, yet criticism ignites even further when considering the energy requirements *per transaction*, as cryptocurrencies only have very limited transaction processing capacity (J. Xie *et al.*, 2019). For instance, Bitcoin processes around four transactions per second; the theoretical maximum (given the currently accepted system parameters) is around seven transactions per second (Georgiadis, 2019). Mathematically this yields around 660 kW h per transaction, more than an average household in Germany consumes in 2.5 months, or as much as the average annual electricity consumption of four Nigerians. Nonetheless, it is important to note that, since transaction fees currently play only a marginal role in the remuneration of miners, increasing the limit of transactions in the Bitcoin protocol would not increase total electricity consumption considerably. Due to this particularity, the *energy per transaction* metric frequently causes misunderstandings (Carter, 2021; Dittmar & Praktiknjo, 2019; N. Lei *et al.*, 2021; Sedlmeir *et al.*, 2020b). In this survey, we will, therefore, consider *annual electricity consumption*, as described in subsubsection 4.3.

Xie, J., Yu, F. R., Huang, T., Xie, R., Liu, J., & Liu, Y. (2019). A survey on the scalability of blockchain systems. *IEEE Network*, *33*(5)

The economic relationship between the total electricity consumption of a PoW-based blockchain and the market price of its native cryptocurrency suggests that increasing the energy efficiency of the mining hardware or reducing the number of transactions will not help much to improve sustainability (Sedlmeir *et al.*, 2020b). Since both mining rewards and transaction fees are paid in the native cryptocurrency (e.g. BTC) of the blockchain, the cryptocurrency market price is the most important factor influencing blockchain electricity consumption systematically (Gallersdörfer *et al.*, 2020). This price, however, cannot be directly controlled

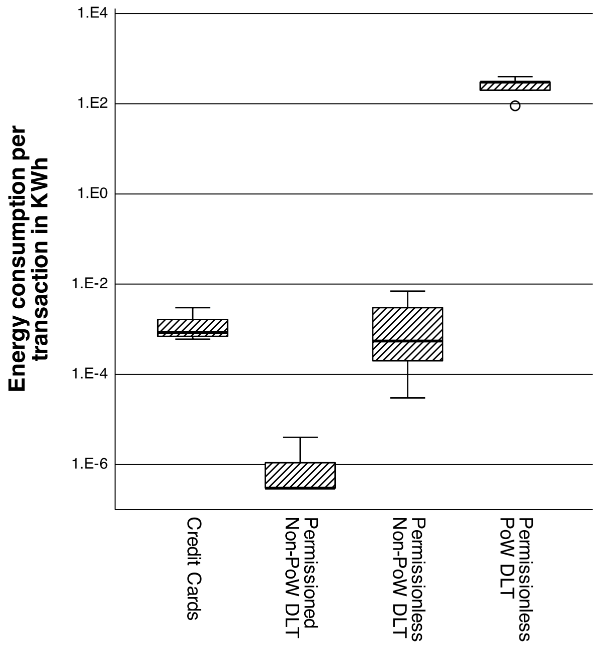

Figure 4.1: A comparison of data concerning the electricity consumption per transaction reported in the wider literature (logarithmic scale) compiled by Agur *et al.* (2022) confirms that the scientific community, despite divergent estimates, considers permissionless PoW DLT systems to consume several orders of magnitude more electricity than other payment systems.

through regulatory means. Instead, a reduction in *adoption* of a given currency is likely to reduce its price (Bhambhwani *et al.*, 2019) and can therefore be considered a sustainability policy tool. For this reason, we focus on evaluating whether consumers who are well-informed about the electricity consumption of a PoW blockchain and its drivers are more willing to support countermeasures, such as abandoning an unsustainable blockchain and moving to a sustainable one.

Bhambhwani, S., Delikouras, S., & Korniotis, G. (2019, May). *Blockchain characteristics and the cross-section of cryptocurrency returns* (Discussion Paper No. 13724). Centre for Economic Policy Research. London, UK

Alternatives to PoW

A popular alternative consensus mechanism used in cryptocurrencies is Proof-of-Stake (PoS), which employs cryptocurrency stake as a scarce resource instead of computational work. As we illustrate in Figure 4.1, systems that rely on this consensus mechanism are several orders of magnitude more energy efficient than those that use PoW (see chapter 3 and Rieger *et al.*, 2022). While PoS-based systems are arguably more challenging to design and implement securely and there are doubts about their incentive compatibility under certain conditions (Brown-Cohen *et al.*, 2019; Houy, 2014): whether PoW or PoS is more secure remains unresolved (Keenan, 2017; Ouyang *et al.*, 2021). In any case, PoS-based systems enable consumer choice: while end users are challenged by the usability of blockchains in general (D. Shin & Ibahrine, 2020), whether systems are operated using PoW or PoS, does not noticeably affect their usability (Jang *et al.*, 2021). Energy-intensive PoW-based currencies, foremost Bitcoin, remain dominant (de Best, 2022) although more sustainable PoS-based alternatives exist and are conveniently available to users through popular centralised exchange websites (Arslanian, 2022) that offer comparatively good usability and low transaction fees (Z. Zhou & Shen, 2022).

Rieger, A., Roth, T., Sedlmeir, J., & Fridgen, G. (2022). We need a broader debate on the sustainability of blockchain. *Joule*, *6*(6)

Arslanian, H. (2022). Crypto exchanges. *The book of crypto*. Springer

Cryptocurrency in Nigeria

Nigeria, distinguished by a high position in the ‘Bitcoin Market Potential Index’ (Hileman, 2015) and considered an attractive environment for

Hileman, G. (2015). The Bitcoin market potential index. M. Brenner, N. Christin, B. Johnson & K. Rohloff (Eds.), *Proceedings of the 19th international conference on financial cryptography and data security*. Springer commercialisation of cryptocurrency activities (Jutel, 2023), has a cryptocurrency usership of approximately 32 % of the population (Adesina, 2020; Lawal, 2021). This high number may be explained by economic hardship associated with the prevailing unemployment, which is believed to be structural (Olubusoye *et al.*, 2022), coupled with a worsening inflation (Lawal, 2021). This climate, linked with the proliferation of mobile and wireless devices (Burns, 2022), allowed many citizens, especially the youth, to interact with and adopt cryptocurrencies as a safe haven from looming inflation (J. Zhao, 2022). Furthermore, high fees for international transfers of funds have established cryptocurrencies as an alternative to traditional banking (BBC News, 2021). Consequently, Nigerians consider international acceptance as one of the key advantages of cryptocurrencies (Onyekwere *et al.*, 2023). In addition to legitimate applications, cryptocurrencies have been found to be used for illicit purposes in Nigeria, including money laundering (Ediagbonya & Tioluwani, 2022) and financing acts of terrorism (Emmanuel & Michael, 2020). Furthermore, they can be used in the context of scams around fabricated cryptocurrencies or cryptocurrency theft (Kothari, 2023), activities that entrenched youth unemployment contribute to (Ewuzie *et al.*, 2023). Nigeria has become infamous for such scams (Ibrahim, 2016; Lazarus & Okolorie, 2019; Okosun & Ilo, 2022) and cryptocurrency transactions, due to their non-reversible nature, hold a special allure for scammers (Butler, 2021). Despite these concerns, cryptocurrency activity in Nigeria is believed to predominantly help people address their daily financial needs (Stringham, 2023), contrary to the portrayals in popular media. This sentiment was echoed by many Nigerians we interacted with in the course of our fieldwork.

Butler, S. (2021). Cyber 9/11 will not take place: A user perspective of Bitcoin and cryptocurrencies from underground and dark net forums. T. Groß & L. Viganò (Eds.), *Proceedings of the 10th international workshop on socio-technical aspects in security and trust*. Springer

Due to the rising popularity of cryptocurrencies concerns arose among the regulatory authorities, especially the Central Bank of Nigeria (CBN): cryptocurrencies were seen as excessively speculative in nature and therefore considered a risk to the financial well-being of Nigerians (Bakare, 2021a, 2021b; Nwanisobi, 2021). Therefore, in an effort to regulate the market, the CBN placed a ban on banks that facilitate cryptocurrency-related transactions in 2017 (Bakare, 2021c). This, however, remained largely unenforced (Adesina, 2022). In another swift move by the CBN, after the initial order was dropped in 2021, an initiative was taken to protect the public and safeguard the country from potential threats posed by 'unknown and unregulated entities' that are 'well-suited for conducting many illegal activities' (Nwanisobi, 2021, p. 8). In this context, the CBN directed banks to stop using their platforms to transact or engage with entities that are involved in cryptocurrency activity (Bakare, 2021c; Uba, 2021). Additionally, they were asked to close accounts of individuals and institutions involved in cryptocurrency transactions (Nwanisobi, 2021). In April 2021, three banks were sanctioned with an 800 million Nigerian naira (NGN)[2] fine for failing to prevent customers from engaging in cryptocurrency transactions (Adesina, 2021). Since then, many Nigerians have reported that their bank accounts have been frozen due to cryptocurrency-related activity. Approximately the same time, the CBN launched a project to improve the efficiency of payment systems (Olowodun, 2021) by implementing a centrally issued and regulated central bank digital currency (CBDC) which is, at the time of writing, the only fully adopted CBDC on the African continent (Ozili, 2022). The resulting system, however, notably lacks decentralisation and decoupling from the fluctuation of the NGN and was therefore not widely recognised as a replacement for cryptocurrencies (Chukwuere, 2021). Therefore, and because it lacked other desirable characteristics such as interest-bearing capacity or feelessness (Ozili, 2023), ultimately, this CBDC achieved only 'disappointingly low' public adoption (Ree, 2023, p. 12). Despite this initiative and the legis-

2: approximately 2.1 million USD

Chukwuere, J. E. (2021). The eNaira – opportunities and challenges. *Journal of Emerging Technologies*, *1*(1)

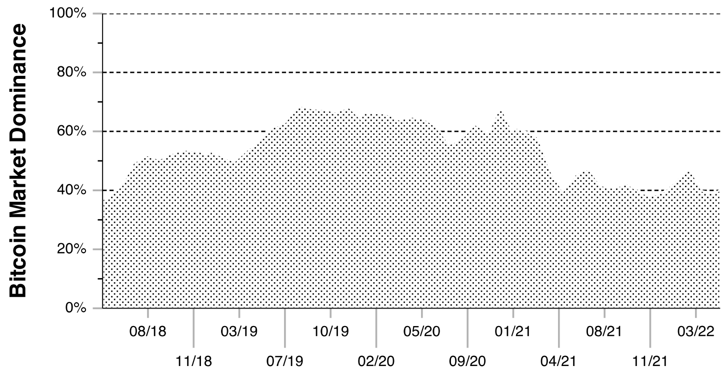

Figure 4.2: Bitcoin's market capitalisation relative to that of all other cryptocurrencies combined between 2018 and 2022 (de Best, 2022).

lative focus on cryptocurrencies, many citizens remain highly committed to them.

Regulating Cryptocurrency Activity

Bitcoin is the first and arguably one of the most relevant applications of blockchain technology. As such, it can be considered the archetype for cryptocurrencies (Ram, 2019), since it served as inspiration for most of the large number of alternative systems in the space (van der Merwe, 2021), including some that are directly derived from its core protocol (e.g., 'Litecoin'). After more than a decade, Bitcoin still accounts for around half of the cryptocurrency market capitalisation (Kulal, 2021), a condition that is known as 'Bitcoin dominance'. Figure 4.2 shows how this measure has fluctuated in recent years with low points below 40 % in 2018 or the second half of 2021 due to the rise of 'altcoins', alternative digital currencies, to peaks exceeding 70 % in times of uncertainty and market volatility.

Bitcoin's dominance may reflect the widespread and growing adoption that ultimately led it to be among the best-performing assets of the last decade, outperforming many stocks, bonds, commodities, and traditional currencies (Grabowski, 2019). There are now proposals at the institutional level to allow banks to keep 1 % of reserves in Bitcoin (Basel Committee on Banking Supervision, 2022). Thus, this chapter focuses on users of this archetypal cryptocurrency.

Grabowski, M. (2019, June). *Cryptocurrencies: A primer on digital money*. Routledge

Despite being remarkably successful, cryptocurrencies have encountered numerous setbacks: contentious issues span from facilitating money laundering (Fletcher *et al.*, 2021; Sicignano, 2021) to concerns regarding their impact on the environment (S. Jiang *et al.*, 2021; Wanat, 2021). While this has sparked regulatory interest, decentralised and transnational cryptocurrency systems challenge more traditional regulatory sandboxing (Ahern, 2021). Consequently, relatively little validation of the regulatory compliance of the processes and practices surrounding cryptocurrencies has been undertaken (Filippi *et al.*, 2022). This has contributed to regulatory gaps and, thus, to legal uncertainty (Ferreira & Sandner, 2021).

Ferreira, A., & Sandner, P. (2021). Eu search for regulatory answers to crypto assets and their place in the financial markets' infrastructure. *Computer Law & Security Review*, *43*, Article 105632

A plethora of different approaches, mostly founded in theory, have been observed in recent years. Some regulators allowed experimentation and showed tolerance; others (see Figure 4.3) opted for implicit or absolute bans (Hendrickson & Luther, 2017). Numerous topical academic works considered regulatory aspects (Silva & da Silva, 2022): regulatory meas-

Silva, E. C., & da Silva, M. M. (2022). Research contributions and challenges in DLT-based cryptocurrency regulation: A systematic mapping study. *Journal of Banking and Financial Technology*, *6*(1) ures in the past focused, for instance, on fiscal interventions addressing miners (S. Jiang *et al.*, 2021; Oğhan, 2022; United States Department of the Treasury, 2023), an approach with relevance beyond cryptocurrencies as evidenced by the increasing attention environmental taxation receives (Patel & Jhalani, 2023). Some proposed measures revolved around introducing sustainability criteria for institutional financial market actors (Gola & Sedlmeir, 2022), or on prohibitive regulations concerning miners (Mathews & Khan, 2019; Truby *et al.*, 2022). Furthermore, design-side policies, such as pushing for voluntary redesigns of PoW protocols, were proposed (Truby *et al.*, 2022). Cryptocurrency developer communities, however, apply decentralised governance and exhibit autonomous characteristics (Luther & Smith, 2020). Therefore, design-side policies are likely to suffer from a lack of enforceability. Consumer-focused policies are rarely suggested, and where they were, often made unrealistic assumptions, such as sovereign control over internet traffic (Fakunmoju *et al.*, 2022). Regulating cryptocurrencies remains challenging (Millard, 2018) with many open legal questions (Ferreira, 2020), and policymakers seem to consistently underestimate the technical complexity involved in efficiently targeting this novel phenomenon (Mezquita *et al.*, 2023).

Mezquita, Y., Pérez, D., González-Briones, A., & Prieto, J. (2023). Cryptocurrencies, survey on legal frameworks and regulation around the world. J. Prieto, F. L. B. Martínez, S. Ferretti, D. A. Guardeño & P. T. Nevado-Batalla (Eds.), *Proceedings of the 4th international congress on blockchain and applications*. Springer

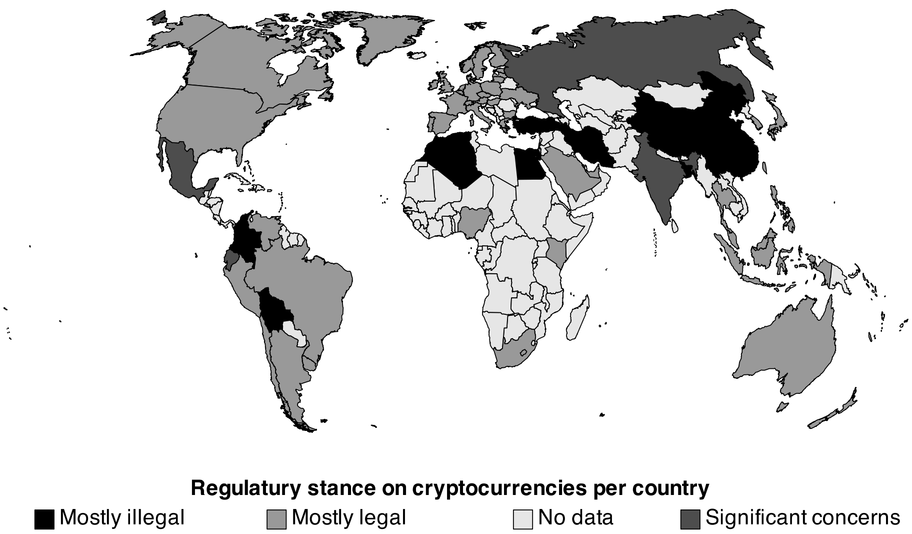

Figure 4.3: The visualisation of the global regulatory stance on cryptocurrencies adapted from the work of Hammond and Ehret (2022) shows that cryptocurrencies are considered 'mostly illegal' by the governments of Algeria, Bangladesh, Bolivia, China, Colombia, Egypt, Iran, Morocco, and Turkey.

The Chinese Example

A case study that illustrates these regulatory challenges is China: in 2017, Chinese authorities started to severely restrict the use of digital currency since the government was concerned they were facilitating capital outflows as well as money laundering and other fraudulent activities. This led to banning initial coin offerings – cryptocurrency-based avenues to raise funds via the issuance of digital tokens and without the participation of a trusted and regulated authority (Okorie & Lin, 2020). Second, China passed regulations to prohibit exchanges of BTC and Renminbi (RMB), effectively cutting the link between traditional financial intermediaries and cryptocurrency markets. At the same time, the People's Bank of China, the country's central bank, issued several warnings concerning Bitcoin and other cryptocurrencies, reminding the public that these do not enjoy

Okorie, D. I., & Lin, B. (2020). Did China's ICO ban alter the Bitcoin market? *International Review of Economics & Finance, 69*

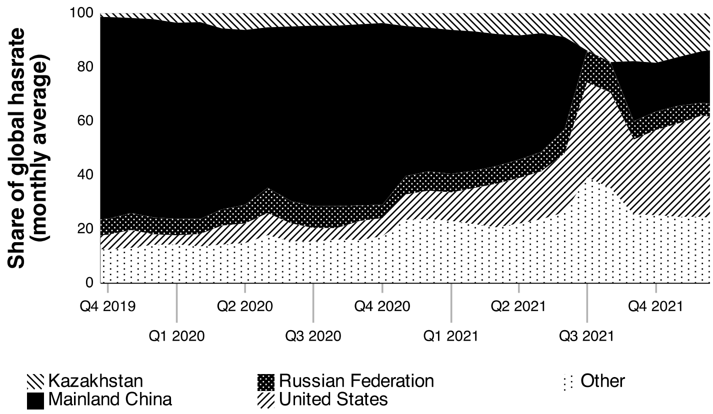

Figure 4.4: The evolution of mining activity between 2019 and 2022 (reproduced from CBECI data).

the same legal status as fiat currencies. This had a major impact on BTC-RMB trading volume and caused spillover effects to geographically close regions shortly after the introduction of more restrictive regulations, including an increase of over 25 % and 20 % in the trading volume of Bitcoin in Korean won and Japanese yen, respectively (Borri & Shakhnov, 2020). Chinese P2P exchanges, where buyers and sellers are matched directly, also registered ample trading increases as they provided a way to bypass regulation.

More recently, China has adopted an even more rejective stance on cryptocurrency-related activities, particularly Bitcoin mining. In 2019, the government termed Bitcoin mining ‘undesirable’ (National Development and Reform Commission, 2019, p. 120), a label used for industries that should be restricted or phased out by local governments. Finally, in 2021, Bitcoin mining was effectively forced to shut down due to environmental concerns brought forward by the government. As illustrated in Figure 4.4, the ban was initially effective: mining activity halted in the summer of 2021, with China’s hash rate, a measure of mining speed in PoW, going to zero. The two countries that benefited most from the ban in terms of share of the global hash rate were the United States and Kazakhstan.

However, given the significant mining capability built in previous years, which at the time accounted for more than 70 % of the global share, some Chinese miners were able to return to their activities despite bans, making China the country with the second-largest share of Bitcoin mining activity globally. Others moved hardware to less strictly regulated regions and continued activities there. Whether miners will continue to elude the ban in the future is unclear, but the events outlined suggest that even for countries with a strong grip on economic activities, such as China, enforcing outright bans is highly challenging. This perspective is consistent with work by C. Chen and Liu (2022) who investigated the impact of government Bitcoin trading interventions on the activities of Chinese investors. They found that, while Chinese participation in the Bitcoin market has decreased, local actors remain deeply involved. Ultimately, the Chinese example shows that it is not sufficient for regulators to ban and discredit cryptocurrencies to effectively prevent adoption (Feinstein & Werbach, 2021).

Feinstein, B. D., & Werbach, K. (2021). The impact of cryptocurrency regulation on trading markets. *Journal of Financial Regulation*, *7*(1)

4.2 Related Work

We consider the following types of studies as relevant related work: first, articles that target cryptocurrency users beyond Nigeria through survey research (see subsection 4), thus exposing patterns of user attitudes, behaviours, and experiences. Second, articles that investigate issues related to Bitcoin in the Nigerian context (see subsection 4) from legal, regulatory, or macroeconomic perspectives.

Search Strategy

We used the database *Web of Science*, which is widely acknowledged in the librarian and research communities for listing highly relevant peer-reviewed content (Mikki, 2009). Initially, we ran the search query[3] to retrieve works on cryptocurrency user attitudes (see subsection 4). Subsequently, we ran the query [4] to obtain relevant literature on cryptocurrencies in the Nigerian context (see subsection 4). Finally, we manually screened the abstracts of the identified manuscripts for relevance to our work and included those we deemed relevant in the corresponding subsections. We also added references to some works as a consequence of addressing reviewer comments.

3: `(cryptocurrency OR cryptocurrencies OR bitcoin) AND attitude`

4: `(cryptocurrency OR cryptocurrencies OR bitcoin) AND (nigeria)`

Cryptocurrency User Attitudes

Most of the relevant prior literature of which we are aware examines predictors of interest in using cryptocurrencies. A notable number of studies apply established psychological techniques, such as the theory of planned behaviour (TOPB) (Ajzen, 1991), one of the most widely used theories of behavioural prediction. Some studies investigate which predictors influence the willingness to adopt cryptocurrencies *in general* (Alaklabi & Kang, 2019; Albayati *et al.*, 2020; Anser *et al.*, 2020; Mazambani & Mutambara, 2019; Schaupp & Festa, 2018). Other studies identify predictors that influence users' decision to consider them as a *form of investment* in particular (Pham *et al.*, 2021; Smutny *et al.*, 2021). Research has been conducted on the circumstances under which users tend to support cryptocurrency as a *means of payment* by M. Kim (2021) and Salcedo and Gupta (2021). Due to the different foci of these surveys and the variety of methods used, the studies come to diverse, and, at times, contradictory conclusions.

Ajzen, I. (1991). The theory of planned behavior. *Organizational Behavior and Human Decision Processes, 50*(2)

For instance, Bashir *et al.* (2016) find that gender and social circle are decisive factors for Bitcoin ownership. Schaupp and Festa (2018), Mazambani and Mutambara (2019), and Pham *et al.* (2021) conclude that attitudes towards the behaviour of using cryptocurrencies are the determining construct in the context of TOPB. Steinmetz *et al.* (2021) conclude that German cryptocurrency users are predominantly young, male, well-educated, and affluent. Gagarina *et al.* (2019) confirm the common belief that a liberal world view correlates with the intention to use cryptocurrencies (Dodd, 2017). Seemingly in conflict with this are the findings of Albayati *et al.* (2020), whose results suggest that users are more interested in adopting cryptocurrencies when their activities are regulated and secured by the government. Alaklabi and Kang (2019) conclude that technological awareness has a positive influence on the intention to use cryptocurrencies. This is consistent with the finding of Smutny *et al.* (2021) that shows that a lack of information on the operating environment is a disincentive to cryptocurrency investment. Anser *et al.* (2020) show that a high level of activity in social media correlates with the willingness to use cryptocurrencies. The findings of M. Kim (2021) similarly present a picture of cryptocurrency users focused on social presence: the dimension 'power-prestige' was established as the most influential factor in the approval of Bitcoin. Salcedo and Gupta (2021) argue that cultural values and norms have a major impact on the willingness to use cryptocurrencies: collectivists, as well as representatives of long-term-oriented cultures, were found to be inclined towards blockchain technology.

Salcedo, E., & Gupta, M. (2021). The effects of individual-level espoused national cultural values on the willingness to use Bitcoin-like blockchain currencies. *International Journal of Information Management*, *60*, Article 102388

In summary, the existing literature portrays users of cryptocurrencies as maintaining interpersonal relationships with their peers, being technically savvy, well-informed, well-networked, and having libertarian world views. This user group is also strongly represented in our work. However, the attitude of this group towards sustainability has not played a significant role in the scientific discourse so far.

Cryptocurrency in Nigeria

Academic coverage of issues related to cryptocurrencies in the Nigerian context is sparse. Many previous works position Bitcoin as a technology discovered by Nigerians in the context of the 2016 recession as a stable alternative to the rapidly depreciating Naira (Nnabuife & Jarrar, 2018). To our knowledge, only two quantitative academic studies based on questionnaires have been conducted with a focus on Bitcoin in the Nigerian context: Eigbe (2018) investigates the level of awareness and adoption of Bitcoin in Nigeria, finding that most of the respondents lacked a proper understanding of the functionalities of Bitcoin, even if they claimed otherwise. A study by Salawu and Moloi (2018) targets Nigerian professional accountants: they were considering offering services in a cryptocurrency environment, although a majority indicated that the enactment of specific legislation would be a prerequisite for doing so.

Eigbe, O. E. (2018). Investigating the levels of awareness and adoption of digital currency in Nigeria: A case study of bitcoin. *The Information Technologist*, *15*(1)

The prevailing sentiment throughout the relevant works is that Bitcoin in the Nigerian context is not a passing fad but is of significant societal importance. This is reflected in a study by Jimoh and Benjamin (2020) that underlines the macroeconomic importance of Bitcoin by showing that the volatility of cryptocurrency returns has a measurable impact on the broader financial markets in Nigeria. Egbo and Ezeaku (2016) underscore the serious disruptive potential of cryptocurrencies by showing that these are threatening the very foundation of the business of commercial banks operating as intermediaries in Nigeria. While the previously outlined works highlight the potentially positive impact of cryptocurrencies on the Nigerian economy, other works focus on negative aspects, such as the risks of using cryptocurrencies for the financing of terrorism (Emmanuel & Michael, 2020), negative effects of cryptocurrencies on the exchange rate (Aberu *et al.*, 2023), or the inability of Nigerian legislation to effectively target cryptocurrency-related activities (Gidigbi *et al.*, 2021; Ukwueze, 2021).

4.3 Method

We designed our survey as an online questionnaire. To minimise the risk of data quality issues, a local research data collection provider was tasked with collecting data by individually approaching potential participants and ensuring that they were members of the target population. Owing to

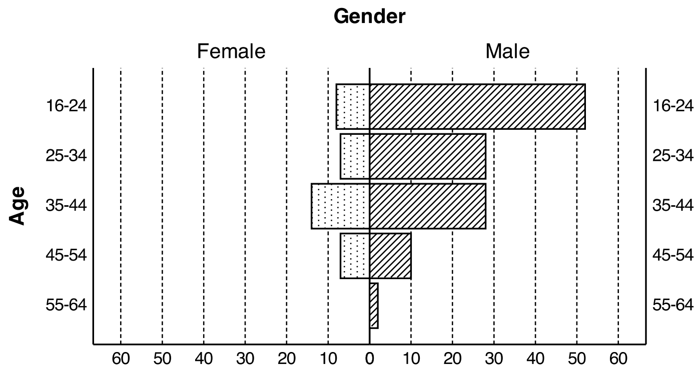

Figure 4.5: Absolute frequencies of distribution of valid responses for gender and age dimension.

the unclear regulatory situation in Nigeria and the fear of legal repercussions that may arise from it, recruiting participants proved challenging, but was ultimately successful as most participants felt reassured about their anonymity. This was helped by the fact that the study was led by a UK institution, as opposed to a local one.

Participants

158 valid responses were collected between the 25th of November 2021 and the 30th of March 2022 by convenience sampling. All participants were 16 years of age or older and resided in Nigeria. All participants reported having undertaken at least one Bitcoin transaction in the last five years at the time of this study.

Ethics approval was obtained before participant recruitment began. Participant recruitment had two avenues. First, Covenant University students from Ota, Nigeria who had a verified interest and background in cryptocurrency, as evidenced by extracurricular activities, were approached and offered opportunities to participate voluntarily. Second, the study was advertised in Nigerian cryptocurrency groups on the Telegram messenger, the 'de facto messaging platform for the cryptocurrency community' (Smuts, 2019, p. 131). For both approaches, the participants were self-selected and did not receive compensation. The sample obtained is biased towards male participants, with 76.9 % of the respondents identifying as male (see Figure 4.5). This imbalance may be traced back to the convenience sampling method in conjunction with a more pronounced interest in cryptocurrencies as investment instruments among younger men (Senkardes & Akadur, 2021; Steinmetz, 2023).

Materials

The questionnaire contained a total of 107 items on eight pages, some of which were conditional. It employed screening questions throughout the survey to ensure that only members of the target population participated.

The questionnaire was designed to measure the degree of expertise in cryptocurrency technology participants possess. It was furthermore designed to measure how accurately participants estimate Bitcoin's electricity consumption. Finally, it measured the degree to which participants believe that Bitcoin's electricity consumption poses an environmental

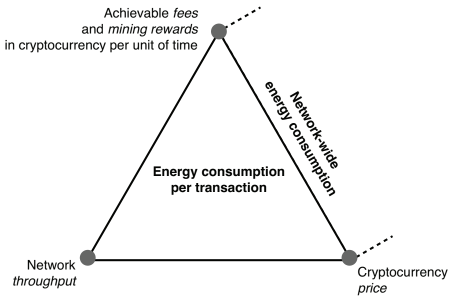

Figure 4.6: Comparison of network-wide electricity consumption models and transaction-based consumption models.

issue, whether measures should be taken, and which stakeholders, if any, they consider responsible for acting against it.

Network-Wide Electricity Consumption as Anchor Point

As described in subsection 4.1 and visualised in Figure 4.6, typically PoW electricity consumption is quantified either on a system-wide basis (taking into account transaction fees, block rewards available to miners, and the price of the cryptocurrency) or per transaction by additionally considering network throughput.

Consequently, when preparing the materials for this survey, the choice arose whether users should be questioned about their assessment of electricity consumption per transaction or about network-wide electricity consumption. We decided on the latter, considering that an increase in the transaction throughput of Bitcoin does not cause a substantial increase in its total electricity consumption. As such, a user who decides to engage in a Bitcoin transaction will not directly contribute to increasing the electricity consumption of the system but only via secondary effects, such as increases in the transaction fee levels and cryptocurrency prices, owing to increased popularity. Yet, in our experience, most users (and even researchers such as Mora *et al.*, 2018) are not aware of this nuance and assume that additional transactions will proportionately increase Bitcoin's total electricity consumption. Therefore, it seems more appropriate to survey users about network-wide electricity consumption. In *Q21*, participants were, therefore, presented with the following question to which six potential answers were provided (see Table 4.1):

Mora, C., Rollins, R. L., Taladay, K., Kantar, M. B., Chock, M. K., Shimada, M., & Franklin, E. C. (2018). Bitcoin emissions alone could push global warming above 2°C. *Nature Climate Change*, *8*(11)

> What do you estimate the electricity requirements of operating the entire Bitcoin network to be?

We assume a value of 121.46 TW h to be close to the actual annual electricity consumption of Bitcoin at the time of conducting the survey. This value is the median of daily estimates of annualised consumption[5] during the data collection period. Estimates fluctuated between 108.08 and 140.11 TW h during data collection. This shows that, during the survey period, option four ('about four times the overall electricity consumption of Nigeria') was the most accurate estimate on all days.

5: See https://ccaf.io/cbeci/index.

We offer a wide range of potential answers that correspond to electricity consumption figures between 600 MW h and 2,845 TW h. The values chosen as potential answers were deliberately extreme to avoid ambiguity. Since participants are unlikely to have a reference point for physical units of measurement, the electricity values were not exposed in the

Question Wording	Annual consumption (TW h)
Similar to the overall electricity consumption of a small town in Nigeria	0.0006
Similar to the overall electricity consumption of the Lagos Metropolitan Area	5.8
Similar to the overall electricity consumption of Nigeria	29
Actual—Total Bitcoin electricity consumption (CBECI)	*121.46*
About four times the overall electricity consumption of Nigeria	116
Similar to the overall electricity consumption of the entirety of the African continent	700
Similar to the overall electricity consumption of the entirety of the African and European continents combined	2,845

Table 4.1: We asked questions to gauge how realistically participants estimate the annual electricity consumption of Bitcoin. We wrote the questions so that the participants could relate electricity consumption to the realities of their lives. The median of estimates obtained from the CBECI is printed as *actual* value for the benefit of the reader only and was not shown to the participants.

questionnaire. Instead, examples that are relatable to the participants' living situation (see Table 4.1) were used.

Experience Assessment and Opinions

Some sections of the questionnaire reuse parts of existing surveys. This is also the case in *Q19*, in which we ask participants for their reasons for acquiring cryptocurrency:

> Why have you acquired Bitcoin in the past?

This question, along with others in the 'cryptocurrency experience' section of the survey, reproduced questions from the Organisation for Economic Co-operation and Development (OECD) consumer insights survey on crypto assets, a questionnaire that 'has been designed to survey consumers/retail investors in order to collect data on their attitudes, behaviours and experiences towards digital financial assets, specifically digital (or crypto) currencies and initial coin offerings' (OECD, 2019, p. 2).

Subsequent parts of the questionnaire assess the degree of concern participants have regarding some effects of climate change. For example, *Q20* asks about participants' areas of concern in the context of climate change:

> How concerned are you about the potential consequences of climate change to your living environment?

The options presented were taken from previous work by Haider (2019), who summarises the likely impacts of climate change in Nigeria based on previous studies. By using this previous work, the options presented to participants were tailored to the effects that were most likely to affect them.

To ensure the quality of measurements, the local research data collection provider conducted a pilot study with 12 participants, assessing the understandability of research materials with members of the target population prior to the commencement of data collection. Some changes to the survey materials were implemented according to the findings of the pilot survey and thereby improved comprehension of the materials in the target population.

Procedure

Of 1,088 participants that started the survey, 158 completed it. Participants completed the questionnaire within 19 min 47 s on average. Deception was not used during survey recruitment. Participants were told that the survey was designed to understand their attitudes towards cryptocurrency use and environmental issues. They were then informed about the fact that their participation is completely voluntary and that they should only take part if they want to. Furthermore, they were educated about the fact that choosing not to participate would not disadvantage them in any way. The research data collection provider then made them aware that they would be provided with an information sheet for participants prior to answering any questions. Those persons that expressed an interest in participating after this introduction by the research data collection provider were given a survey link, either in the form of a printout, via e-mail, or via Telegram message. The research data collection provider had no knowledge of whether the potential participants indeed followed the link.

Some participants raised a serious concern that their identity could be revealed to Nigerian authorities. This concern stemmed from the fear of facing legal repercussions by the Nigerian government which has taken a rejective stance towards cryptocurrencies (see subsection 1). The research data collection provider was able to alleviate some concerns by pointing to the applicability of the United Kingdom General Data Protection Regulation, however, some potential participants were not convinced by this argument and remained disinterested in participation.

Once participants followed the link provided, informed consent was obtained using the online survey system through a series of approved questions. Participants were informed that the data would be converted to an anonymised format and that the data collected might be subject to publication. After completing the survey, participants received a written debrief through the online survey tool, were thanked for their participation, and were dismissed.

Data Analysis

To test the hypotheses (see subsection 4), we applied statistical methods to the collected survey data. To begin, descriptive statistics were employed to illustrate both the user profile of participants and their attitudes toward Bitcoin. No gender-based analysis was conducted on the data.

Next, we examined three crucial variables: awareness, actionability, and responsibility. To assess the correlation between actionability and awareness, considering their nominal nature, we conducted a Chi-Square test.

Responsibility for addressing Bitcoin's electricity consumption was assessed using a 5-point Likert scale that measured participants' attitudes towards the responsibility of six different actors: Bitcoin miners, Bitcoin users, Bitcoin developers, intergovernmental organisations, the legislature, and federal agencies/regulators. We divided the six different actors into two distinct groups: non-government actors and government actors. Based on this assessment, we computed the average Likert scores by analysing participants' responses on the 5-point Likert scale.

4.4 Results

The results generated from the questionnaire provide insights into the environmental attitudes of the surveyed Bitcoin users and provide information on the key hypotheses in the areas of awareness, actionability, and responsibility.

User Profiles

When analysing the user profile of participants (see Figure 4.7), we found no relationship between transaction volume and familiarity with Bitcoin. We, however, observed some outliers that reported large numbers of Bitcoin transactions and self-reported being extremely familiar with this cryptocurrency. We found that the median number of transactions conducted in the last five years for all levels of experience was lower than 10. When analysing how participants obtained Bitcoin (see Figure 4.8), we found that online platforms were by far the most popular method, with 79.1 % of participants having used them to acquire Bitcoin in the past. Few participants (2.5 %) used dedicated kiosks (i.e., machines resembling cash machines) to acquire Bitcoin. The main motivation to acquire Bitcoin (reported by 40.5 % of the participants) was as a long-term investment or retirement fund. Only 3.8 % mentioned avoiding government regulation as a reason for obtaining Bitcoin[6].

6: This is likely under-reported because of a 'chilling effect': a condition in which prospective participants refrain from behaviour that deviates from the perceived rules, norms and guidelines of a powerful supervisor for fear of negative consequences (Schüll, 2018): in this case, Nigerian authorities.

In addition, we analyse the concerns that participants reported about the possible effects of climate change on their environment. Here, we found great consternation among participants with the mode of responses being 'extremely concerned' for all the effects provided. Participants expressed significant concerns about freshwater resources, rising temperatures, and extreme weather events, while they were less troubled by variable rainfall. 56.3 % of participants believed that Bitcoin's electricity consumption contributes significantly to global CO_2 emissions, with almost all of these (93.3 %) also believing that the CO_2 emissions caused by Bitcoin contribute to climate change. 65.2 % of overall participants felt that measures to reduce the CO_2 footprint of Bitcoin should be taken now. A minority of 42.7 % of the participants who answered the relevant question supported the view that Bitcoin users should move away from Bitcoin to other cryptocurrencies in the interest of reducing CO_2 emissions. Some of these participants provided the names of alternative blockchain-based cryptocurrencies (e.g. Dogecoin and Ethereum). Others suggested alternative payment infrastructure tokens such as Ripple's XRP.

Awareness

One of the key purposes of this questionnaire was to assess how realistic the estimates of the total Bitcoin electricity consumption made by the participants were. Here, we found that most participants (68.4 %) significantly underestimated the energy demand of Bitcoin, while only a

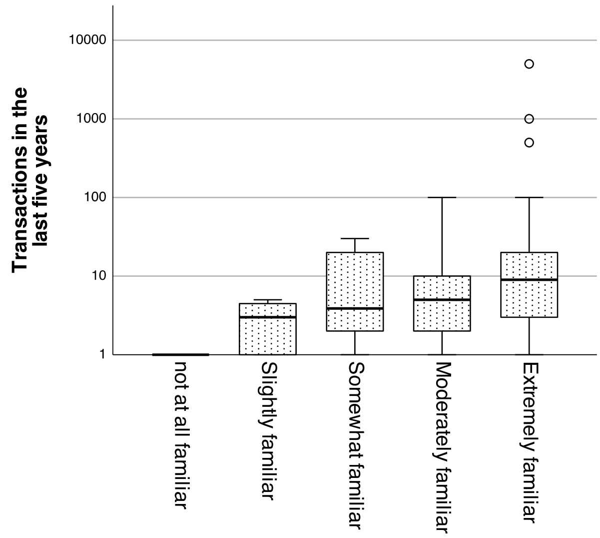

Figure 4.7: Respondents that report extreme familiarity with Bitcoin on average report the highest number of transactions (logarithmic scale). This group also includes outliers that report participation in very large numbers of transactions.

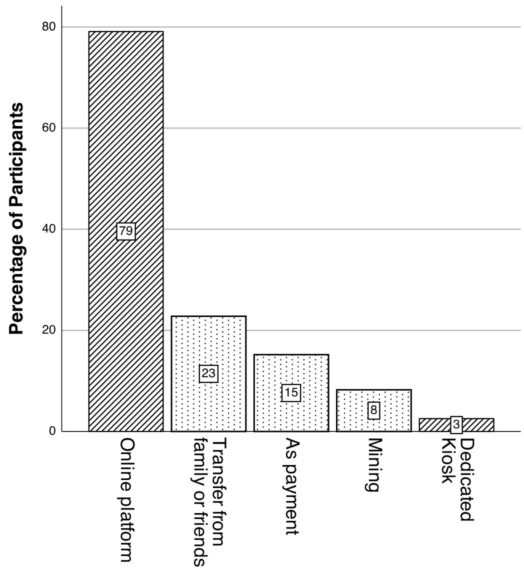

Figure 4.8: Participants have acquired Bitcoin through different channels. Online platforms and dedicated kiosks lend themselves well to digital energy labelling (see section 4.5) while others do not.

minority (16.5 %) overestimated it (see Figure 4.9). This goes beyond what is expected under random conditions: 50 % of participants randomly selecting would underestimate electricity consumption, approximately 16.7 % would accurately assess it, and approximately 33.3 % would overestimate it.

Actionability

We provided a variety of reference points to participants (see subsection 4.3) to assess actionability. Based on the participants' estimates of the overall Bitcoin electricity consumption, we separated the participants into two groups: those who estimated energy demand correctly and those who did not. Thus, being supportive of measures and estimating the electricity consumption of Bitcoin correctly both constitute dichotomous variables.

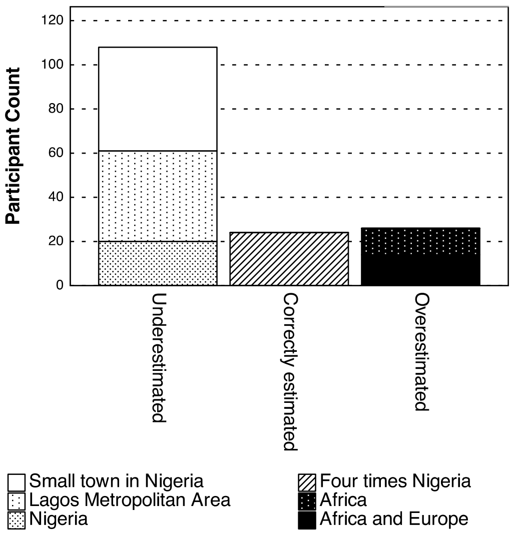

Figure 4.9: Most participants underestimate the overall energy demand of Bitcoin.

Subsequently, we conducted a Chi-Square test to analyse the relationship between supporting measures and estimating the electricity consumption of Bitcoin correctly: we found a medium correlation between these two variables ($\chi^2(1, N = 158) = 4.105$, $p = 0.043 < 0.05$, $\phi_c = 0.16$, see Table 4.2). Specifically, a *post-hoc* comparison test with correction showed that under $\alpha = 0.05$, the proportion of participants supporting measures in the correct estimates group (83.3 %) is higher than in the incorrect estimates group (61.9 %). Consequently, the proportion of the non-supporting measures group in the correct estimates group (16.7 %) is lower than that of the incorrect estimates group (38.1 %).

Supportive	Estimates		Total	χ^2
	Correct	Incorrect		
Yes	83	20	103	4.105*
	61.9 %	83.3 %	65.2 %	
No	51	4	55	
	38.1 %	16.7 %	34.8 %	
Total	134	23	158	

* Correlation is significant at the 0.05 level.

Table 4.2: We found a medium correlation between estimating the electricity consumption correctly and being supportive of measures.

Responsibility

Where participants did see the need for action, they felt that Bitcoin miners, Bitcoin users, and Bitcoin developers should be taking action instead of intergovernmental organisations, the legislature, and federal agencies or regulators (see Figure 4.10).

To further examine participants' notion of the responsible actors, we rendered paired sample t-tests to compare the mean Likert scores in support of non-government actors (averaged by Bitcoin miners, Bitcoin users, and Bitcoin developers) and government actors (averaged by intergovernmental organisations, the legislature, and federal agencies or regulators). We observe a significant difference in mean Likert scores in support of non-government actors and government actors ($t = 2.943$,

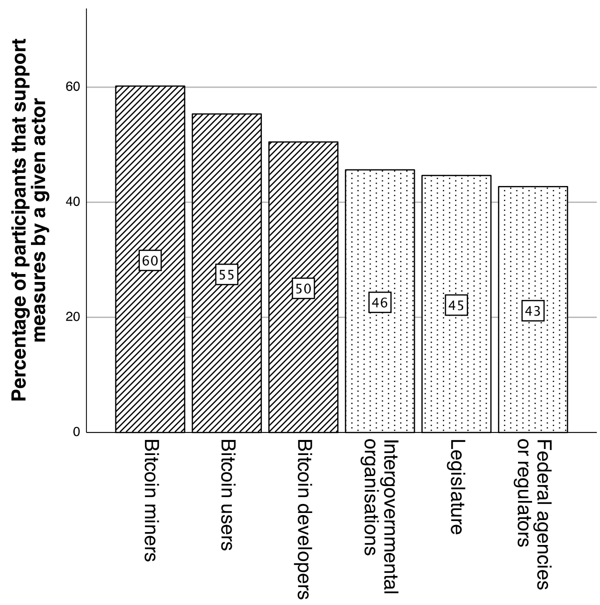

Figure 4.10: 60 % of those participants that support measures in principle find that Bitcoin miners should act, while only 43 % find that federal agencies or regulators should. A divide between non-governmental actors (top 3) and governmental actors (bottom 3) is noticeable.

$p = 0.004$, see Table 4.3). On average, participants expressed a stronger expectation of responsibility towards non-government actors ($M = 3.490$) compared to government actors ($M = 3.260$).

Table 4.3: On average, participants hold private sector actors more accountable.

Group	μ	σ	t
Non-government actors	3.49	1.02	2.943**
Government actors	3.26	1.15	

μ: Mean, σ: Standard deviation; ** Significant at the 0.01 level.

Some participants named alternative actors they felt were responsible: these included Bitcoin exchanges, wealthy individuals, and activists. Where participants did not feel that action to reduce the CO_2 footprint of Bitcoin should be taken now, they predominantly articulated two reasons for this perspective (both with 30.1 %): they brought forward the view that future technological improvements would reduce Bitcoin's electricity demand and/or that the environmental impact of Bitcoin is acceptable for the benefit it provides. No meaningful alternative reasons were provided in the free-text fields.

4.5 Discussion

Our results are largely corroborated by previous research. Specifically, they support the results of Eigbe (2018) who previously pointed out gaps in the technical expertise of Nigerian Bitcoin users, as well as the findings of Steinmetz *et al.* (2021) and Duggan (2022) which yield similar conclusions, although outside of Nigeria. The results furthermore align with consumer knowledge assessments in the broader financial products space that showed that consumers often had little knowledge of the key properties of the products they were using (Ramchander, 2016; Sukumaran *et al.*, 2022). Although previous research has focused on technical or financial dimensions of user attitudes alone (see section 4.2), the results of this chapter demonstrate that, throughout the user base, the concern over the effects of climate change is significant. These results should be taken into account when designing policies to respond to the high electricity consumption of cryptocurrencies.

To develop effective strategies to reduce the popularity of PoW cryptocurrencies, and therefore, ultimately, their electricity demand, decision-makers must first realise that such strategies cannot be targeted at miners alone. While miners are, in fact, almost solely responsible for the energy footprint of cryptocurrencies (see subsection 4.1), they can quickly relocate their activities to other regions where there are fewer legal restrictions (see subsubsection 2). Relocating allows them to evade regulatory access without affecting the end users of the respective cryptocurrency since those are oblivious to where mining hardware is operated. A more effective strategy, instead, focuses on the end users of cryptocurrencies by empowering them to make more sustainable choices. This increase in transparency is a potential enabler for a consumer movement away from unsustainable cryptocurrencies. Such a consumer movement may result in a systematic reduction of the carbon footprint of unsustainable cryptocurrencies, beyond the individual user, should the expected price effects described earlier (see subsection 4.1) materialise.

The finding that users who correctly assess the sustainability parameters of cryptocurrencies tend to show more support for measures indicates that consumer education is a promising tool for policymakers. Care must, however, be taken that cryptocurrencies are not portrayed in an all-encompassing and overly negative way: after all, our results do neither support nor rule out a correlation between *overestimating* electricity consumption and supporting measures. Rather, policymakers should initiate measures that achieve basic consumer education and provide users of cryptocurrencies with a realistic view of their electricity consumption and economic parameters.

Energy labelling, i.e. providing key sustainability metrics to cryptocurrency users at the point of exchange, is one potentially suitable measure to achieve customer education. Such labels would allow users to compare the electricity consumption characteristics in this vast market, thereby allowing them to take sustainability into consideration when making cryptocurrency purchasing decisions. The concept of energy labelling aligns with the broader discussion of the importance of transparency and adequate disclosure in the blockchain and cryptocurrency industry (Liebau, 2021). While little is known about the effectiveness of this intervention in the context of cryptocurrencies, the assessment of a protocol's consensus algorithm has previously been contemplated as a key environmental metric (Liebau & Krapels, 2021) and early research into measures to reduce the carbon impact of digital behaviours has produced promising results (Seger *et al.*, 2023). Furthermore, results from the field of household appliances, where energy labels are common, give cause for optimism: here it was found that customers are aware of the information on labels (Waechter *et al.*, 2015) and comprehend it (Jeong & Kim, 2014), albeit being confused by changes in labelling schemes (Stasiuk & Maison, 2022). Ultimately, consumers were found to make better decisions when guided by labels (Davis & Metcalf, 2016). Furthermore, consumers were found to attach a value to energy efficiency beyond the prospect of reducing costs (Andor *et al.*, 2020). Even though sustainability awareness may differ between countries (Schallehn & Valogianni, 2022), it seems conceivable that energy labelling initiatives present an effective long-term energy efficiency policy for cryptocurrencies that may promote green innovation (Z. Li *et al.*, 2022).

Seger, B. T., Burkhardt, J., Straub, F., Scherz, S., & Nieding, G. (2023). Reducing the individual carbon impact of video streaming: A seven-week intervention using information, goal setting, and feedback. *Journal of Consumer Policy, 46*

4.6 Conclusions

The data obtained suggest that most Bitcoin users underestimate its electricity consumption. Our work also demonstrates a correlation between participants' ability to estimate the electricity consumption of Bitcoin correctly, and their support of measures to counteract Bitcoin's CO_2 footprint. Furthermore, we find that users predominantly hold private actors (e.g. Bitcoin miners, users, and developers) responsible for addressing Bitcoin's energy demand. Subsequently, the empirical results lend support to all three hypotheses posited.

Taking into account the current trajectory of CO_2 emissions, regulators face unprecedented pressure to introduce policies to avoid a climate catastrophe. Cryptocurrencies based on PoW consume large amounts of electricity, while, arguably, providing very similar benefits to those built on alternative consensus mechanisms that are orders of magnitude less energy-demanding. The counteracting of the enormous electricity consumption of PoW-based cryptocurrencies must therefore be urgently attended to by policymakers, not least since cryptocurrencies are now ubiquitous and no longer exclusive to users with specific demographic or regional characteristics. Improving customer knowledge about cryptocurrency sustainability could lead to more sustainable consumer behaviour.

Therefore, in this work, we recommend a specific course of action to promote customer knowledge: confronting users with the consequences of their cryptocurrency choices through energy labelling. Although this proposal has not yet been tested, the key results of this work suggest that it may improve sustainability.

4.7 Limitations

It is important to note that this work is based on a small sample with a narrow scope, since it focused solely on one asset and country and was obtained by convenience sampling. This sampling method may introduce biases from self-selection, inadvertent selection of specific groups, and recruitment channel preferences. Furthermore, the reliability of the data we collected is impacted by the challenging legal situation in Nigeria that may prevent cryptocurrency users from publicly acknowledging their activities. These factors warrant caution when generalising our findings to larger populations.

Improving the Energy Footprint of Blockchains 5

This chapter follows the previous survey (see chapter 4), which established that individuals are unaware of some core technological properties of the systems they use, specifically their energy consumption. Therefore, in this chapter, we investigate whether technology choices can be influenced by making individuals aware of these core technological properties. This is a crucial prerequisite to understanding whether users might change their opinions on which type of system to use once they are made aware of the underlying benefits. This insight is essential for establishing a novel democratic consensus mechanism that is acceptable by users of other systems.

Cryptocurrencies are positioned as technologies to transact without state intervention (Jarvis, 2021) and out of reach of sovereign monetary policy (Butler, 2021). They are adopted as speculative forms of investment (Ante, 2022; Glaser *et al.*, 2014) and as means of payment (Radic *et al.*, 2022) throughout the world (F. Fang *et al.*, 2022). While Bitcoin remains dominant (Kulal, 2021), numerous alternative products or 'altcoins' have emerged (Gatabazi *et al.*, 2022), turning the cryptocurrency landscape into a competitive market (Fernandez-Vazquez & de la Fuente, 2022). Cryptocurrencies enable users to conduct payments without central oversight and without the need for intermediaries (G. P. Kumar, 2022). From an end-user perspective, cryptocurrencies provide such benefits as decentralization, security, pseudonymity, convenience, and low transaction fees (Alqaryouti *et al.*, 2020). However, despite investors becoming more sensitive to considerations of environmental, social, and corporate governance (ESG) (Steblianskaia *et al.*, 2023), the cryptocurrency landscape remains exposed to strong criticism for its excessive electricity consumption (Agur *et al.*, 2022; de Vries, 2022; George, 2021; Rieger *et al.*, 2022; Schinckus, 2021; Sedlmeir *et al.*, 2020a, 2020b; Truby *et al.*, 2022). There is a wide range of consensus mechanisms (see chapter 2) that result in blockchain systems with an equally wide range of energy demands. Some of these systems are particularly excessive in terms of electricity consumption, while others have only modest requirements, as shown in subsection 5.

Jarvis, C. (2021). Cypherpunk ideology: Objectives, profiles, and influences (1992–1998). *Internet Histories*, *6*(3)

Steblianskaia, E., Vasiev, M., Denisov, A., Bocharnikov, V., Steblyanskaya, A., & Wang, Q. (2023). Environmental-social-governance concept bibliometric analysis and systematic literature review: Do investors becoming more environmentally conscious? *Environmental and Sustainability Indicators*, *17*, Article 100218

In chapter 4 we have speculated that users' lack of awareness of electricity consumption characteristics contributes to the popularity of energy-inefficient cryptocurrencies, and that raising consumers' awareness could motivate them to adopt more sustainable cryptocurrencies. Comparative energy labels that show vital information on electricity consumption have been widely adopted for consumer education (Andor *et al.*, 2020). To understand the educational effect of comparative energy labels on users' cryptocurrency consumption intentions, we conduct a control/treatment test with the following hypothesis.

Andor, M. A., Gerster, A., & Sommer, S. (2020). Consumer inattention, heuristic thinking and the role of energy labels. *The Energy Journal*, *41*(1)

Hypothesis Cryptocurrency users are less likely to acquire energy-inefficient cryptocurrencies when presented with energy labels during acquisition.

Table 5.1: Categories of cryptocurrency electricity consumption according to Agur *et al.* (2022).

Type	Permissioning	Consensus	Avg. Electricity Demand (kW h/tx)
I	Permissioned	Non-PoW	0.00000145
II	Permissionless	Non-PoW	0.00202
III	Permissionless	PoW	273

Background Literature

Three lines of background literature are relevant to this chapter: (1) cryptocurrency technology, especially those concerning the interplay between technology choices and electricity consumption; (2) human interaction with cryptocurrencies, especially technology preferences, attitudes, and expertise; (3) energy labelling schemes in other contexts.

Cryptocurrency Technology

In addition to chapter 3, an extensive body of work has investigated the electricity consumption characteristics of cryptocurrencies (Agur *et al.*, 2022; Gola & Sedlmeir, 2022), showing that these differ fundamentally (see Table 5.1). Early permissionless cryptocurrencies, building on PoW consensus (type *III* in Table 5.1), have been criticised for their extreme electricity consumption (Agur *et al.*, 2022; de Vries, 2022; George, 2021; Rieger *et al.*, 2022; Schinckus, 2021; Sedlmeir *et al.*, 2020a, 2020b; Truby *et al.*, 2022). Bitcoin, the most popular PoW cryptocurrency, was found to produce up to 65.4 Mt CO_2 annually, as of 2022, which equated to the overall emissions of Greece (de Vries, 2022). Despite this enormous electricity consumption, the Bitcoin network only delivers 5 to 7 tx/s (L. Zhang *et al.*, 2022). Certain subsequent permissionless cryptocurrencies, such as Cardano, Solana, and Ethereum apply alternative consensus mechanisms (type *II* in Table 5.1) and have been found to be more energy-efficient in terms of orders of magnitude (see chapter 3 and George, 2021). Permissioned systems (type *I* in Table 5.1) like Hedera Hashgraph have undercut the electricity consumption of type *II* systems even further. This may be attributed to their smaller replication factor (see chapter 3).

de Vries, A. (2022). Cryptocurrencies on the road to sustainability: Ethereum paving the way for Bitcoin. *Patterns*, *4*, Article 100633

Cryptocurrency Users

Studies from different regions support the notion that users of cryptocurrencies have a limited understanding of the underlying technology (Eigbe, 2018; Steinmetz *et al.*, 2021). This limitation extends to their knowledge of electricity consumption (see chapter 4). Some previous work investigated the pathways of acquisition of cryptocurrencies. It was found that the prevalent avenue for users to acquire cryptocurrencies is via CEXs (Z. Zhou & Shen, 2022) such as Binance and Coinbase, which are already targeted by regulators (Brasse & Hyun, 2023). This has motivated us to investigate energy labelling on CEXs.

Brasse, A., & Hyun, S. (2023, January). Cryptocurrency exchanges and the future of cryptoassets. H. K. Baker, H. Benedetti, E. Nikbakht & S. S. Smith (Eds.), *The Emerald handbook on cryptoassets: Investment opportunities and challenges*. Emerald

Energy Labelling

Methods to increase consumer knowledge have a long history. A tried and tested method of consumer education, which has commonly been applied to durable goods, such as appliances (Harrington, 1999), is energy

Harrington, L. (1999). Appliance energy labels from around the world. P. Bertoldi, A. Ricci & B. H. Wajer (Eds.), *Proceedings of the 1st international conference on energy efficiency in household appliances*. Springer labelling. Energy labels were found to be effective tools in helping consumers make more sustainable purchasing decisions (Davis & Metcalf, 2016) and adopt more sustainable behaviours (Seger *et al.*, 2023). They are recognised throughout the European Union (EU), including Romania, where the survey is conducted. Energy labels are recognised by 86 % of Romanian consumers (European Comission, 2019, p. T23) and 74 % of them refer to energy labels when buying energy-efficient products (European Comission, 2019, p. T25). A seven-point (A-G) scale, such as that applied in the EU since March 2021 (Boogen, 2022), was found to result in high perceived importance of energy efficiency (Heinzle & Wüstenhagen, 2011).

5.1 Method

A control/treatment test with 1/2 users in each group was conducted. Participants took part in the test via an online survey.

This method is an application of participatory design, a technique to 'actively [involve] users throughout the design process' (Mackay & Beaudouin-Lafon, 2023) that encompasses different methods that can be applied throughout the stages of social systems design (M. J. Muller & Kuhn, 1993, p. 26). Specifically, the way in which user interfaces are presented to users as mock-ups with the goal of measuring user reactions is a typical example of participatory design.

Participants

The survey targeted Romanian internet users, 18 years of age and older, with an interest in holding cryptocurrencies.

Survey Design

To ensure a relevant group of participants, a screening question was applied. Participants were asked to indicate all financial products they considered holding in future: stocks, cryptocurrencies, indices, exchange-traded funds, commodities, and/or foreign currencies. Only participants who selected cryptocurrencies were considered. These participants were presented with the questions described in the following sections.

Section on Cryptocurrency Usage

The first survey section borrowed key elements from the 'Consumer Insights Survey on Cryptoassets' (OECD, 2019) of the OECD. Specifically, participants were asked to declare how well they understood cryptocurrencies and whether they had experience of holding them.

Section on Environmental Attitude

A question regarding the participants' attitude towards environmental regulation was adopted from the sustainability consciousness questionnaire of Gericke *et al.* (2018) to allow for the categorization of participants in terms of their sustainability attitudes.

Gericke, N., Pauw, J. B.-d., Berglund, T., & Olsson, D. (2018). The sustainability consciousness questionnaire: The theoretical development and empirical validation of an evaluation instrument for stakeholders working with sustainable development. *Sustainable Development*, *27*(1)

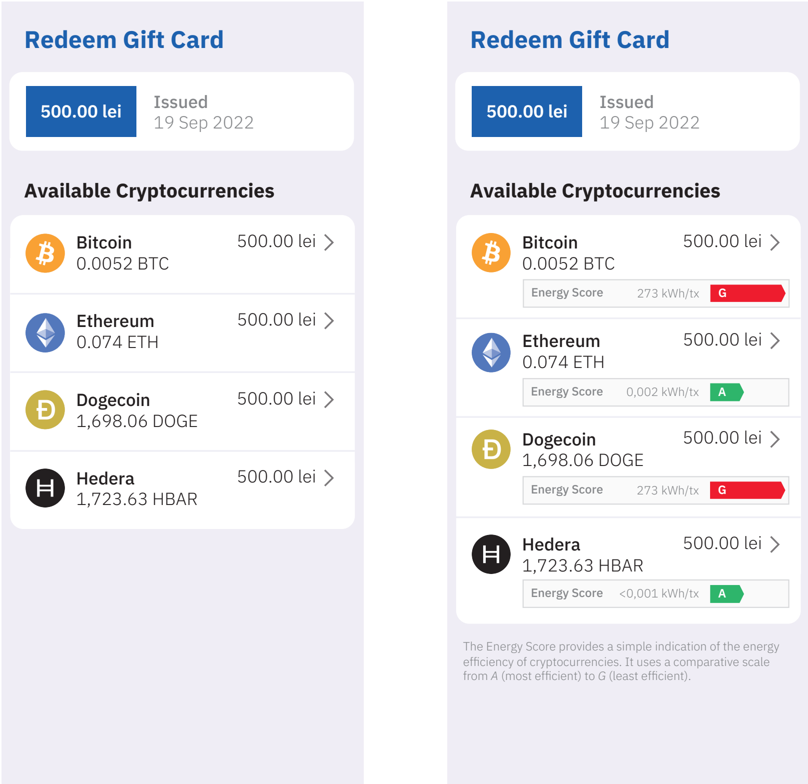

Figure 5.1: The UIs shown to the control group (left) and treatment group (right).

Acquisition Scenario

Finally, subjects were presented with the control/treatment test: this revolved around an imaginary scenario in which they had received a gift card[1] that could only be redeemed at a fictitious CEX. Subjects in both groups were shown a user interface (UI) that mimicked standard CEXs (see Figure 5.1 for translations of the UIs that were originally shown to the participants in the Romanian language). Both UIs offered a choice between four different cryptocurrencies. These represented a selection of highly capitalised cryptocurrencies, covering the entire spectrum of electricity consumption characteristics shown in Table 5.1: Bitcoin, Ethereum, Dogecoin, and Hedera Hashgraph[2]. However, there were slight variations in the quantity of information that was displayed in the respective UIs: the UI presented to the *control group* (see Figure 5.1) showed the name of the cryptocurrency (e.g., Bitcoin), its ticker symbol (e.g., BTC), and how many currency units were redeemable for the gift card balance (e.g., 0.0052). In addition, the UI presented to the *treatment group* (see Figure 5.1) showed an 'Energy Score', a fictitious energy label intended to express how much electricity a cryptocurrency transaction on a given blockchain consumes. Most participants were likely to be familiar with energy labels, due to their popularity in the EU. Nevertheless, a brief description of the concept was provided to this group in the UI (see Figure 5.1).

1: The gift card amount was set at 500 Romanian lei, an amount roughly equivalent to 100 US Dollars at the time of conducting the survey.

2: We use the name of the Hedera Hashgraph distributed ledger technology synonymously with the HBAR cryptocurrency throughout this chapter.

To determine whether energy labels have a significant effect on the acquisition decision, cryptocurrencies of vastly different sustainability characteristics were selected. We show energy labels which follow the 2021 EU energy label scale that allows for values ranging from *A* (most energy-efficient) to *G* (least energy-efficient). We consider type *I* and type *II* systems to be energy-efficient in the sense of the hypothesis and consider type *III* systems to be energy-inefficient. To provide polarizing choices, we presented two energy-efficient cryptocurrencies (Ethereum

Table 5.2: Key dimensions of the cryptocurrencies used in the control/treatment test.

Name ^{c,t}	Symbol ^{c,t}	Type	Units acquirable ^{c,t}	Consumption ^{t} (kW h/tx)	Label ^{t}
Bitcoin	**BTC**	III	0.0052	273	G
Ethereum	**ETH**	II	0.074	0.002	A
Dogecoin	**DOGE**	III	1,698.06	273	G
Hedera Hashgraph	**HBAR**	I	1,723.63	<0.001	A

The dimensions marked with ^{c} are displayed to subjects in the control group.
The dimensions marked with ^{t} are displayed to subjects in the treatment group.
Electricity consumption data were estimated by calculating the mean of values aggregated by Agur *et al.* (2022).
Price as per https://coinmarketcap.com on 19 September 2022.

and Hedera Hashgraph) and two energy-inefficient contenders (Bitcoin and Dogecoin). Using established estimates (see Table 5.1), we identified a range of 1.45 mW h/tx to 273.14 kW h/tx to be broad enough to capture all popular products on the cryptocurrency market at the time. While an actual labelling scheme would likely be more complex, we assumed a linear relationship between electricity consumption and energy label: here, types *I* (1.45 mW h/tx) and *II* (2.02 W h/tx) represented the least electricity-consumptive currency types. Thus, label *A* was assigned to the Hedera Hashgraph and Ethereum cryptocurrencies. Consequently, type *III* (273.14 kW h/tx) currencies (Bitcoin and Dogecoin) were allocated the energy label *G*.

Measures

The key observation to gauge the effect of energy labelling on cryptocurrency purchase intention is the self-declared likelihood of a participant acquiring a given cryptocurrency. We measure this using a Likert scale: after being presented with the scenario details (see subsubsection 5.1), users are asked to select how likely they are to redeem their gift card for a given cryptocurrency. The indicated likelihood is measured using a four-point scale ('very unlikely' to 'very likely'). We assume that a strong likelihood of acquiring a given cryptocurrency (i.e., option 'very likely') can be interpreted as an indication of an acquisition intention. Applying this interpretation, we measure the number of energy-*inefficient* cryptocurrencies participants intend to acquire. We equally measure the number of energy-*efficient* cryptocurrencies participants intend to acquire. These two measures are central to the result (see section 5.2).

Procedure

The survey was conducted using the *Pollfish* platform: participants were selected via organic random device engagement (RDE) sampling, a voluntary response sampling method that relies on advertising networks in which participants receive minor, non-monetary rewards, such as benefits in mobile games (Rothschild & Konitzer, 2019). The method was found to produce accurate results (Goel *et al.*, 2015) despite the known shortcomings of non-probability sampling (Schreuder *et al.*, 2001). Participants were initially presented with the terms of the survey platform and informed consent was obtained. Participants were then presented with a screening question to ensure that only those who considered holding cryptocurrencies for investment purposes were selected. If qualified, they were presented with the questions described earlier (see subsection 5.1)

Goel, S., Obeng, A., & Rothschild, D. (2015, November 1). *Non-representative surveys: Fast, cheap, and mostly accurate.*

in sequence. After answering the relevant questions, an attention check was administered. On completion of the survey, the participants were thanked for their participation and were dismissed.

5.2 Results

Demographic Information

Two hundred valid responses were collected in November 2022. The mean time to completion was 2 min 15 s with participants accessing the survey from mobile devices and web browsers. Of the participants, 59.5 % identified as male, with the remainder identifying as female. Participant gender was not considered in the analysis. The average age of participants was 32.7 years ($\sigma = 12.97$). Apart from demographic characteristics, experience with and attitude towards cryptocurrencies were assessed: we classify most participants (76 %) as novice users, based on their self-declared understanding of cryptocurrencies, as shown in Figure 5.2.

Cryptocurrency Preferences

Figure 5.3 shows the distribution of participants' acquisition intention across different cryptocurrencies in the treatment and control groups. For energy-inefficient cryptocurrencies, we noticed that participants were more likely to express an acquisition intention for Bitcoin when in the control group ($N = 48$ vs. $N = 33$ in the treatment group). The same effect applied to Dogecoin ($N = 16$ in the control group vs. $N = 10$ in the treatment group). As we can see, with energy labelling, fewer participants expressed an acquisition intention towards energy-inefficient cryptocurrencies.

Treatment Effect

We ran Spearman's correlation test on our measurements in relation to the number of energy-inefficient and energy-efficient cryptocurrencies participants intended to acquire. We found that there was a significant negative correlation between being exposed to energy labels and expressing a strong preference for energy-*inefficient* cryptocurrencies ($r = -0.220$, $p = 0.002$). In other words, energy labelling discouraged

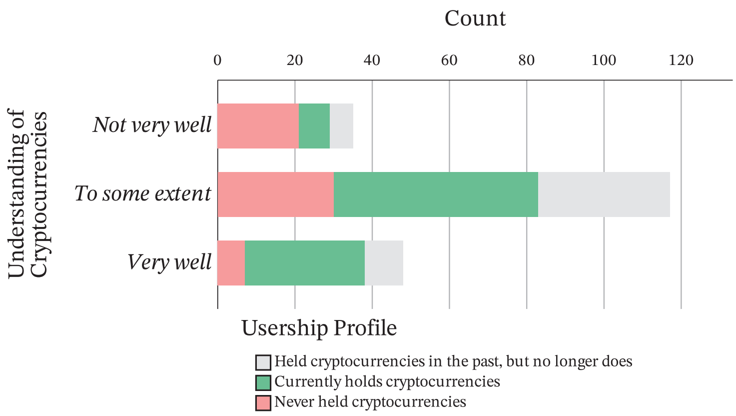

Figure 5.2: Participant knowledge and user profile.

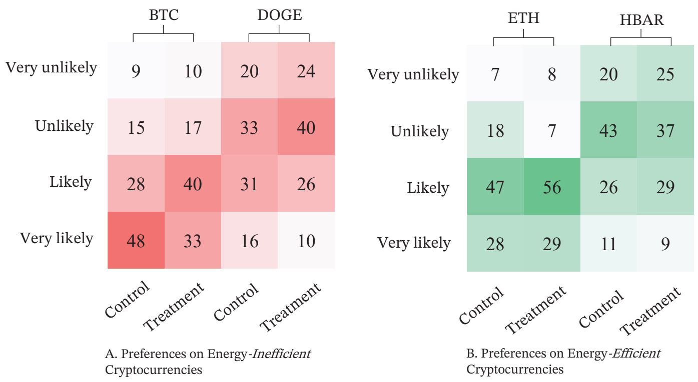

Figure 5.3: Preferences for energy-inefficient and energy-efficient cryptocurrencies in the control and treatment groups.

users from buying energy-*inefficient* cryptocurrencies. However, surprisingly, there was no significant correlation between being exposed to energy labels and expressing a strong preference for energy-*efficient* cryptocurrencies ($r = -0.010$, $p = 0.888$).

To further determine the extent of the treatment effect, an inferential statistics method, such as an independent sample t-test can be applied. We perform a Shapiro-Wilk test to investigate the normality of the number of energy-*inefficient* and energy-*efficient* cryptocurrencies with the result that neither of the two variables are normally distributed ($p < 0.001$). Therefore, we fitted a Mann-Whitney U test, which can be considered equivalent to the t-test without assumptions regarding the normal distribution of the data, instead of the t-test to determine the treatment effect. As shown in Table 5.3, there is a significant treatment effect ($Z = -3.097$, $p = 0.002$) for the number of energy-*inefficient* cryptocurrencies and no significant treatment effect ($Z = -0.142$, $p = 0.887$) for the number of energy-*efficient* cryptocurrencies. To be specific: in the treatment group, the number of energy-*inefficient* cryptocurrencies ($M = 0.43$, $\sigma = 0.62$) is smaller than in the control group ($M = 0.64$, $\sigma = 0.52$), confirming our hypothesis.

5.3 Discussion

Users presented with energy labels are found to be less likely to acquire energy-inefficient cryptocurrencies according to this work. Our work, therefore, confirms the hypothesis and produces a result that is congruent with the wider literature that establishes a positive relationship between energy labelling and sustainability (Davis & Metcalf, 2016; Jeong & Kim, 2014). Although a significant correlation between energy labelling and users' intention to acquire energy-efficient cryptocurrencies is not seen, the results suggest the potential of energy labelling in nudging users away from purchasing unsustainable cryptocurrencies.

Table 5.3: Results of the Mann-Whitney U test for treatment effect.

Group	N	μ	σ	M	MR	U	Z
Control	100	0.64	0.52	1.00	111.69	3,881	-3.097 **
Treatment	100	0.43	0.62	0.00	89.31		

μ: mean, σ: standard deviation, M: median, MR: mean rank
** significant at the $p = 0.01$ level.

This result has potential practical implications: regulatory bodies with a remit of sustainability and consumer education may contemplate the introduction of energy labelling standards for cryptocurrencies. Additionally, operators of CEXs may pre-empt the potential introduction of mandatory energy labels by exposing electricity consumption information to their customers voluntarily. The findings suggest that both courses of action may lead to more sustainable consumer behaviour.

Conclusion

The results demonstrated in this work provide a new perspective on reducing the environmental impact of cryptocurrency market activities. Many previous regulatory initiatives in pursuit of sustainability have focused on *operators of mining hardware*. These, however, have rarely been successful, due to the supranational nature of cryptocurrencies. While some limitations remain, this work has confirmed that displaying energy labels on centralised exchange websites, thereby targeting the *buyers of cryptocurrencies*, influences consumer choice.

Energy labelling was shown to be a viable method to deter users from acquiring highly electricity-consumptive cryptocurrencies, thus offering assistance to ESG-conscious investors. Energy labelling should, therefore, be considered by regulators when establishing cryptoasset sustainability standards. Educational efforts should, however, be based on models that reflect the full energy costs of blockchain operations, as the narrow focus of current models may be misleading (Platt *et al.*, 2024).

Platt, M., Platt, D., & McBurney, P. (2024). Sybil attack vulnerability trilemma. *International Journal of Parallel, Emergent and Distributed Systems, 39*

Limitations

This chapter presents several limitations. Firstly, the group of participants was limited to residents of Romania, thus our results may not be generalised to other regions, especially those outside of the EU where people may be less familiar with or mindful of energy labelling. Secondly, product decisions were made in a mock environment: participants might be more conscious of their acquisition decision if money was at stake. Finally, the RDE sampling method might have influenced the results: most participants declared to have at least a basic understanding of cryptocurrencies.

Reputation Systems for Democratic Consensus

A New Consensus Mechanism for Democratic Self-Governance

6

This chapter follows the previous empirical work (chapter 4, chapter 5) by describing a consensus mechanism – identity-augmented proof-of-stake (IdAPoS) – intended to address the identified shortcomings, namely high energy consumption and the centralisation of control. Subsequently, the vulnerability of IdAPoS to Sybil attacks is evaluated.

6.1 Background

DLT systems, including blockchains which constitute a specialisation of those, can be considered decentralised record-keeping systems. Blockchain technology, introduced by the cryptocurrency 'Bitcoin' (Nakamoto, 2008) and further popularised by the application platform 'Ethereum' (Buterin, 2013), brought with it a variation of the centrally governed systems previously studied in distributed computing. Both Bitcoin and Ethereum, along with many contenders, take inspiration from anti-authoritarian concepts (W. Dai, 1998) interlinked with the 'Cypherpunk' movement and its objective to enable anonymous economic transactions (Jarvis, 2021), thereby emphasizing decentralisation and the absence of supervision.

Dai, W. (1998). *B-money.*

Fault Tolerance and Byzantine Behaviour

While the formal study of faults in distributed systems reaches back to the inception of data networking theory in the 1960s (Kleinrock, 1961), this early work assumed participants behave non-maliciously, with predictable communication malfunctions. Only in the 1980s was the notion of *malicious* participants formalised, describing participants that *lie* to disrupt systems (Pease *et al.*, 1980). This led to Lamport *et al.* (1982), who demonstrated the behaviour of decentralised systems when actors spread incorrect, or conflicting information, or withhold information, coining the term *byzantine* faults for such behaviour. A common strategy to mitigate these faults were voting-based leader selection protocols (Castro & Liskov, 1999; Lamport, 2004).

Pease, M., Shostak, R., & Lamport, L. (1980). Reaching agreement in the presence of faults. *Journal of the ACM*, *27*(2)

Validator Selection

In any decentralised system, selecting trustworthy verifiers that adhere to the protocol by verifying and replicating data without interference is crucial. In a trustless decentralised environment, devoid of hierarchies, selecting actors to replicate and evolve records is non-trivial and comes with risks. Multiple replicating actors must form consensus regarding the validity of proposed records, necessitating access to these records either openly or confidentially (Platt, Bandara *et al.*, 2021). A fundamental assumption of voting-based protocols is, however, that the number of participants in an election is reliable. This becomes problematic when Sybil actors, gain control by 'presenting multiple identities' (Douceur, 2002). Without appropriate consensus mechanisms, attackers controlling many identities can disrupt trustless systems entirely.

To counteract such attacks, decentralised systems research has developed various consensus protocols aimed at limiting the impact a single entity can achieve. Common protocols, such as PoW or PoS, use scarce resources, such as computing power or cryptocurrency holdings, to approximate legitimacy. While subject to criticism (see chapter 3), the absence of disastrous security events concerning PoW or PoS systems and their high availability (Weber *et al.*, 2017) is an indication of their effectiveness. Their incentive compatibility in economic contexts encourages compliance with system rules, making them foundational for significant public blockchains today.

Weber, I., Gramoli, V., Ponomarev, A., Staples, M., Holz, R., Tran, A. B., & Rimba, P. (2017). On availability for blockchain-based systems. *Proceedings of the 36th Symposium on Reliable Distributed Systems*

The main goal of early blockchain protocols is to 'allow benign nodes, through a shared consensus mechanism, to agree on an (almost) immutable record of transactions despite byzantine failures and eventually achieving consistency' (Sunyaev, 2020). Early blockchain systems typically follow leader-based protocols to realise this. Such systems make use of asymmetric cryptography to allow users to obtain lightweight identities by generating pairs of public and private keys. Public keys (or derivations thereof) serve as *addresses* on ledger. This is particularly important in the context of cryptocurrencies where transfers of value between different holders is a key requirement. However, signed transaction proposals alone are insufficient to conclusively finalise transactions due to potential double-spend attempts (Camenisch *et al.*, 2006). To prevent this, network-wide consensus on transaction order and permissibility must be established. The latter can be achieved using 'smart contracts' (Szabo, 1997), deterministic programmes that describe permissible changes to blockchain data.

Camenisch, J., Hohenberger, S., & Lysyanskaya, A. (2006). Balancing accountability and privacy using e-cash: Extended abstract. R. D. Prisco & M. Yung (Eds.), *Proceedings of the 5th international conference on security and cryptography for networks.* Springer

In the case of leader-based protocols, mechanisms to determine a leader that are resistant to Sybil attacks must be applied. The leader determined is responsible for impartially selecting from the pool of transaction proposals – though the ideal of impartiality is not always achieved in practice (Eskandari *et al.*, 2020) – verifying proposals, combining them in a *block*, and then approving this block by means of a cryptographic signature. Following approval, the block contents are passed on to other network participants. Finally, the selected leader may receive a reward for their activity.

Bitcoin is distinguished from earlier distributed systems by being strongly permissionless (see chapter 7) in the sense that 'any network participant has the ability to create a candidate record' (Rauchs *et al.*, 2018, p. 61). This paradigm laid the foundation for a variety of similar consensus protocols. PoW-based mechanisms underpin the most relevant public decentralised systems today, namely the cryptocurrency Bitcoin (Nakamoto, 2008) and, at the time of writing of this chapter, the application platform 'Ethereum' (Buterin, 2013). While this is testament to their effectiveness in the context of cryptocurrencies and decentralised finance (Werner *et al.*, 2022), they have been widely criticised for their environmental impact (see chapter 3), for poor throughput characteristics when compared to centralised systems (Croman *et al.*, 2016), and for their undemocratic nature (Tozzi, 2019).

Buterin, V. (2013). *Ethereum whitepaper.*

Recognising these shortcomings, alternative approaches for selecting participants in public decentralised systems have been proposed (see section 6.2), typically differentiated by their trust assumptions (Natoli *et al.*, 2019, p. 7). PoS, a concept discussed first in the context of Bitcoin (QuantumMechanic, 2011), reduces electricity consumption and, thereby, improves cost-efficiency. In PoS, a stochastic selection process considering the duration and amount of cryptocurrency held determines who can gain the privilege of block generation on the network. This, arguably (Y. Huang *et al.*, 2021), brings with it a *rich getting richer* effect correlated with incentive system parameters (Fanti *et al.*, 2019).

Given that consensus mechanisms aim at providing members with a fair chance of participation in a decentralised system, parallels to the political realm are obvious. It can be speculated that member selection in decentralised systems follows mechanisms similar to those present in archetypes of self-governance. In this chapter we, therefore, apply principles of self-governance to blockchain consensus, thereby deriving a novel overlay over PoS member selection that takes inspiration from principles of self-governance. It combines a numeric representation of trust in the identity of a participant with the simple arithmetic properties of PoS.

6.2 Related Work

The absence of established standards for blockchains makes a systematic treatment of their properties difficult and, consequently, no canonical taxonomy of Blockchain consensus protocols exists today (Hyland-Wood & Khatchadourian, 2018). Nonetheless, surveys have been undertaken to categorise protocols. Emerging categories are verifier selection approach, block addition protocol, and transaction confirmation method (X. Fu *et al.*, 2021). Considering highly capitalised cryptocurrencies as archetypal applications, it is evident that PoW and PoS are the most prominent algorithms for public blockchains (see chapter 2).

From a broader application perspective, DPoS and PBFT are additionally relevant (Sutherland, 2019). Most private blockchains employ proof of authority (PoA) or similar algorithms that rely on a set of authorised validators (Leal *et al.*, 2020; Polge *et al.*, 2021). Collaborative leader-based consensus algorithms, exemplified by Raft (Ongaro & Ousterhout, 2014) and Paxos (Lamport, 1998), rely on mechanisms to select leaders from the known set of participants, making provisions to address Sybil attacks unnecessary.

Consensus Mechanisms and Democratic Principles

Majority rule is central to decision-making in democratic systems. Blockchain-based voting applications for large-scale (Xiao *et al.*, 2019) and small-scale (McCorry *et al.*, 2017) elections have been discussed extensively, and strategies for democratically resolving contentious decisions via weighted voting have been proposed (Merrill *et al.*, 2020). Yet, underlying consensus mechanisms are rarely debated.

Whether an algorithm is appropriate for a given use case depends, among other factors, on its hierarchicality, accessibility, and verifiability requirements (Platt, Bandara *et al.*, 2021). Application developers often do not actively choose a particular consensus mechanism but follow one prescribed by the targeted platform (A. Dhillon *et al.*, 2021).

Platt, M., Bandara, R. J., Drăgnoiu, A.-E., & Krishnamoorthy, S. (2021). Information privacy in decentralized applications. M. H. U. Rehman, D. Svetinovic, K. Salah & E. Damiani (Eds.), *Trust models for next-generation blockchain ecosystems.* Springer

Wall, S. (2007). Democracy and equality. *The Philosophical Quarterly*, *57*(228)

The aspect of equality, i.e., distributing political standing equally to all citizens, is essential for consensus mechanisms to be considered democratic (Wall, 2007). Thus, democratic consensus mechanisms must ensure a $0..1 : 1$ relationship between participants and citizens, meaning a single citizen operates at most one participant account. Once satisfied, a protocol must distribute political standing equally.

Blockchain as a Democratic Technology

The intersection of blockchain and democracy has been explored previously. While some works emphasise blockchain's potential to reduce electoral fraud (A. Dhillon *et al.*, 2021), others challenge its suitability for facilitating democratic processes, citing vulnerabilities to centralisation and unknown attacks (Hermstrüwer, 2019; Park *et al.*, 2021).

Dhillon, A., Kotsialou, G., McBurney, P., & Riley, L. (2021). Voting over a distributed ledger: An interdisciplinary perspective. *Foundations and Trends in Microeconomics*, *12*(3)

Racsko (2019) finds existing protocols inadequate for democratic use. Tozzi (2019), on the other hand, considers blockchain technology potentially suitable for democratic applications and concludes that DPoS and PBFT may be considered more democratic than PoW by reducing political advantages conferred by wealth. DPoS resembles liquid democracy by allowing vote delegation (Blum & Zuber, 2015), inheriting potential elitist divisions. Redshaw (2017) highlights that PoW, exemplified by Bitcoin, lacks ethical justification for democratic systems due to inherent inequalities. Consequently, consensus protocols significantly influence the democratic evolution of decentralised systems, with PoS-like mechanisms appearing most suitable for identity-based governance.

Non-Economic and Alternative Consensus Protocols

Protocols like Proof-of-Personhood require physical presence verification (Borge *et al.*, 2017; Pournaras, 2020). Biometrics-based methods, such as Worldcoin's iris verification system, also aim to establish uniqueness (Gent, 2023). However, hardware-based approaches inherently require trust in the hardware manufacturer, making them quasi-permissioned. Reputation-based consensus mechanisms, like the one proposed by de Oliveira *et al.* (2020), that randomly selects judges to reward cooperation, generally achieve only limited Sybil attack resistance (see chapter 7). Therefore, there remains a gap for protocols that enable multiple constituencies, governed by independent identity authorities, to collaboratively operate decentralised systems strongly resistant to Sybil attacks.

Proof-of-Work and Proof-of-Stake Protocols

Bitcoin formalised PoW, a consensus mechanism to counteract double-spending without central trusted parties (Nakamoto, 2008). While PoW protocols incentivise honest participation through computational effort rewarded by block rewards (Chiu & Koeppl, 2019), they remain vulnerable to some attacks and only exhibit probabilistic finality (Conti *et al.*, 2018; X. Yu *et al.*, 2019).

Avoiding the computational waste caused by PoW motivated the development of PoS (Deuber *et al.*, 2020). 'PPCoin' is one of the earliest cryptocurrencies using PoS that was discussed academically. 'PPCoin' was introduced by King and Nadal (2012), drawing inspiration from initial discussions among the Bitcoin community (QuantumMechanic, 2011). As per this original community contribution, being able to prove possession of currency, along with demonstrating how long it has been retained, should determine the complexity of creating a block, thereby making established participants with greater holdings more likely to create new

King, S., & Nadal, S. (2012, August). *PPCoin: Peer-to-peer crypto-currency with proof-of-stake.*

blocks. Compared to PoW, this shifts the responsibility of maintaining and evolving decentralised data from those who invest computing resources to those who hold currency.

Proof-of-Authority and Permissioned Protocols

Permissionless member selection is unsuitable for environments requiring restricted participation due to privacy or regulatory concerns. PoA, used by protocols like Ripple, Corda, and Diem, employs predefined lists of trusted validators for member selection and validation (Chase & MacBrough, 2018; Hearn & Brown, 2019; Libra Association Members, 2020).

Cross-Chain Interoperability

In multi-constituency environments interoperability is crucial, especially regarding information about the reputation of participants (Hesse & Teubner, 2019). Even though recent advances in cross-chain transactions make synchronous inter-blockchain function calls more easily implementable (Robinson *et al.*, 2020), connecting two disjoint chains remains subject to strong assumptions (Zamyatin *et al.*, 2019). This is particularly prevalent as lock-based techniques pose the risk of arbitrage in scenarios in which a participant can gain monetary advantage from aborting a cross-chain protocol prematurely (Platt *et al.*, 2020). The protocol proposed within this chapter is designed to circumvent these issues by integrating multiple constituencies within a unified blockchain for atomic transactions.

Hesse, M., & Teubner, T. (2019). Reputation portability – quo vadis? *Electronic Markets*, *30*(2)

6.3 Contribution

Despite the variety of consensus mechanisms available, no voting-based protocol exists that allows for the delegation of decisions around legitimacy to elected authorities. IdAPoS addresses this shortcoming.

Table 6.1 shows how the protocol presented here differs from existing ones. While computational effort and wealth determine participation in PoW, PoS, and DPoS, IdAPoS uses voting tokens for this purpose. Unlike PoW and PoS, these tokens are not subject to inflation control and can be freely issued onto the system. This fact allows IdAPoS to represent multiple constituencies within the system without the need for

Table 6.1: A comparison of PoW, PoS, DPoS, and IdAPoS.

Protocol	Entitling Resource	Inflation	Requirements for Diverse Representation	Removal of Validators
PoW	Computational Effort	Controlled	Willingness to expend computational resources	By competing on entitling resource
PoS	Wealth	Controlled	Agreement on underlying asset	By competing on entitling resource
DPoS	Wealth	Controlled	Agreement on underlying asset	By competing on entitling resource
IdAPoS	Voting Tokens	Unbounded	None	Via reputational messages

computational effort or agreement on a cryptocurrency underlying the system. Furthermore, through the use of reputation messages, IdAPoS allows validators to be dismissed from the system through a democratic process. This is not feasible using conventional consensus mechanisms.

6.4 Problem

Some parallels can be drawn between member selection methodologies in decentralised systems and processes of political representation that can be observed in the analogue world. The virtual constituency of those who can create candidate records in a decentralised system (see Figure 6.1), can be compared to a constituency in the political sense, albeit, not a stable, state-level constituency, but a transnational fast-evolving one, where elites and influencers have a disproportionate impact on the selection of candidates (DiCaprio, 2012). The constituency can rely on the legal framework of the governing system to gain a high degree of certainty that its codified rules will be enforced by the executive branch.

The group of participants who validate candidate records, approve, and evolve them, can be compared to a government, particularly its legislative and executive branches. Depending on the underlying consensus protocol, miners are appointed or elected via probabilistic methods that roughly resemble a majority vote (e.g. the majority computing power in the case of PoW, or the majority of funds in the case of PoS). Participants who create candidate records make these available to miners, who, in turn, validate that these are compliant with the legal framework. Should a candidate record be found to be compliant, miners will endorse it publicly. Should the candidate record found to be in breach of the framework, participants may be penalised. It can be assumed that the legislative framework is deterministic, meaning all honest miners will come to an identical conclusion on compliance when evaluating a record. This can be achieved via Deterministic, automatically evaluable, ‘smart contracts’ (Szabo, 1997), a concept at the intersection of law and computer science, subject to further formal validation and verification (Harz & Knottenbelt, 2018; Magazzeni *et al.*, 2017). In contrast to natural language contracts, smart contracts are designed not to require interpretation (Cannarsa, 2018), even though common forms of smart contract programming are prone to bugs (Delmolino *et al.*, 2016; C. Liu *et al.*, 2018).

Szabo, N. (1997). Formalizing and securing relationships on public networks. *First Monday*, *2*(9)

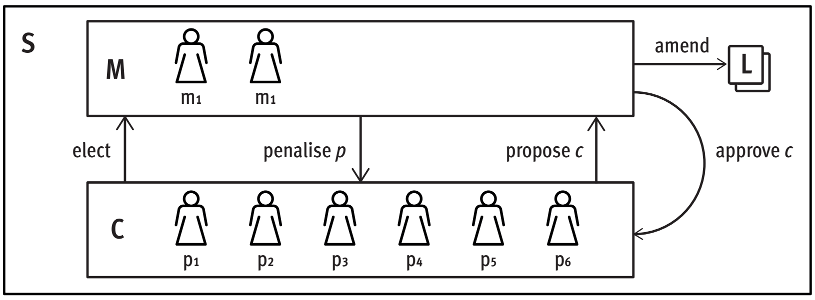

Figure 6.1: A decentralised system S, comprised of regular participants ($p_{1..n}$) and participants with additional duties (‘miners’ $m_{1..n}$) who are appointed or elected to fulfil these duties. Participants propose candidate records, c, to be included in the entirety of public records. Miners decide, based on a legislative framework, L, whether a candidate record is permissible and, based on this evaluation, either approve it or, penalise the responsible participant for proposing an impermissible record.

6.5 Approach

To overcome the limitations of other consensus protocols, an identity-based consensus mechanism centred around PoS, IdAPoS, is proposed in subsection 2. This realises the democratic principle of 'one person, one vote' in decentralised member selection. Instead of using capital as a deciding factor in the selection of participants, as it is common in PoS, the protocol incorporates the idea of publicly elected authorities that provide voting tokens for the establishment of PoS consensus in non-economic environments, an approach dubbed 'identity-augmentation'. This approach goes beyond various existing consensus methods that resemble majoritarian member selection (see section 6.2), by enforcing democratic principles and giving the participants in the system mechanisms to counteract those who violate them.

This section demonstrates why existing algorithms are unsuitable for implementing 'one person, one vote'. The difficulty here is to determine which participant should be considered eligible to vote, rather than the implementation of the voting protocol itself. The IdAPoS protocol achieves this by employing the concept of voting tokens as an approximation of eligibility. Building on the definition of Borge *et al.* (2017), voting tokens express the degree of certainty the issuer has in the recipient having a unique identity in the real world, a concept that is commonly referred to as 'personhood'. There is no canonical definition of legal personhood (Foster & Herring, 2017) and, consequently, calls for a fundamental revision of its doctrine have been voiced (Garthoff, 2018). Fittingly, in IdAPoS, authorities are free to apply a custom definition of personhood according to the social context they operate in, thereby potentially even allowing artificial agents to assume personhood. To implement this, authorities can issue voting tokens to participants.

One person, one Vote

In a DPoS system S members $p_{1..n}$ form a constituency C. They can delegate participation in the consensus protocol to other parties on the network, miners ($m_{1..n}$), thus approximating a vote. Since delegation privileges in this protocol are aligned with the currency holdings of the delegating party, this pattern corresponds to 'One Share/One Vote', a paradigm with a long-standing tradition in corporate government (Ratner, 1970). Conceptually, here, voting power is tied to the holdings of a shareholder. Although this approach is by no means democratic, it can serve as a conceptual building block to develop a democratic system.

Ratner, D. L. (1970). Government of business corporations critical reflections on the rule of one share one vote. *Cornell Law Review, 56*

Since DPoS conceptually follows a 'One *Share*/One Vote' paradigm (voting power is aligned with cryptocurrency holdings), it can be restructured with the goal of supporting a 'One *Person*/One Vote' paradigm. This can be done by introducing additional constraints to limit the number voting tokens and how they can circulate to 'system contracts'[1]. With the following additional restrictions in place, DPoS, a paradigm that uses 'One *Share*/One Vote' mechanics, can be evolved to IdAPoS, which provides 'One *Person*/One Vote' characteristics:

1: The term refers to the fundamental functionality of the core protocol that is not modifiable by individual users, as compared to user-defined smart contracts

1. DPoS is performed using the market value of voting tokens as stake
2. Every participant holding voting tokens is recognised as a person with voting rights on the network
3. Voting tokens cannot be traded and are not given out as a 'block reward'

It should be noted that the restriction on the tradability of tokens does not imply that economic activity cannot take place in the system. Rather, this restriction merely implies that the system does not provide a *native*, tradable cryptocurrency. Participants on the network will remain free to issue and trade digital tokens, similar to how common blockchain protocols allow for the issuance and transfer of arbitrary user-defined tokens (Vogelsteller & Buterin, 2015). Consequently, these restrictions are designed to decouple economic activity from the consensus protocol itself.

Vogelsteller, F., & Buterin, V. (2015). EIP-20: ERC-20 token standard

Establishing Personhood

The requirements *i* and *iii* of the previous section can easily be satisfied through minor modifications of existing protocols. An implementation can fork existing DPoS protocol implementations and change validation code so that token movement becomes impermissible. The second requirement, however, raises a more significant problem. Here, it needs to be considered who should be admitted to the decentralised system, by whom and on what basis.

Identity Authority

A trivial solution to this problem is the introduction of a central 'identity authority' A who assigns voting tokens. They could simply assign all participants they believe to represent a valid identity a token of value 1. A hypothetical protocol that implements this paradigm could require prospective participants to generate an address and to subsequently submit it to the authority along with proof of their identity through an off-ledger protocol. Should the identity information provided satisfy the requirements of the central authority, they would allocate 1 voting tokens by funding the address of the applicant.

This approach would effectively recreate a proof of authority protocol with one centralised admitting entity. The shortcomings of this approach align with the shortcomings of any proof of authority protocol, namely, the fact that the governance structure needs to be determined at the inception of the network and can only be amended by the initiative of the admitting entity itself. This approach, therefore, requires an invariable degree of trust in the admitting entity.

Self-Governed Evolving Constituencies

While a central approach is well-suited for cases in which no fundamental changes to the constituency of a system are to be anticipated, and in which the central authority is irrefutably trusted, it is inappropriate in a scenario in which the constituency is changing dramatically. An evolving constituency is conceivable in self-governance scenarios, e.g. when a group of constituents chooses to administer common resources through a joint governance process (Ostrom, 1999; Townsend, 1995), or when a group decides they require self-governance, for resolving grievances or making political decisions outside a wider context governed by an external authority (Ostrom *et al.*, 1992). It is also appropriate when a system operator plans to evolve from a centralised governance model to a decentralised model at a later point (Baird *et al.*, 2021; Shehar *et al.*, 2019), or where conventional centralised systems exist but do not reach users

Ostrom, E., Walker, J., & Gardner, R. (1992). Covenants with and without a sword: Self-governance is possible. *American Political Science Review*, *86*(2) operating outside of common societal structures (Khurshid & Gadnis, 2019; Madianou, 2019).

As with any process building on majority rule, the risk of a tyranny of the majority in which 'society collectively [tyrannises] the separate individuals who compose it' (Mill, 1859) needs to be recognised. Such a situation can arise where the subsidiarity principled is violated and decisions that should be made in a subunit are made on a higher hierarchical level (Føllesdal, 1998), or, more generally, where majority decisions lack rationality (de Tocqueville, 1835). Netanel (2000) echoes this observation in the context of digital systems by arguing that self-governance of large and diverse digital environments will not be conducive to liberal democratic ideals but will instead free majorities to defeat minorities. This thinking can be used both as an argument for the self-governance of smaller autonomous groups in general, as well as an argument for the need for flexibility around growing and shrinking of constituencies and their need for flexible reorganisation.

Mill, J. S. (1859). *On liberty.* John W. Parker & Son

To replicate a self-governing structure in a decentralised system, the previously described central identity authority approach (see subsubsection 1) is a suitable point of departure. Conceptually, self-governance entails going beyond individualist ideology, and, consequently, the existence of at least one, potentially self-adapting (Rekkas, 2017), constituency. These constituencies can be represented by authorities $A_{1..n}$ that may operate in parallel and make decisions about their powers subject to a democratic process. DPoS protocols already bring all necessary technological primitives for voting, since delegation is a voting process in itself. Building on that, a second voting layer can be proposed. This voting layer would be responsible for the election of identity authorities into and out of the decentralised system. A decentralised system making use of evolving constituencies would be initialised with a 'genesis identity authority'[2]. This authority would act in the same way as described above, i.e. they would assign voting right tokens to members of the genesis constituency, following their principles of identity validation. The self-governing aspect would manifest itself by introducing reputational signals as new capabilities to the system:

Rekkas, S. (2017, January 2). *A behaviourist framework for describing open self-organising systems* [Doctoral dissertation, King's College London]

2: This naming is inspired by the term 'genesis block', the initial Block of Bitcoin.

1. Members of a constituency can endorse an identity authority on the network
2. Members of a constituency can discourage an identity authority on the network

The IdAPoS Protocol

IdAPoS approaches personhood on a PoS network through voting tokens, a numerical representation of confidence in the holder expressed by the issuer. It recognises two kinds of participants: Participants p_n, who partake in common activities of a decentralised system, e.g. by transacting with other participants on matters that do not concern identity, and authorities A_n, who may issue voting tokens to certain members that they believe to be trustworthy. The influence of authorities is in turn determined by network participants who express their trust or mistrust through reputational signals (see Figure 6.2).

Voting tokens replace tradeable cryptocurrency balances common in PoS. Rather than being exchangeable between participants, they act as numerical representations of trust in the holder. Voting tokens have a nominal value $nv \in (0, 1]$ that represents the trust the authority has in the recipient

Variable	Description
B	The 'block'; a set of all mined transactions
H	A map of maps containing the nominal holdings of voting tokens of a participant (Participant ↦ (Authority ↦ Nominal voting tokens value)) as established in previous blocks
R	A map of maps containing the trust scores expressed by participants (Authority ↦ (Participant ↦ Nominal reputational value)) as established in previous blocks

Table 6.2: Inputs to the IdAPoS protocol.

of the voting tokens. A value of $nv = 1$ would be commonly assigned in situations where authorities have full trust in the recipient. Smaller values may be appropriate where the issuing authority undertakes a risk assessment to quantify the actual trustworthiness of the recipient. This approach aligns with existing methods for quantifying trust in identity information and can be understood as a level of assurance (LoA) assigned by the issuer.

Endorsements and discouragements, so-called reputational signals, are used by regular participants to signify trust or mistrust in an authority. Participants are expected to emit such signals once they have learned positive or negative information about an authority. By default, they are expected to have a favourable view on the authorities that have issued tokens to them as those provide them with access to the system and voting power. They can, however, emit signals towards any authority known on the network, including the ones that have issued voting tokens to them. Participants can furthermore use endorsements to add new authorities to the network. These signals, dampened by an attenuation function, generate a numerical value for the trustworthiness $t_a \in [0, 1]$ of an authority. This numerical value is similar to other LoA: An authority that has received a large proportion of discouragement messages will have a low score, whereas an authority that has received a large proportion of encouragement messages will enjoy a high rating.

The trustworthiness, in turn, informs the market value $mv \in [0, 1]$ of any voting tokens issued by said authority. The market value acts as another LoA, this time based on the trustworthiness of the authority. Authorities associated with a high degree of trustworthiness can issue valuable tokens, whereas tokens issued by untrustworthy authorities have low value. The average market value of voting tokens a participant holds can be seen as a network-wide numerical approximation of whether the holder represents a unique identity. This stands in contrast to the nominal value of voting tokens that only approximates the issuing authorities' trust. The choice of averaging the values might seem somewhat arbitrary, after all, it is not obvious how two trust values should interact arithmetically. It can, however, be motivated by ensuring that the overall voting power of a participant remains unaffected by whether tokens issued are pooled under one participant identity, or separate identities are created.

Ultimately, the average market value of voting tokens is used as a basis for staking, the process underlying PoS miner selection. The exact mechanics of the underlying PoS protocol itself, such as the minutiae of miner selection, are secondary to IdAPoS. It merely demands the basic principles of PoS are being adhered to, namely that participants obtain voting rights as a result of producing 'proof of ownership of the currency' (King & Nadal, 2012), specifically, proof of ownership of voting tokens. In a system utilising voting tokens, participants would likely delegate their

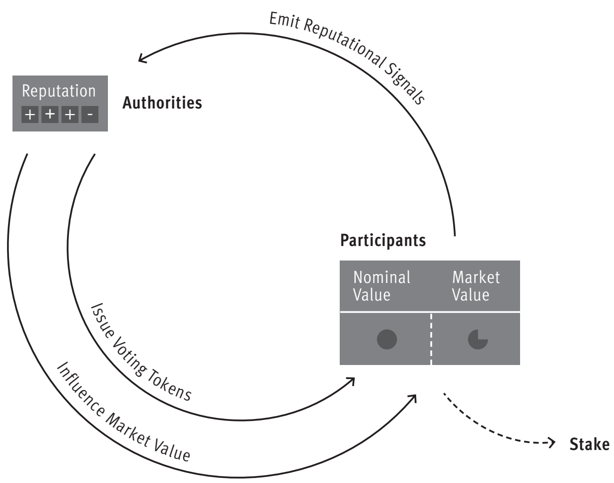

Figure 6.2: Interactions in IdAPoS follow a circular pattern. Authorities issue tokens of a given nominal value to participants. In turn, these become eligible to emit reputational signals towards authorities. Reputational signals influence the reputation of an authority. The reputation value influences the market value of tokens issued. The market value is then used for PoS miner selection.

continuous participation in voting processes to agents acting on their behalf. Therefore, these systems would effectively constitute open agent societies that are likely to implement deliberative democracies, in which participants are both consumers of political information (i.e. the voting token market values of others) and producers of political information (by endorsing or discouraging authorities) (McBurney & Parsons, 2004).

Protocol Sequence

Algorithm 1 formalises the protocol: it takes the inputs described in Table 6.2. These are all transactions selected to be included in the current block (*B*), and data structures containing nominal holding information (*H*) and reputational information (*R*). Both *H* and *R* can be reconstructed by parsing all transactions contained within previous blocks, and are included here as dedicated variables for reasons of computational efficiency and readability only. As IdAPoS is an augmentation of PoS, the leader selection aspect of an PoS protocol is omitted. What is presented in Algorithm 1 is merely the stake calculation part of any wider, generic, PoS algorithm.

Transaction Analysis Phase Each protocol step begins with an analysis phase in which all reputational signals are parsed. This can be achieved by iterating over all transactions (tx) that are contained within the current block (*B*). Should the algorithm detect a transaction pertaining to the issuance of voting tokens, endorsement, or discouragement, the previously described data structures *H* (holding information related to participants), and *R* (holding information related to authorities), will be updated. In case of endorsements, positive values are written to *R*. In case of discouragements, negative values are stored. Since *H* stores *nominal* voting tokens issued to participants, the result of an issuance transaction can be applied directly without the need for additional arithmetic. In the case of endorsements or discouragements, however, the nominal holdings need to be determined in order to populate *R* with appropriately calculated values.

Calculation of Nominal Holdings Nominal holdings for a participant p are calculated using *H*, a data structure holding information related to voting tokens issued to participants. Using this data structure, the sum of nominal values of voting tokens issued to p can be calculated.

By dividing this by the overall number of voting tokens issued to p an average nominal holding ($(0, 1]$) can be calculated. The data structure capturing information on authorities (R) will be updated with the results of this calculation.

Market Value Calculation Phase After the reputational messages contained within a block have been analysed and the results of the analysis have been applied to the data structures H and R, the market value of voting tokens issued by all issuers on the network can be calculated. To achieve that, the algorithm iterates over all participants p known to the system as evidenced by them having received at least one voting token previously. For each of these participants the algorithm iterates over all authorities (a) that have issued voting tokens to them in the past.

Calculation of Trustworthiness For each of these authorities a a trustworthiness value can be calculated. This is done by first calculating the sum of reputational values a has received via analysis of R. Should this sum be positive, an attenuation function is applied to it. Should it be negative, 0 is returned.

Calculation of Token Market Value Multiplying this trustworthiness score with the previously calculated *nominal* token value provides a token *market* value. Performing this calculation for all combinations of issuers and participants allows generating a data structure that holds the token market values for all participants p. On the basis of this information, that is contained in the return variable V, an arbitrary PoS protocol can be applied. Here, the market value serves as a stake.

Hierarchical Propagation of Trust

As shown in Figure 6.3, participants can create further identity authorities beyond the genesis authority ($A1$) via reputational signals (dashed lines) and identity authorities can endorse arbitrary participants. This means that the trust relationships on an IdAPoS-backed network can be formulated as a graph with the genesis authority as a root, bearing similarity to the *PageRank* (Page *et al.*, 1999) algorithm.

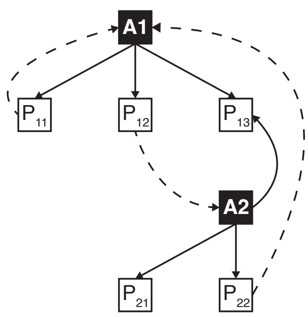

Figure 6.3: Authorities ($A1$, $A2$) can issue voting tokens (solid lines) to participants ($P_{11}..P_{22}$) who can emit reputational signals (dashed lines) towards authorities, thereby creating an implicit hierarchical structure for propagating trust.

This allows Sybil attackers to create deep, self-endorsing, hierarchies, which can be treated through mitigation strategies, some of which are contemplated in subsection 2 and evaluated in section 6.7.

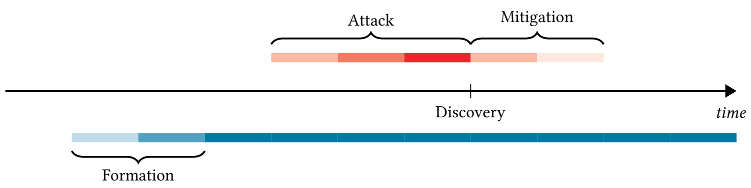

Figure 6.4: The anatomy of a thwarted Sybil attack on an IdAPoS system. The bottom bar symbolises a trustworthy constituency, the top bar a malicious one.

Power Dynamics

Since authorities have the power to assign arbitrary amounts of voting tokens to participants, they could withhold them from undesirable participants or dissidents, thereby abusing tokens as a quantification of obedience rather than trust. While more work is needed to understand the repercussions of an autocratic government or private entity acting as authority on an IdAPoS network, safeguards to stop malicious authorities from enforcing the political will through voting tokens issuance need to be discussed. Safeguards may take the form of legal and constitutional measures (Tirole, 2021). Additionally, from a protocol design perspective, enforcing diversity may counteract centralistic tendencies that accompany autocratic efforts. It, however, needs to be acknowledged that IdAPoS may not be a suitable protocol for scenarios threatened by autocratic actors in which participants have to fear real-live repercussions based on their disapproval of despotic authority nodes in the virtual space.

Tirole, J. (2021). Digital dystopia. *American Economic Review, 111*(6)

Sybil Attacks on IdAPoS

While PoS brought with it a variety of previously unknown attacks, Sybil attacks are not an attack vector commonly considered applicable to them. Commonly, the total wealth of cryptocurrency available within a PoS network is subject to a systemic limit. This limit can be static or increase continuously through inflation incurred by mining rewards (Irresberger & Yang, 2023). Since validator selection is contingent on the share of wealth a participant holds, the wealth limit acts as an inbuilt mechanism to prevent Sybil attacks. In an IdAPoS system, where issuers can bring new tokens into circulation at will, such an upper bound is, however, not conceivable. Smaller authorities might only issue voting tokens infrequently to small numbers of participants while larger ones might issue them in high frequency. A reasonable default could be to introduce no upper bound at all, allowing authorities to issue arbitrary amounts of voting tokens to the participants they trust. This would allow for the explosive growth of wealth, potentially leading to a concentration of it originating from a single authority. While this is unproblematic in a scenario where all authorities are trustworthy, it introduces the risk of Sybil attacks: Malicious authorities can issue large numbers of voting tokens to accomplice accounts, allowing them to inflate their influence in the IdAPoS protocol, thereby creating enough voting tokens to conduct a successful majority attack. In the absence of protocol-level limitations on the issuance of voting tokens, a malicious authority would only be constrained by technical parameters (e.g. maximum block sizes that limit how many transactions can be included per period). This limitation would provide no substantial protection from Sybil attacks, making the system vulnerable to a takeover within a short time.

Despite these blatant risks of a takeover, it can be assumed that an attack spans at least several periods (blocks). Therefore, the following picture of an attack emerges: A Sybil attack on an IdAPoS system follows three stages (see Figure 6.4). A formation stage, in which the attacker has not started the attack, but one or more trustworthy constituencies are established and develop. An attack stage, starting with the undermining of an identity authority leading to the creation of malicious authorities that begin issuing voting tokens. And finally, should the attack be discovered by the wider network, a mitigation phase in which the wider network takes action to remove the malicious authority.

Mitigation Strategies for Sybil Attacks

Some mitigation strategies for these attacks, all revolving around limiting the growth of wealth of voting tokens originating from a single malicious issuer, are further discussed in this section. Their effectiveness is evaluated in section 6.7.

Attenuation Function

The design of an effective universal attenuation function for reputational signals is a first step towards limiting the impact of Sybil attacks. In section 6.7, several attenuation functions, including $f(r) = \frac{2}{1+e^{-0.01r}} - 1$, a logistic function with a small growth rate, are applied. This function is applied to the raw trustworthiness score r, which is computed as the number of encouragement messages from participants minus the number of discouragement messages from participants. This way, the effect of a new encouragement message from a participant on $f(r)$ is generally small if many encouragement messages have been sent previously by other participants.

Temporal Normalisation of Reputational Signals

When an untrustworthy authority joins the network, an attack conducted by them will be characterised by a sudden influx of malicious voting tokens. While an attenuation function can address this, it only affects the market value of voting tokens, and therefore the mechanics of PoS miner selection, but not the reputational signals. This can lead to voting tokens of trustworthy constituents rapidly losing market value. Temporal normalisation is a strategy that seeks to mitigate these sudden changes. This strategy mandates that participant's reputational signals mature in value. Similar to market value normalisation, this strategy can apply any function $f : \mathbb{N} \mapsto [0, 1)$, such as a sigmoid function, to the age of the participant. The age of the participant is defined as the number of steps that have passed since they have first received voting tokens. The result of this normalisation is then multiplied with the nominal value of the voting tokens issued to provide a normalised signal strength $[0, 1]$.

Concepts similar to the proposed normalisation strategy have been discussed in the wider literature, e.g., by Xiong and Liu (2004) who contemplate a reputation system in which more weight is assigned to 'larger, more important, and more recent transactions'. Countermeasures to behaviour fluctuations, e.g., by an attacker who initially behaves in accordance with the protocol to gain more influence later, have been researched in the context of network security and reliability (Srivatsa *et al.*, 2005). In the same domain, the issue of 're-entry' (lowly rated participants leaving a system to later re-enter it under a different identity) has been discussed (Jøsang & Golbeck, 2009), emphasizing the need for strategies like temporal normalisation.

Wealth Ceilings

Assuming an attacker conducts attacks that take the time-normalisation of voting token market values into account, this safeguard alone cannot be effective. When a many Sybil identities are created by one authority, this authority can, as a result, issue numerous voting tokens in very short time. Even when using a defensive temporal normalisation function, their cumulative market value will be significant. Similarly to the unmitigated protocol, this renders the efficacy of temporal normalisation contingent solely on the pace at which accomplice accounts can be funded by the attacking authorities. To pre-empt such attacks, a constituency wealth ceiling that limits the total funds issuable by one authority, can be enforced. This could be a static maximum number of voting tokens issuable by a single authority or a dynamic function that limits the permissible growth of voting tokens depending on the age of the issuing authority. Both can be implemented through ceiling functions $f : \mathbb{N} \mapsto \mathbb{N}$, such as $f(a) = 100 \cdot s$ in the case of a dynamic ceiling or $f(a) = 10,000$ in the case of a static ceiling, where a denotes the steps passed since an authority was first mentioned.

Wealth ceilings are inspired by fiscal policy and can be thought of as extreme forms of progressive taxation (Mirrlees, 1971) that cap a participants potential stake to prevent disproportionate influence. Quadratic voting schemes (Lalley & Weyl, 2018) can be considered de-facto wealth ceilings that ensure each additional unit of wealth yields less additional influence.

Authority Pruning

Once authorities on the network receive increased rates of discouragement messages, they are likely to pose a danger to the system's stability. For this reason, a declining identity authority should inevitably lose all influence over the network, even when the voting tokens distributed by them still hold value in arithmetic terms. To enforce this, an upper bound for the maximum number of discouragement messages can be introduced. Once an authority reaches the threshold, the protocol would fully devalue the voting tokens provided by them.

This approach constitutes a technique of network centrality reduction, a domain researched by Albert *et al.* (2000). They find that scale-free networks – networks characterised by a certain degree distribution (Lenaerts, 2023), such as those IdAPoS would likely exhibit in realistic deployments – are particularly susceptible to the selection and removal of central nodes. It can be assumed that attacker nodes may be considered such central nodes and their removal, or *pruning*, may constitute an attack mitigation strategy.

Factional Trust

S. Chen *et al.* (2024) propose some improvements to the IdAPoS protocol presented within this chapter. One of them is a mitigation strategy in which honest participants establish trust by endorsing authorities introduced by directly connected entities, assuming these belong to the same faction. Thus dynamically forming trust networks that reduce vulnerability to Sybil attacks without relying on off-ledger information. However, this approach could make IdAPoS networks less accessible to newcomers, as it implicitly favours established participants and authorities, thereby creating barriers for entry for entities without existing factional connections. This reflects a broader trade-off between security, decentralisation, and openness in blockchain systems, earlier described by Mssassi and Abou El Kalam (2024).

Algorithm 1 The IdAPoS protocol determines the compound market value of voting tokens per participant.

```
Require: See Table 6.2
  for tx ∈ B do
      if tx is IssuanceTransaction then
          H{tx.recipient}{tx.sender} ← tx.value
      else if tx is Endorsement then
          R{tx.recipient}{tx.sender} ← GetNominalHoldings(tx.sender)
      else if tx is Discouragement then
          R{tx.recipient}{tx.sender} ← − GetNominalHoldings(tx.sender)
      end if
  end for
  V ← ∅
  for p ∈ H.keys do
      vt ← ∅
      for a ∈ H{p}.keys do
          tr ← GetTrustworthiness(a)
          vt ← vt ∪ H{p}{a} · tr
      end for
      V{p} ← mean(vt)                ▷ x̄ denotes the mean of values in x
  end for
  return V
  procedure GetNominalHoldings(p)
      nh ← ∅
      for h ∈ H{p}.keys do
          nh ← vt ∪ H{p}{h}
      end for
      return mean(nh)                ▷ x̄ denotes the mean of values in x
  end procedure
  procedure GetTrustworthiness(a)
      tr ← 0
      for r ∈ R{a}.keys do
          tr ← tr + R{a}{r}
      end for
      if tr < 0 then
          return 0
      else
          return Attenuate(tr)       ▷ An attenuation function, e.g. 2/(1+e^(−0.01tr)) − 1
      end if
  end procedure
```

6.6 Model

The attackability of IdAPoS can be evaluated using an agent-based approach. While approaching it through mathematical proof might provide a more reliable answer, agent-based modelling allows framing Sybil attacks on IdAPoS as an emergent phenomenon, the effects of which are not derivable from the inspection of the model parameters alone. This allows for the relaxing of assumptions that may be difficult to accurately specify in the context of mathematical analysis (Bankes, 2002). For this, a system model (see subsection 3) is designed and implemented in the 'Kotlin' programming language. The system model is executed stepwise with steps representing blocks. At the end of a step, a number of proposed transactions are randomly sampled from the memory pool and applied to the ledger atomically, similar to how transactions would be included in a block through mining in an actual blockchain system.

Bankes, S. C. (2002). Agent-based modeling: A revolution? *Proceedings of the National Academy of Sciences*, *99*(Supplement 3)

Environmental Values Table 6.3 shows the environmental parameters of the system simulation along with the values assumed. Apart from the value for the block size c, that roughly aligns with the average number of transactions per block on the Bitcoin blockchain, the environmental parameters are curated to provide an *illustrative* environment for a simulation, meaning, they should provide underlying conditions in which mitigation strategies are observable well. In extremis, i.e. where a feeble attacker (characterised by low g_m, me, mn, md) is met by a potent group of existing vigilant constituencies (characterised by high g_h, lf, p_{vig}, p_{he}, p_{ho}, p_{hd}), or vice versa, the simulation cannot produce meaningful outputs. This stems from the very short attack phases scenarios with such parameters suffer from that make it impossible to meaningfully analyse the interplay between opposing actors.

In addition to the environmental parameters which, in a system operating in the wild, would be dictated by the environment it operates in, several aspects of an IdAPoS system are explicitly configurable as part of the protocol. Table 6.4 shows the parameters with which the simulation can be configured, as well as the variations of candidate values that are tested during simulation. Section 6.7 describes in more detail how combinations of the configurable parameters influence the system's resistance to Sybil attacks.

The outputs of the simulation, the ledger, are written to a log file[3]. The model avoids randomness where possible. Where unavoidable, it uses seeded pseudorandom number generators (PRNG) to make the results of the simulation deterministic and reproducible. PRNG are used where the model demands probabilistic selection, i.e. p_{vig}, p_{he}, p_{ho} and p_{hd} as described in Table 6.3, as well as random selection from the memory pool.

3: The log files for the different mitigation strategies are published as supplementary dataset (Platt, 2021b).

System Simulation

The simulation (see Algorithm 2) is initialised with an empty ledger, an empty memory pool and an empty set of participants. It follows the sequence of instructing the agents to act (Act), which results in those agents emitting transaction proposals to the memory pool, and subsequently selecting a number of transaction proposals from said pool randomly. The selected proposals are then committed to the ledger (Mine). Any effects of the selected proposals on the constituency of the system, i.e.

Variable	Description	Value
lf	The number of steps the formation phase in which honest constituencies build up is comprised of	200
li	The number of steps the attack remains undiscovered is comprised of	5
la	The number of steps the mitigation phase in which the malicious activity is discovered is comprised of	100
c	The number of transactions a block holds	3000
p_{vig}	The probability with which an honest participant is created as a 'vigilant' participant, i.e. they discourage malicious authorities	2%
vig	The number of malicious authorities a 'vigilant' participant tries to discourage in a step	5
p_{he}	The probability with which an honest participant endorses a new honest authority in a step	0.01%
p_{ho}	The probability with which an honest participant endorses their own authority in a step	2%
p_{hd}	The probability with which an honest participant discourages their own authority in a step	0.1%
g_h	The growth rate of an honest constituency (endorsement messages emitted/period)	2
g_m	The growth rate of a malicious constituency (endorsement messages emitted/period)	10
me	The number of endorsement messages a malicious participant emits in support of existing malicious authorities	5
mn	The number of endorsement messages a malicious participant emits in support of new malicious authorities	2
md	The number of discouragement messages a malicious participant emits towards existing honest authorities	3

Table 6.3: Environmental parameters of the system simulation.

Value	Description	Domain
at(r)	The attenuation function that is applied to the sum of reputational values (r)	$f : \mathbb{R}_+ \mapsto [0, 1]$
tn(a)	The temporal normalisation function that dampens the value of reputational signals depending on the age of the emiter (a)	$f : \mathbb{N} \mapsto [0, 1)$
wc(a)	The wealth ceiling that is applied to the total issuable amount of voting token as a function of the issuer's age (a)	$f : \mathbb{N} \mapsto \mathbb{N}$
pr	Whether authority pruning is applied	$\mathbb{B}$

Table 6.4: Configurable aspects of IdAPoS and their domains.

where new participants or authorities are introduced, will be registered by the simulation. New agents will be created accordingly and references to them will be stored within the set of participants. Starting with the following iteration, these newly created agents will participate in the simulation and will be called during the Act phase of a step.

Within the simulation programme, assumptions are made regarding both the environmental parameters and the strategies that the agents pursue. Strategies that agents adopt are based on intuition and no work has been done to optimise strategies of either type of agent. Despite this, the simulation–even where reasonable but suboptimal strategies are employed–should allow for the effectiveness of countermeasures to be compared. The system model is free of any strategic considerations and merely acts as a layer to orchestrate agent actions. Furthermore, it provides data structures to hold all transactions and, in the interest of optimisation, additional data structures that provide convenient access to voting tokens holdings and reputational signals. These can be thought of as caches.

Memory Pool A Sybil attack on an IdAPoS system is accompanied by an explosion in the number of transactions proposed. Common blockchain protocols, however, impose significant limits on the number of transactions included in a block. Optimal block sizes in PoW systems are severely limited (S. Jiang & Wu, 2019). This limitation is imitated in the simulation by introducing the concept of a naïve memory pool that stores proposed transactions locally until they are included in a valid block. Since considerations of the economic properties of transaction mining are beyond the scope of this chapter, a simplified strategy for selecting transactions from this pool is applied: In each step of the protocol, transactions, up to the 'block size' defined, are chosen randomly from the set of uncommitted transactions.

Jiang, S., & Wu, J. (2019). Bitcoin mining with transaction fees: A game on the block size. *Proceedings of the 2019 International Conference on Blockchain*

Agents

To inform their actions, all agents have visibility of the current ledger (L). Through this, they have knowledge of all participants, along with their voting tokens holdings, and all authorities, along with the reputational signals received by them. They also understand whether their co-actors are honest or malicious. Based on this information, they execute a predefined strategy, encoded in their respective Act methods, which leads to a number of transaction proposals being emitted to the memory pool.

Honest Authorities Honest authorities facilitate a steadily growing constituency by following a strategy in which they issue a steady stream of voting tokens to new honest participants up to the maximum number of messages configured (see Algorithm 3).

Malicious Authorities Malicious authorities follow a strategy (see Algorithm 4) in which they issue voting tokens to new malicious participants up to the maximum number of messages configured.

A more complex strategy might provide benefits in a scenario where the malicious nature of an actor is not easily discoverable and actors have to analyse the behaviour of their co-actors to it. In such a scenario, malicious

Algorithm 2 The algorithm simulating an IdAPoS system.

```
Require: See Table 6.3
  i ← 0
  L ← ∅                      ▷ The ledger containing committed transactions
  M ← ∅        ▷ The memory pool containing uncommitted transactions
  A ← ∅                            ▷ The agent instances in the simulation
  M ← GenesisTx
  Mine
  repeat                  ▷ Formation phase, attack phase, mitigation phase
      Act
      Mine
      i ← i + 1
  until i ≥ lf + li + la
  procedure Mine
      txs ← RandomSample(c, M)  ▷ Picks c elements from M randomly
      for tx ∈ txs do
          L ← L ∪ tx
          if tx is GenesisTransaction then
              A ← A∪ HonestAuthority()
          else if tx is IssuanceTransaction then
              if tx.sender = honest then
                  A ← A∪ HonestParticipant()
              else
                  A ← A∪ MaliciousParticipant()
              end if
          else if tx is ReputationalTransaction then                ▷
  Endorsement/discouragement
              if tx.sender = honest then
                  A ← A∪ HonestAuthority()
              else
                  A ← A∪ MaliciousAuthority()
              end if
          end if
      end for
      M ← ∅
  end procedure
  procedure Act
      for a ∈ A do
          M ← M∪ A.Act
      end for
  end procedure
```

Algorithm 3 The programme an agent representing an *honest* authority executes.

```
Require: The current ledger L, the current memory pool M, and all
  configuration parameters described in Table 6.3
  procedure Act
      i ← 0
      repeat
          M ← M∪ Issuance(HonestParticipant())
          i ← i + 1
      until i ≥ g_h
  end procedure
```

Algorithm 4 The programme an agent representing a *malicious* authority executes.

Require: The current ledger L, the current memory pool M, and all configuration parameters described in Table 6.3
  procedure ACT
    $i \leftarrow 0$
    repeat
      $M \leftarrow M \cup$ Issuance(MaliciousParticipant())
      $i \leftarrow i + 1$
    until $i \geq g_m$
  end procedure

authorities might disguise as honest ones and issue an artificially low number of tokens to avoid being discovered.

Honest Participants Honest participants are mostly passive. Their strategy is defined based on the assumption that they make use of an IdAPoS system for their individual purposes but are less concerned with participating in its governance. Algorithm 5 shows how they infrequently endorse authorities, and, if they are instantiated as 'vigilant' actors, how they discourage malicious authorities. Honest participants don't explicit coordinate but execute the described strategy, a shortcoming that could be improved upon by modelling their strategy as a flag coordination game (Marzagao, 2019)

Marzagao, D. K. (2019). *Flag coordination games* [Doctoral dissertation, King's College London]

It is not specified how honest participants become aware of malicious activity on the network. Conceivably, in an actual implementation of a protocol, they would become aware through off-ledger communication. It is also not modelled here how an honest participant would form an understanding of whether another actor on the network is trustworthy or malicious. In an actual implementation of the protocol, where this information would have to be derived from the ledger alone, identification of malicious actors would be imprecise. Imitating this behaviour in the context of a simulation requires deep game-theoretical consideration that goes beyond the scope of this chapter.

Malicious Participants Malicious participants follow a strategy in which they emit messages endorsing existing and new malicious authorities as well as discouraging honest authorities (see Algorithm 6). The number of messages emitted is defined in environmental parameters. Malicious participants, just like honest ones, do not coordinate, although the former represent a centralised group. A future extension of the presented model could therefore be to consider it as an example of a system in which centralised and decentralised groups compete with each other (Agamennone, 2020).

Agamennone, M. (2020, December). *Riots and uprisings: Modelling conflict between centralised and decentralised systems* [Doctoral dissertation, King's College London]

This strategy is assumes that malicious agents do not communicate directly. Conceivable improvements would entail selecting endorsing authorities with high trustworthiness instead of picking the authorities to be endorsed randomly.

Algorithm 5 The programme an agent representing an *honest* participant executes.

Require: The current ledger L, the current memory pool M, and all configuration parameters described in Table 6.3
  procedure ACT
    shouldEndorseNew ← GETBOOLEAN(P_{eh}) ▷ Returns `True` with $P = P_{eh}$
    shouldEndorseOwn ← GETBOOLEAN(P_{eo})
    shouldDiscourageOwn ← GETBOOLEAN(P_{ed})
    isVigilant ← GETBOOLEAN(P_{vig})
    if shouldEndorseNew **then**
      $M \leftarrow M \cup$ Endorsement(HonestAuthority())
    end if
    if shouldEndorseOwn **then**
      $M \leftarrow M \cup$ Endorsement(ownAuthority)
    end if
    if shouldDiscourageOwn **then**
      $M \leftarrow M \cup$ Discouragement(ownAuthority)
    end if
    if isVigilant **then**
      MA ← MALICIOUSAUTHORITIES(L)
      targets ← RANDOMSAMPLE(vig, MA)
      for $target \in targets$ **do**
        $M \leftarrow M \cup$ Discouragement(target)
      end for
    end if
  end procedure

Algorithm 6 The programme an agent representing a *malicious* participant executes.

Require: The current ledger L, the current memory pool M, and all configuration parameters described in Table 6.3
  procedure ACT
    $M \leftarrow$ GETMALICIOUSAUTHORITIES(L)
    $mas \leftarrow$ RANDOMSAMPLE(mn, M) ▷ Picks c elements from M randomly
    for $ma \in mas$ **do**
      $M \leftarrow M \cup$ Endorsement(ma)
    end for
    $H \leftarrow$ GETHONESTAUTHORITIES(L)
    $has \leftarrow$ RANDOMSAMPLE(md, H)
    for $ha \in has$ **do**
      $M \leftarrow M \cup$ Discouragement(ha)
    end for
    $i \leftarrow 0$
    repeat
      $M \leftarrow M \cup$ Endorsement(MaliciousAuthority())
      $i \leftarrow i + 1$
    until $i \geq mn$
  end procedure

6.7 Results

A metric to evaluate the effectiveness of mitigation strategies is the sum of market values of voting tokens (*mv*) that participants hold. This allows to compare the influence that both honest and malicious participants have on the mining process. A system can be considered overwhelmed at the latest once the sum of market values of voting tokens held by malicious participants exceeds that of voting tokens held by honest participants (i.e., the total value held by malicious participants exceeds 1/2). While this limit can be considered to be optimistic under the assumption that common proof-of-stake protocols require 2/3 of the network to be in agreement in order for it to remain deadlock-free (Thin *et al.*, 2018), 1/2 can serve as a firm upper bound. The values are evaluated per protocol step (*s*), i.e. after selecting and, subsequently, applying transaction proposals selected from the memory pool.

Thin, W. Y. M. M., Dong, N., Bai, G., & Dong, J. S. (2018). Formal analysis of a proof-of-stake blockchain. *Proceedings of the 23rd International Conference on Engineering of Complex Computer Systems*

Table 6.5 shows how the different mitigation strategies, described in more detail later, influence the takeover time of an IdAPoS-based system. An 'unmitigated' simulation run serves as the baseline to compare strategies against. This is the result of IdAPoS consensus as formalised in Algorithm 1 being executed.

Unmitigated Protocol

The visualisation of the sum of market values of voting tokens provided in Figure 6.5 shows that, once the attack commences, the market value of honest participants' voting tokens declines steeply while the value of voting tokens held by malicious actors appears to grow linearly.

When no mitigation strategies are applied, the system can be taken over within a brief period prior to the attack being discovered (see Table 6.3). This confirms the intuitive assumption made earlier (see subsection 2) that IdAPoS is highly vulnerable to Sybil attacks if no mitigation strategies are applied.

Temporal Normalisation

To evaluate the temporal normalisation strategy, the effect of a temporal normalisation function $f(r) = \frac{2}{1+e^{-d \cdot r}} - 1$ with varying growth rates d is simulated. Table 6.5 shows that $d = 0.0001$ yielded the best results. Here, t represents the age of the holder which is defined as the number of periods between the holder receiving the first endorsement message and

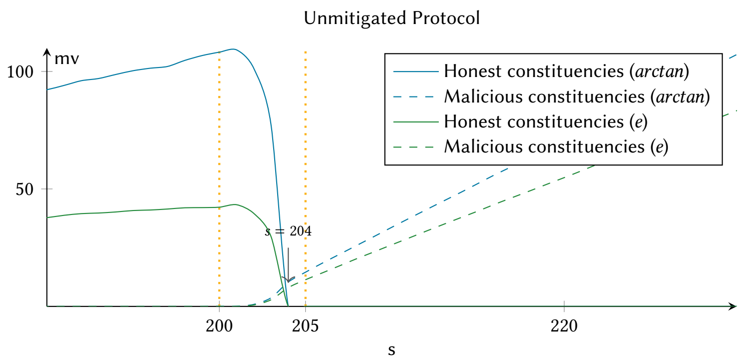

Figure 6.5: After 204 periods (*s*) the market value (*mv*) of tokens held by malicious authorities exceeds that held by honest ones.

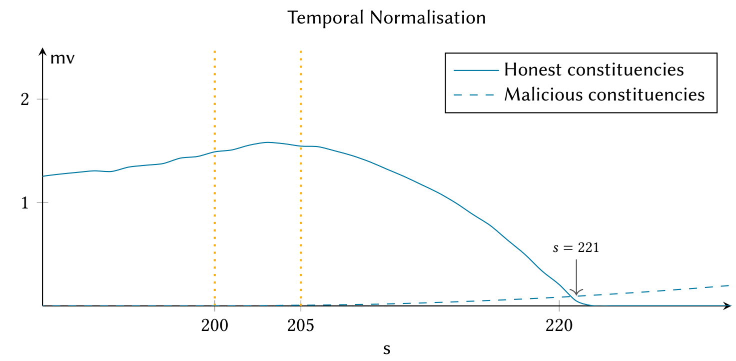

Figure 6.6: When applying temporal normalisation with the temporal normalisation function $f(r) = \frac{2}{1+e^{-0.0001r}} - 1$ only after 221 periods (s) the market value (mv) of tokens held by malicious authorities exceeds that held by honest ones.

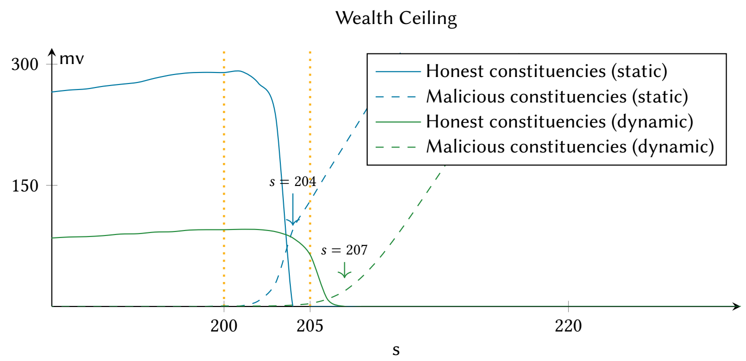

Figure 6.7: The simulation of static and dynamic wealth ceiling strategies shows that static strategies are less effective, with $wc(a) = 10$ leading to the system being overwhelmed after only 204 periods (s) when the market value (mv) of tokens held by malicious authorities exceeds that held by honest ones. Dynamic strategies (e.g. $wc(a) = a$) lead to minor improvements in system stability.

the current step. This modifier is subsequently applied to all reputational signals emitted by the holder.

The diminishing effect this has on the sum of market values of voting tokens held by honest participant is evident from the plot shown in Figure 6.6. In the context of temporal normalisation, the attenuation function used is of particular importance. When using a logistic function $f(x) = \frac{1}{1+e^{-k(x-x_0)}}$ as attenuation function, as in the experiment, the logistic growth rate influences the point of overwhelming the system. A steeper curve will lead to a system being overwhelmed more quickly, while a flatter curve will push out the time at which the system is overwhelmed. In the extreme case ($k = 0$) will never be overwhelmed, but new constituencies will not be able to join the system. Consequently, where $k = \infty$, temporal normalisation will have no effect.

Wealth Ceilings

Wealth ceilings define the maximum number of voting tokens an issuer can bring into circulation up to a certain age t. Here, t is defined as the number of steps passed between the authority receiving their first endorsement message and the current step. Ten experiments are executed with different modalities of wealth ceilings: First, static wealth ceilings are applied, second dynamic wealth ceilings. Figure 6.7 shows how the choice of wealth ceiling parameters influences takeover time. While the most severely limiting constant function tested ($wc(a) = 10$) has a negative effect on takeover time, likely to due the capping of the growth of the trustworthy authority in the formation phase, dynamic wealth ceilings, particularly those that map directly to the constituency size (i.e. $wc(a) = a$), have a small positive effect.

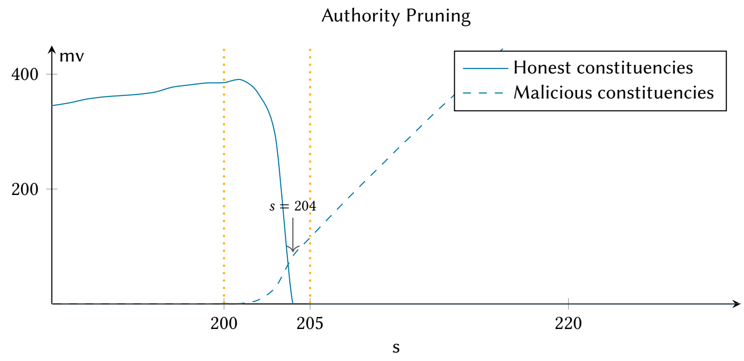

Figure 6.8: Authority pruning does not provide a measurable improvement over the unmitigated protocol: similar to the this, the market value (*mv*) of tokens held by malicious authorities exceeds that held by honest ones after only 204 periods (*s*).

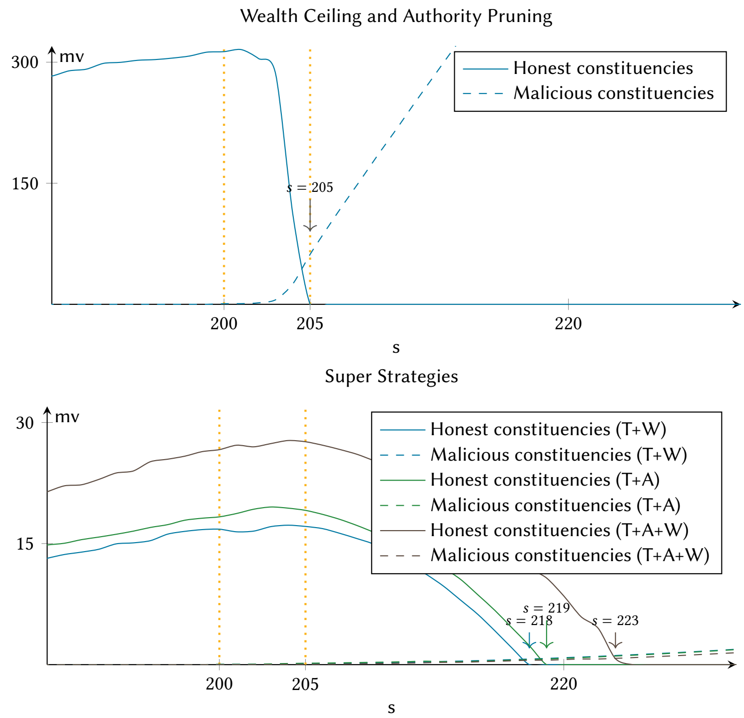

Figure 6.9: Combining the individual mitigation strategies of temporal normalisation (*T*), wealth ceilings (*W*) and authority pruning (*A*) creates super strategies that can prolong the time (*s*) it takes until the market value (*mv*) of tokens held by malicious authorities exceeds that held by honest ones.

Authority Pruning

A threshold is reached where discouragement messages an authority has received exceed the number of endorsement messages an authority has received. During each step, it can be calculated whether this threshold has been reached per authority. Should an authority cross it, it will be removed from the pool of actors. Figure 6.8 shows how this strategy in its current form has no effect on takeover time. This is likely because trustworthy authorities have already received large numbers of endorsement messages before the start of the attack.

Combining Strategies

The strategies previously presented in isolation are not mutually exclusive. Therefore, 'super strategies' can be devised. To create those, representative configurations of strategies previously tested are combined. Figure 6.9 shows how super strategies improve the performance of the underlying individual strategies.

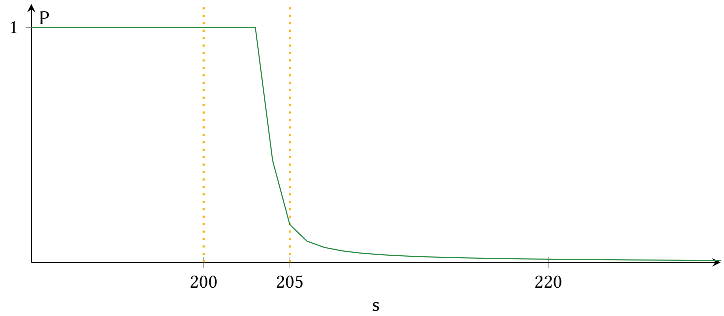

Figure 6.10: The probability P of a message submitted into the naïve memory pool being mined decreases steeply after an attack commences in step $s = 200$.

The most performant super strategy simulated is the 'TAW' (temporal normalisation, authority pruning, wealth ceiling) strategy. Here, a logistic attenuation function ($at(r) = \frac{2}{1+e^{-0.01r}} - 1$) is used in combination with temporal normalisation ($tn(a) = \frac{2}{1+e^{-0.001a}} - 1$), authority pruning, and a dynamic wealth ceiling ($wc(a) = 4a$).

Memory Pool Oversaturation

Simulation reveals that the number of messages submitted to an IdAPoS system under attack is very large. This is due to malicious participants aggressively endorsing accomplices. As discussed before (see subsection 3), the simulation imitates common blockchain protocols by severely limiting the maximum number of transactions included in a block.

As shown in Figure 6.10, this leads to the vast majority of transactions proposed to the memory pool being discarded in the later phases of an attack. The probability of a proposed transaction being executed in a block is still 100% in step 200 when the attack commences, but drops to under 1% by step 227. While economic strategies for miners that lead to including some transactions while excluding others have been researched in the context of cryptocurrencies (Eyal & Sirer, 2014; Luu *et al.*, 2015), the simplistic random selection method applied to the naïve memory pool does not consider these. Transactions that are not selected by the end of the simulation are discarded.

Table 6.5: Comparison of mitigation strategies (S) using temporal normalisation functions (*tn*(a)), wealth ceiling functions (*wc*(a)), authority pruning (*pr*) and various attenuation functions (*at*(r)). The table shows when the attack started (p_s), when the system was overwhelmed (p_o) and the number of periods in between (Δ).

S	$at(\mathbf{r})$	$tn(\mathbf{a})$	$wc(\mathbf{a})$	pr	p_s	p_o	Δ
i	$\frac{2}{1+e^{-0.1r}}-1$	-	-	no	200	203	3
	$\frac{2}{1+e^{-0.01r}}-1$	-	-	no	200	204	4
	$\frac{2}{1+e^{-0.001r}}-1$	-	-	no	200	204	4
	$\frac{2}{1+e^{-0.0001r}}-1$	-	-	no	200	204	4
	$\frac{2}{1+e^{-0.00001r}}-1$	-	-	no	200	204	4
	$\frac{2arctan(0.1r)}{\pi}$	-	-	no	200	204	4
	$\frac{2arctan(0.01r)}{\pi}$	-	-	no	200	204	4
	$\frac{2arctan(0.001r)}{\pi}$	-	-	no	200	204	4
	$\frac{2arctan(0.0001r)}{\pi}$	-	-	no	200	204	4
	$\frac{2arctan(0.00001r)}{\pi}$	-	-	no	200	204	4
ii	$\frac{2}{1+e^{-0.01r}}-1$	$\frac{2}{1+e^{-0.1a}}-1$	-	no	200	210	10
	$\frac{2}{1+e^{-0.01r}}-1$	$\frac{2}{1+e^{-0.01a}}-1$	-	no	200	217	17
	$\frac{2}{1+e^{-0.01r}}-1$	$\frac{2}{1+e^{-0.001a}}-1$	-	no	200	218	18
	$\frac{2}{1+e^{-0.01r}}-1$	$\frac{2}{1+e^{-0.0001a}}-1$	-	no	200	221	21
	$\frac{2}{1+e^{-0.01r}}-1$	$\frac{2}{1+e^{-0.00001a}}-1$	-	no	200	217	17
iii	$\frac{2}{1+e^{-0.01r}}-1$	-	10	no	200	202	2
	$\frac{2}{1+e^{-0.01r}}-1$	-	100	no	200	203	3
	$\frac{2}{1+e^{-0.01r}}-1$	-	1000	no	200	204	4
	$\frac{2}{1+e^{-0.01r}}-1$	-	10000	no	200	204	4
	$\frac{2}{1+e^{-0.01r}}-1$	-	100000	no	200	204	4
iv	$\frac{2}{1+e^{-0.01r}}-1$	-	a	no	200	207	7
	$\frac{2}{1+e^{-0.01r}}-1$	-	$2a$	no	200	206	6
	$\frac{2}{1+e^{-0.01r}}-1$	-	$4a$	no	200	205	5
	$\frac{2}{1+e^{-0.01r}}-1$	-	$8a$	no	200	204	4
	$\frac{2}{1+e^{-0.01r}}-1$	-	$16a$	no	200	204	4
v	$\frac{2}{1+e^{-0.01r}}-1$	-	-	yes	200	204	4
vi	$\frac{2}{1+e^{-0.01r}}-1$	-	4a	yes	200	205	5
vii	$\frac{2}{1+e^{-0.01r}}-1$	$\frac{2}{1+e^{-0.001a}}-1$	$4a$	no	200	218	18
viii	$\frac{2}{1+e^{-0.01r}}-1$	$\frac{2}{1+e^{-0.001a}}-1$	-	yes	200	219	19
ix	$\frac{2}{1+e^{-0.01r}}-1$	$\frac{2}{1+e^{-0.001a}}-1$	$4a$	yes	200	223	23

6.8 Conclusion

Permissionless blockchain technology, by virtue of being trustless and therefore not relying on central authorities, is an appropriate building block to enable self-governance of systems, small and large. Yet, established consensus protocols for such systems do not exist, and common protocols, albeit popular, can be considered inherently undemocratic. IdAPoS targets this shortcoming, but is highly susceptible to Sybil attacks. In this chapter we apply agent-based simulation to show that mitigation strategies exist that can effectively prolong the time it takes for an attacker to take over an IdAPoS system. Temporal normalisation, a technique through which established participants are given more weight, proved to be the most effective individual mitigation strategy simulated. Individual strategies were outperformed by *TAW*, a super strategy that combines temporal normalisation with authority pruning, a strategy that removes authorities that have received predominantly negative feedback, and wealth ceilings that limit the total number of identities a single authority can control. Using this super strategy, the system was able to withstand an attack for significantly longer (23 periods instead of 4 periods in the unmitigated protocol). The findings of this chapter may be considered a further validation of the viability of identity-based consensus protocols.

A Sybil Attack Vulnerability Trilemma

7

This chapter increases the analytical depth from chapter 6, which employed an agent-based modelling approach, by more rigorously modelling the problem of Sybil attack vulnerability. Fault tolerance is crucial for both centralised and distributed computer systems, yet it is particularly vital for the latter: in distributed systems, reliable orchestration of shared computing resources over communication networks is essential (Rennels, 1980), as all distributed systems may, at times, face physical or human-made faults of varying durations and extents (Avižienis, 1983). Generally, such systems can withstand a maximum of 1/3 faulty participants (Pease *et al.*, 1980), a limit that is sufficient to allow the operation of common centrally governed systems in which the number of – potentially faulty – participants can be limited by preselection.

Rennels. (1980). Distributed fault-tolerant computer systems. *Computer*, *13*(3)

Blockchain technology introduced a new class of systems, secure decentralised systems, by allowing public and permissionless access and thus eliminating the need for a dominant operator. Due to this change in permissioning and other technological innovations, secure decentralised blockchain systems provided some novel benefits; namely, resistance to censorship, immutability, and pseudonymity. This led some to consider them suitable systems for democratic or participatory decision-making. However, with these benefits came vulnerabilities, particularly to 'Sybil attacks' (Douceur, 2002) in which 'a single faulty entity [...] present[s] multiple identities [so] it can control a substantial fraction of the system' (Douceur, 2002)[1]. As the first large-scale system of its kind, Bitcoin (Nakamoto, 2008) counteracted Sybil attacks by applying a PoW mechanism in which the likelihood of being selected as a system validator is proportional to the computational effort invested.

1: See subsection 1 and Definition 7.1.4 for a more comprehensive description.

In response to challenges around energy demand (Sedlmeir *et al.*, 2020b), security (R. Zhang & Preneel, 2019), fairness (Poux *et al.*, 2022), performance (Gervais *et al.*, 2016), and suitability for democratic processes (Racsko, 2019) of PoW, numerous 'second generation' blockchain systems have been proposed (De Angelis *et al.*, 2023). *Permissionlessness*, i.e. 'free entry' (Gans & Gandal, 2019), remained a central characteristic for these systems. *Sybil attack resistance* was equally important for such systems, even though this characteristic was often implied, as without it, permissionless systems cannot function effectively. A third characteristic many new blockchain systems strove for was *freeness*, i.e. the absence of an 'explicit monetary cost' (Saleh, 2020, p. 1157) to participation.

While all three characteristics are desirable, it is unclear whether they are achievable: observations of existing blockchain systems, particularly around tendencies towards centralisation (Sai *et al.*, 2021) and documented instances of Sybil attacks (Z. Yang *et al.*, 2014), cast doubt on the feasibility of a system embodying all three characteristics. This research gap is particularly relevant to designers of public blockchain systems, who need to develop protocols accordingly. We therefore set out to investigate the compatibility of permissionlessness, Sybil attack resistance, and freeness in blockchain systems. We do so by providing background on blockchain technology, with a particular focus on consensus mechanisms (see subsection 1). We then define a formal model of a blockchain system (see section 7.1) that culminates in an impossibility theorem (see subsection 7.1) and apply it to some real-world blockchain systems (see section 7.2). We close with conclusions (see section 7.3). Examples from

Sai, A. R., Buckley, J., Fitzgerald, B., & Gear, A. L. (2021). Taxonomy of centralization in public blockchain systems: A systematic literature review. *Information Processing & Management*, *58*(4), Article 102584 the blockchain space used in subsection 1 and section 7.2 are provided to illustrate practical applications of the trilemma for reader comprehension, without claiming comprehensive coverage of all aspects of these systems. We recognise that these examples may extend beyond immediate definitions and assumptions.

Background

As early as the 1960s, the foundations for distributed systems research were laid in the form of multiprocess networking theory (Kleinrock, 1961). It has since evolved into a robust field that focusses on improving system reliability by developing and integrating methods to detect, mask and recover from operational and design faults (Aviziens, 1976). Contrary to blockchain technology, in early distributed systems, a *central* authority controlled the admission of participants through attribute-based access control mechanisms (V. C. Hu *et al.*, 2018) and by encoding their permissions in policies (Bacon & Moody, 2004).

Kleinrock, L. (1961, May 31). *Information flow in large communication nets* [PhD Thesis Proposal]. Massachusetts Institute of Technology

Motivations for Decentralisation and Impact Beyond Finance

Blockchain systems, which are positioned as *decentralised* transactional technologies that can facilitate the censorship-resistant replication of pseudonymous data between distrusting peers, potentially even without human intervention (Hartwich *et al.*, 2023), allow the preservation of system history without exposure to the risk of nation-state censorship or regulatory overreach (Rocco, 2019). They were, therefore, welcomed by those who wanted to evade governmental influence (Jarvis, 2021) and by those who sought a decentralised payment mechanism that contests traditional financial hierarchies (Bohr & Bashir, 2014). Through cryptocurrencies, blockchain has gained widespread adoption, serving as the underlying technology for both speculation (Glaser *et al.*, 2014) and payments (Jonker, 2019), thereby fulfilling the traditional functions of money at least partially (Gurguc & Knottenbelt, 2018). Blockchain technology is commercially leveraged in trading platforms and payment applications for retail and institutional audiences (Beinke *et al.*, 2018). Due to their decentralised and, therefore, difficult-to-regulate nature (Aquilina *et al.*, 2024), blockchains, particularly when applied to permissionless cryptocurrencies, continue to attract criticism for their role in facilitating the financing of criminal activities (Piazza, 2017; Trozze *et al.*, 2022). The use of this technology, however, extends beyond the financial realm: some notable applications are evident in fields such as IoT (particularly adhoc networks Hamdan *et al.*, 2019), healthcare, energy, public services, artificial intelligence, and big data (Paulavičius *et al.*, 2019).

Hartwich, E., Rieger, A., Sedlmeir, J., Jurek, D., & Fridgen, G. (2023). Machine economies. *Electronic Markets*, *33*(1), Article 36

Blockchain Characteristics

Tai *et al.* (2017) characterise blockchain systems as symmetric, admin-free, ledgered, and time-consensual. These key characteristics are useful to differentiate blockchain systems from prior, centrally governed, distributed systems. Most relevant for Sybil attacks is the *admin-free* property: it describes systems that possess 'no concept of a system administrator who is responsible for maintenance, infrastructure provisioning or access control' (Tai *et al.*, 2017, p. 758). Tai *et al.* (2017, p. 758), furthermore, state that updates to a blockchain system happen 'on an individual basis [governed by] community consensus'. Although, as described earlier, such

Tai, S., Eberhardt, J., & Klems, M. (2017). Not ACID, not BASE, but SALT - a transaction processing perspective on blockchains. *Proceedings of the 7th Conference on Cloud Computing and Services Science*

	Permissioned	**Permissionless**
Public	*i*	*ii*
Private	*iii*	*iv*

Table 7.1: Tezel *et al.* (2020, p. 549) categorise four archetypes of blockchain architectures.

consensus was not necessary for previous distributed systems due to the availability of centralised controls, it becomes critical in open systems where these controls are absent. Therefore, it is evident that, for successful community consensus, the group forming the community must be known and that this group, by and large (Tholoniat & Gramoli, 2022), must follow accepted rules to reach consensus.

Blockchain systems are commonly classified along the anonymity and trust continuums (see Table 7.1). From an anonymity perspective, they can be classified as public, displaying data openly, or private, with centrally enforced access controls. From a trust perspective, they are either permissioned, with pre-selected validators, or permissionless[2], if participation in the use, development, and governance of the system is possible without needing permission from an authority, simply by following publicly stated procedures (Nabben & Zargham, 2022).

2: Notwithstanding Definitions 7.1.2 and 7.1.3.

Therefore, the many permissioned blockchain systems (types *i* and *iii* in Table 7.1), in which only a limited and controlled group of participants can take part (Nadir, 2019; Polge *et al.*, 2021; Valenta & Sandner, 2017), are not of interest in the context of this chapter: we focus on permissionless systems (types *ii* and *iv* in Table 7.1), as only these are commonly threatened by Sybil attacks.

Sybil Attack Resistance

All distributed systems, but particularly permissionless systems, face the consensus problem: 'a problem in distributed computing wherein nodes within the system must reach an agreement given the presence of faulty processes or deceptive nodes' (Bach *et al.*, 2018, p. 1545). Malicious participants may worsen this problem via Sybil attacks, where large numbers of bogus processes are created to overpower genuine ones. Under the assumption that the number of malicious processes executed by an attacker is not limited, achieving consensus in systems under Sybil attacks is not possible using simple voting-based techniques (Anceaume *et al.*, 2021; Meir *et al.*, 2022). Specifically leader-based mechanisms, such as Practical Byzantine fault tolerance (Castro & Liskov, 1999) or Raft (Ongaro & Ousterhout, 2014), while well suited to centrally managed distributed systems, are ineffective in public and permissionless systems, where an attacker can create arbitrary processes. There are many models to reach agreements in the presence of Sybil attacks (see chapter 2), with PoW being the most popular.

Ongaro, D., & Ousterhout, J. (2014). In search of an understandable consensus algorithm. *Proceedings of the 2014 USENIX Technical Conference*

Proof of Work Bitcoin (Nakamoto, 2008) introduced PoW, a mechanism that fundamentally employs an 'efficiently verifiable, but parameterisably expensive to compute' (Back, 2002), cryptographic puzzle to prevent Sybil attacks. Solving this puzzle entitles the solver to carry forward the system log as a validator and grants them a cryptocurrency reward that is automatically distributed. This ensures that those investing significant computational effort are selected as validators, irrespective of the total number of candidates, thereby making Sybil attacks ineffective. Furthermore, the underlying reward mechanism ensures the anonymity of system validators in data replication, while simultaneously disincentivising them from consolidating (Leshno & Strack, 2019). This approach to consensus and data replication has been remarkably effective for maintaining system stability but requires the expenditure of large amounts of electricity (Sedlmeir *et al.*, 2020b) and is thus facing resistance (see chapter 3).

Leshno, J., & Strack, P. (2019, October). *Bitcoin: An impossibility theorem for proof-of-work based protocols* (Cowles Foundation Discussion Paper No. 2204R). Yale University. New Haven, CT, USA

Alternative Consensus Mechanisms Consequently, alternative blockchain systems have been introduced. These often employ less resource-consumptive consensus mechanisms (see chapter 2 and chapter 3), thereby lending themselves to sustainable innovation (Treiblmaier, 2024). Some of these novel systems make participation free by relying on reputation systems to determine the probability of being selected as a system validator (Z. Li *et al.*, 2022). Although these novel systems have demonstrated the ability to withstand small-scale Sybil attacks (see chapter 6), there are doubts about their resistance to Sybil attacks in fully permissionless contexts (see chapter 2). We aim to address such doubts with this work.

Consensus Mechanisms as Votes on System State

Both proof of resources and majority voting can be considered forms of votes on the canonical state of a decentralised system. For example, a simple PoW scheme, like the one used by Bitcoin, can be considered a probabilistic weighted voting scheme: in it, the voting power of a participant aligns with how much computational effort they invest into solving the PoW puzzle. A miner contributing 1/10 of the system-wide computational effort for solving a given PoW puzzle would be selected as a block proposer with an approximate likelihood of 10 %. Likewise, in a simple PoA scheme based on random miner selection, the likelihood of being selected as a block proposer for a system with 10 participants is 10 %.

The concept of *blocks* is central to drawing parallels between blockchain consensus and voting. Blocks are compiled by miners, who gather transactions over time. Blocks, furthermore, group 'a set of transactions [and are] used as the unit of consensus' (Tai *et al.*, 2017, p. 759) in blockchain systems. Since the system state of a blockchain is built up by monotonically chaining such blocks: every instance of a block proposal that is subject to a vote contributes to the overall system state, and thus to the decision on the version of events that is to be considered canonical. Therefore, even if common consensus protocols do not purposively implement democratic principles, the mechanisms they apply to derive system state may show strong similarities with voting processes.

Consensus and Social Choice Theory

Based on the realisation presented in the previous section, it can be shown that findings concerning collective decision-making relate to blockchain consensus mechanisms: in particular, questions of social choice theory and blockchain consensus have an overlap. For example, Arrow's impossibility theorem (Arrow, 1950) and the Gibbard–Satterthwaite theorem (Gibbard, 1973; Satterthwaite, 1975) can be applied to blockchain consensus to analyse how non-objective miners behave when they seek to advance a non-canonical version of the system's history.

Arrow, K. J. (1950). A difficulty in the concept of social welfare. *Journal of Political Economy*, *58*(4)

Context

Many researchers have discussed blockchain systems as foundations for democratic self-governance (Cila *et al.*, 2020; Mukhametov, 2020; Poupko *et al.*, 2019; Racsko, 2019; Tozzi, 2019), some emphasising their suitability for democratic or participatory decision-making (Wright & De Filippi, 2015). Indeed, it appears reasonable to assume that systems that are resistant to bad actors attempting to take over control while allowing anyone to participate freely would be ideally suited for self-governing communities (Razzaq *et al.*, 2019). Any type of community activity, such as elections (A. Dhillon *et al.*, 2021), trade (Wongthongtham *et al.*, 2021), or the administration of the commons (Rozas, Tenorio-Fornés & Hassan, 2021), could be supported by such systems. However, despite the 15-year history of blockchain technology, there is little evidence that it is being used in this way (Cengiz, 2023; DuPont, 2017; Ziolkowski *et al.*, 2020), with governance challenges remaining widespread (Rikken *et al.*, 2019). On the contrary, there is reason to suspect that public blockchain technology, especially in conjunction with cryptocurrencies, might replicate existing social disparities (Chohan, 2021; Walsh, 2021) or lead to centralisation of power (K. Heo & Yi, 2023; N. Leonardos *et al.*, 2020; Sai *et al.*, 2021). An important research gap is, therefore, to analyse whether democratic self-governed systems are implementable, and simply haven't manifested yet, or whether they cannot exist. Addressing this research gap is the purpose of our work.

Previous Work

Douceur (2002) introduced the term *Sybil attack* in a paper of the same name. In Lemma 2 of that paper, it is shown that in a distributed system in which participation is free, an attacker may present arbitrarily many Sybil identities. It is stated informally that in such a free system without trusted, centralised authority, Sybil attacks are always possible.

Common Sybil Attack Mitigation Strategies

Z. Yang *et al.* (2014) show that Sybil attacks are not merely a theoretical threat but are observable in existing networks. The most prevalent Sybil attack mitigation strategies for blockchains are PoW, defined by Golle *et al.* (2003) as 'primitives which enforce either high communication or high storage complexity on some party', and PoS: 'mechanisms that extend voting power to the stakeholders of the system' (Bentov *et al.*, 2016).

Bentov, I., Gabizon, A., & Mizrahi, A. (2016). Cryptocurrencies without proof of work. J. Clark, S. Meiklejohn, P. Y. Ryan, D. Wallach, M. Brenner & K. Rohloff (Eds.), *Proceedings of the 20th international conference on financial cryptography and data security.* Springer

Reputation Systems for Sybil Attack Mitigation

Beyond these, a common Sybil Attack mitigation strategy is the application of reputation systems, in which the relationships between participants in a system are used to derive a measure of trustworthiness for participants that, in turn, determines how much influence these participants have in collaborative tasks (Seuken & Parkes, 2014). In particular, the work of H. Yu *et al.* (2006), which introduces such a protocol by building on established trust relationships, predates blockchain technology. Agent-based analysis of reputation systems for Sybil attack resistance has been undertaken by Platt and McBurney (see chapter 6) who find that simple Sybil attacks can be repelled by reputation systems. This result is

Seuken, S., & Parkes, D. C. (2014). Sybil-proof accounting mechanisms with transitive trust. *Proceedings of the 2014 International Conference on Autonomous Agents and Multi-Agent Systems* in line with findings by Seuken and Parkes (2011, Theorem 3), who show that systems may achieve 'K-sybil-proofness', i.e. resistance against up to K Sybil identities. However, they further prove that reputation systems alone cannot yield a fully Sybil-proof decentralised system.

Physical-World-Linking to Address Sybil Attacks

An alternative mitigation strategy is that of *physical world linking* (K. Zhang *et al.*, 2014), in which information from the physical world, such as sensor readings, is used to ensure the individuality of participants (K. Zhang *et al.*, 2014; Y. Zhang *et al.*, 2006). As discussed well before the era of blockchain (Yee & Tygar, 1995), such an approach commonly requires trusted hardware, in which case it needs to be considered quasi-permissioned, with trust being anchored in the maker of said trusted hardware. Furthermore, implementations of physical world linking can be vulnerable to attacks in which groups of attackers act in concert, combining device signals to circumvent protocols.

Zhang, K., Liang, X., Lu, R., & Shen, X. (2014). Sybil attacks and their defenses in the internet of things. *IEEE Internet of Things Journal*, *1*(5)

Contribution

We formalise the notions of *free* and *Sybil attacks* for blockchain systems. In Proposition 7.1.2, we formalise and prove the statement by Douceur (2002, p. 254), that in a free system without trusted authority, Sybil attacks are always possible. Note that a correct formalisation of *without trusted authority* must also exclude other, potentially very complicated, organisational structures, such as consortia or consensus systems that obfuscate centralisation. We achieve this, for blockchain systems only and not for distributed systems in general, in our definition of *(strongly) permissionless*.

7.1 Formalising Blockchain Systems

In this section, we will formally define a model for blockchain systems and will rigorously define what it means for such a system to be permissionless, Sybil attack resistant, and free. In the end, we will show that, in our definition of a blockchain system, there can be no system satisfying all three properties.

This is similar to classical theorems in social choice theory: Arrow's impossibility theorem states that there is no voting system satisfying three distinct fairness criteria (Arrow, 1950); the Gibbard–Satterthwaite theorem states that voting systems are susceptible to tactical voting if there is more than one voter and more than two options to choose from (Gibbard, 1973; Satterthwaite, 1975). In these two cases, as in our application, three desirable properties for systems are defined, and it is shown that no system can satisfy all three simultaneously.

Formal Model

We approach decentralised systems from a transactional perspective: we assume that the system state is derived from a set of temporally ordered *actions* that comply with the rules of the underlying system protocol and form a *history*. Actions can change the system from one state to another, possibly identical, state. We assume them to be instantaneous and deterministic, and they are taken at discrete time steps. This aligns with common blockchain implementations, in which transactions are sequentially ordered by compiling them into blocks (S. Gupta & Sadoghi, 2019). We assume that, for any blockchain system, rules are in place that allow it to be determined whether an action is in accordance with the underlying protocol. Typically, protocols require that the content of every block satisfies some cryptographic property; for example, that it contains a hash of the content of the previous block. The definition of 'action' is intentionally abstract to encompass all rule-compliant measures a participant in a blockchain network might initiate. Common actions under existing protocols are to propose transactions and to compile transactions into blocks.

Gupta, S., & Sadoghi, M. (2019). Blockchain transaction processing. S. Sakr & A. Y. Zomaya (Eds.), *Encyclopedia of big data technologies*. Springer

Definition 7.1.1

(a) *Let A be a non-empty set of actions, containing an element* none $\in A$. *none is called* no action. *Let*

$$\mathrm{HIST}_0 \subset \left\{ f \;:\; \mathbb{N}\times\mathbb{N} \to A \;:\; \begin{array}{r} \text{ex. at most finitely many } s \in \mathbb{N} \\ \text{with: } \exists t \in \mathbb{N} \text{ s.t. } f(s,t) \neq \textit{none} \end{array} \right\}$$

and $\mathrm{HIST} \subset \mathrm{HIST}_0$ *be defined as those histories in which every action is in accordance with the underlying protocol.* HIST *is called the set of possible histories of the protocol. An element* hist $\in$ HIST *is called* finite *if $f(s,t) \neq$ none for at most finitely many $(s,t) \in \mathbb{N}\times\mathbb{N}$. For $(s,t) \in \mathbb{N}\times\mathbb{N}$ we call s a* participant *and t a* point in time.

(b) *Let B be the set of all possible blocks. Then B is contained in A in the sense that for every block B there is the action of proposing it. An element* hist $\in$ HIST *is a collection of actions, and it defines a tree of blocks in which each branch obeys the rules of the protocol. This tree is called* blockchain defined by hist.

(c) *A decision function is a function*

$$\mathrm{dec} \;:\; \mathrm{HIST} \to \{(b_1, b_2, \ldots, b_k) \in B^k \text{ for any } k \geq 0\} \cup \{\emptyset\}$$

satisfying the following property: if $\mathrm{dec}(\mathrm{hist}) = (b_1, b_2, \ldots, b_k)$, *then* $(b_1, b_2, \ldots, b_k)$ *must be a branch in the blockchain defined by* hist. *If* $\mathrm{dec}(\mathrm{hist}) = \emptyset$ *we say that* dec *makes no decision for the history* hist.

A minor point is that, in blockchain systems, views on what is the current state of the system may differ between system participants for short timeframes (e.g. due to replication delays). In our simplified model, we assume that all participants have an aligned view of all actions. The difficulty that remains is that participants may be presented with a history exhibiting multiple branches, each of which is in accordance with the underlying protocol. Decision functions, as illustrated in Figure 7.1, allow participants to resolve such conflicts.

Commonly, system protocols employ decision functions that apply probabilistic techniques to this problem, such as the 'longest chain' rule in Bitcoin or the Ethereum fork-choice algorithm that takes into account the weight of a branch. In some scenarios, a decision function may select none of the existing branches as correct; for example, where multiple branches of equal length exist in Bitcoin. In our formalism, the decision function in this case would return $\emptyset$ instead of a preferred branch.

Zorn's lemma (Kuratowski, 1922; Zorn, 1935) may be used to construct decision functions. However, it is never needed if only deciding on finite

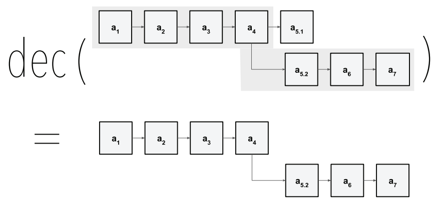

Figure 7.1: A decision function dec takes an ambiguous history (e.g. one with forks) as input and derives a canonical history as output.

histories (which is the only case considered in this chapter), and additional choices would need to be made if deciding on infinite histories.

Definition 7.1.2 (Weakly Permissionless) *A protocol is called weakly permissionless if for all finite* hist $\in$ HIST *there exists a branch* $(b_1, \dots, b_k) \in B^k$ *such that for any future behaviour of the participants in* hist *and for any participant* s^* *not in* hist*, there exists a continuation* $\widetilde{\text{hist}}$ *of* hist *defining a branch* $(b_1, \dots, b_{k+n})$ *such that for all legal blocks* b^* *we have*

$$\text{dec}(\widetilde{\text{hist}} + (s^*, b^*)) = (b_1, \dots, b_{k+n}, b^*).$$

Here, $\widetilde{\text{hist}} + (s^*, b^*)$ *denotes the history that is equal to* $\widetilde{\text{hist}}$ *but has one additional action added to it, namely the action of a participant* s^* *to propose the block* b^* *at time* 1 *after the last action in* $\widetilde{\text{hist}}$*.*

Informally, this means the following: hist is the state of the system of a blockchain at a given time, including all actions that led to its current state, potentially including forks. If the system is weakly permissionless, then a new participant can get some future block b^* accepted by the decision function, and no one can stop them. If there is even a small chance that the new participant cannot get their block accepted, no matter their actions, then that protocol would not be weakly permissionless. Of course, we do not expect that the new participant can get all blocks they want accepted: before their block b^* is accepted, there may come other accepted blocks $b_{k+1}, \dots, b_{k+n}$ that were not suggested by the new participant. There are no conditions on the content of b^*, except that it must be legal according to the blockchain protocol.

Definition 7.1.3 (Strongly permissionless) *A protocol is called strongly permissionless if for all finite* hist $\in$ HIST *and for any future behaviour of the participants in* hist*, there exists an infinitely long continuation* $\widetilde{\text{hist}}$ *of* hist *and a positive integer* N_0 *such that the following is true: for all* $N > N_0$*, the majority of accepted blocks in* $\widetilde{\text{hist}}$*, cut off at time N, have been proposed by participants who did not have any activity in* hist.

In contrast to *weakly* permissionless protocols, *strongly* permissionless protocols are characterised not only by allowing new participants to *occasionally* propose transactions but by, eventually, allowing them to propose a *majority* of transactions.

The adherence of Bitcoin, the archetypal strongly permissionless protocol, to Definition 7.1.3 can be illustrated as follows: hist $\in$ HIST could be the current state of the Bitcoin blockchain. This includes the full blockchain, including forks. A new participant s^* can immediately propose any legal block a^*, containing the required PoW, to be added to the longest chain

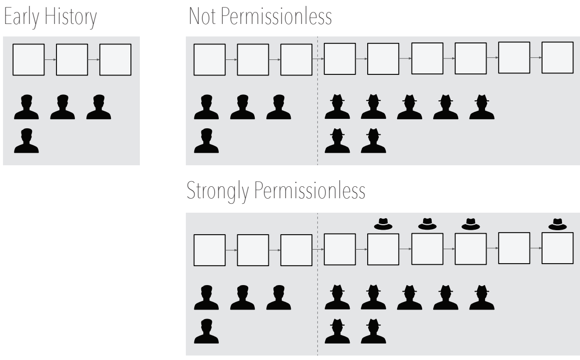

Figure 7.2: Left: a blockchain created by some participants. Top right: an example of a blockchain that is not weakly permissionless and not strongly permissionless. A large number of new participants (shown with a hat) joined but they are not allowed to create blocks. All blocks going forward are created by the original participants. Bottom right: an example of a blockchain that is strongly permissionless. New participants joined and are creating a majority of new blocks (shown with a hat) going forward, which cannot be prevented by the original participants.

which will be accepted by the decision function dec, the 'longest chain' rule. In this case $n = 0$. That is, the participant s^* does not need to take any actions to improve their own standing in the protocol, since a correct PoW immediately entitles them to propose a block. Given appropriate resource expenditure, the participant could propose new blocks in perpetuity. This is illustrated in Figure 7.2: in the strongly permissionless system shown, a majority of new blocks were created by new participants.

Definition 7.1.4 (Sybil attack vulnerable) *A system with finite history* hist *is vulnerable to Sybil attacks if for any behaviour of the participants in* hist, *there exist arbitrarily long continuations* $\widetilde{\text{hist}}$ *of* hist *such that a majority of accepted blocks in* $\widetilde{\text{hist}}$ *have been proposed by participants not in* hist *and these participants have expended only negligible explicit monetary cost.*

In the context of decentralised systems, an attack is understood as the realisation of a threat, representing a harmful action aimed at exploiting vulnerabilities within the system (Paulauskas & Garsva, 2006). A type of attack that has received particular attention in the blockchain community is the Sybil attack: Informally, in accordance with Douceur (2002, p. 251), we define a Sybil attack on a decentralised system as a series of protocol-conforming actions through which an attacker presents multiple identities with the intent of seizing control of a substantial fraction of the system. Attackers achieve the seizure of control by skewing collective decisions, such as voting, in their favour. A potential result of a successful Sybil attack on a strongly permissionless system is the attacker increasing the probability of proposing new blocks, thereby being able to censor block contents and, ultimately, centralising power over the system. It is intuitively clear that a Sybil attack is only possible if a malicious actor can take many actions while expending only negligible explicit monetary cost. In accordance with Saleh (2020, p. 1157), we formally define this as:

Definition 7.1.5 (Free) *A system is* free *if all actions can be taken by incurring only negligible explicit monetary cost.*

Notably, staking requirements common in cryptocurrencies, which are perceived not as monetary costs but as commitments for potential financial gain, are excluded by this definition.

An Impossibility Theorem for Sybil Attack Resistance

We define three properties of interest for blockchain systems: permissionless in Definition 7.1.2 and Definition 7.1.3, Sybil attack vulnerable in

Definition 7.1.4, and free in Definition 7.1.5. Here is how these notions are related to each other:

Proposition 7.1.1 *Any Sybil attack vulnerable system is strongly permissionless.*

Proof. Definition 7.1.3 and Definition 7.1.4 are the same, except that Sybil attack vulnerability has an additional restriction that is not present in the definition of strongly permissionless. ☐

Most importantly, there is no system that satisfies all three requirements:

Proposition 7.1.2 (Impossibility Theorem for Sybil Attack Resistance) *A blockchain protocol cannot be strongly permissionless, Sybil attack-resistant, and free.*

Proof. By definition, a system that is *free* and *strongly permissionless* must be *Sybil attack vulnerable*. Therefore, such a system cannot be *Sybil attack-resistant*, which illustrates the claim. ☐

7.2 Illustration of the Trilemma

In consideration of Proposition 7.1.2, an analysis of existing blockchain systems has been performed for illustrative purposes, acknowledging that the described Trilemma may not encompass all aspects of these real-world systems in their entirety. To this end, we selected two particularly popular blockchain systems with associated cryptocurrencies to illustrate common combinations of characteristics, potentially extending beyond the immediate scope of defined assumptions. Both systems are representative of many others that use PoW (see subsection 7.2) or PoS (see subsection 7.2) in similar ways. The remaining combination – strongly permissionless and free but not Sybil attack resistant – is rare, as systems with these characteristics can be disrupted as a result of targeted Sybil attacks and are, therefore, mostly of theoretical interest. To shed light on this combination of characteristics, we selected a technique that has so far been discussed only theoretically (see subsection 7.2). Figure 7.3 and Table 7.2 show the different possible combinations, visualising Proposition 7.1.2.

Figure 7.3: Diagram showing how free, Sybil attack vulnerable, and strongly permissionless blockchain systems are related. By Proposition 7.1.1, Sybil attack vulnerable systems are a subset of strongly permissionless systems. By Proposition 7.1.2, the intersection of the two circles 'Free' and 'Strongly Permissionless' and the complement of 'Sybil Attack Vulnerable' is empty.

Sec.	System	Mechanism	SP	SAR	F
7.2	Bitcoin	Proof-of-Work (PoW)	∘	∘	
7.2	Ethereum 2	Proof-of-Stake (PoS)		∘	∘
7.2		Proof of lucky ID (PoL)	∘		∘

Table 7.2: A comparison of common blockchain systems. The ∘ symbol indicates whether they meet the definition of strongly permissionless (*SP*), Sybil attack-resistant (*SAR*), or free (*F*).

Proof of Work

Proof-of-Work (PoW) is the archetypal approach to preventing Sybil attacks. While it predates Bitcoin (Narayanan & Clark, 2017), its use in the context of decentralised networks dates back to this cryptocurrency (Nakamoto, 2008). Strong permissionlessness is central to Bitcoin's design, in which messages are exchanged on a best-effort basis between nodes that are able to leave and join the network at will (Nakamoto, 2008). Arbitrary nodes can participate in the consensus mechanism in a censorship-resistant manner that can recover from network partitions. Nakamoto (2008) motivates the use of PoW by emphasising that it enables *one CPU, one vote* (Nakamoto, 2008, p. 3) mechanics that, under the assumption that 'a majority of CPU power is controlled by honest nodes' (Nakamoto, 2008, p. 3), can secure the existence of the Bitcoin system in perpetuity. While attackers can easily create arbitrary numbers of accounts, PoW prevents these from interfering with consensus. However, despite the potential of block rewards offsetting the costs, participation in Bitcoin's consensus mechanism is costly, since it requires significant computational effort and, consequently, physical resources to provide it.

Narayanan, A., & Clark, J. (2017). Bitcoin's academic pedigree. *Queue*, *15*(4)

Proof of Stake

To the best of our knowledge, the first mention of Proof-of-Stake (PoS) occurred in Bitcoin circles: here, it was proposed to align the influence in majority-based activities with 'the number of bitcoins you can prove you own' (QuantumMechanic, 2011). The most popular implementation of this concept is Ethereum, conceived as a PoW system (Buterin, 2014) and recently successfully transitioned to PoS during the Ethereum 2 merge (Kapengut & Mizrach, 2023). Note that the restrictiveness of Definition 7.1.3 excludes Ethereum 2: the current Ethereum 2 stakeholders possess the ability to collude and effectively bar external participation by virtue of the requirement to deposit cryptocurrency into the protocol's deposit contract to qualify as validator. Consequently, those without access to the Ethereum cryptocurrency are perpetually excluded from the validator pool, making them unable to propose blocks. Ethereum's PoS system exhibits strong Sybil attack resistance: similar to other blockchain systems, users can create arbitrary accounts for the Ethereum 2 blockchain. However, due to the prerequisite of staking cryptocurrency to act as a validator, unfunded accounts would not be selected for participation in some network activities. Participation in Ethereum 2 (including in the role of validator) is free according to our unrestrictive Definition 7.1.5, which ignores transaction fees and cryptocurrency staking requirements.

Proof of Lucky ID

Ogawa *et al.* (2018) introduced proof of lucky ID (PoL), a mechanism that can be used to illustrate the class of strongly permissionless and free systems that are not Sybil attack-resistant. In PoL, a miner is selected pseudorandomly using their ID. We assume a simplified ID supply mechanism that allows users to self-generate IDs[3]. The random selection of nodes from the pool of participants can be considered an approach that achieves strong permissionlessnes. Depending on the ID supply mechanism, this mechanism is, however, highly susceptible to Sybil attacks: where a self-generated ID is used (Ogawa *et al.*, 2018, p. 1216) and no further restrictions on ID generation are made, an attacker can easily conduct a Sybil attack by creating many IDs, thereby increasing the likelihood of being randomly selected. Note that it is conceivable to alter the protocol by applying other ID supply mechanisms that are less free or less permissionless, but may in turn lead to Sybil attack resistance. Under the assumption of a free ID supply mechanism, PoL can be considered equally free.

3: This configuration is somewhat artificial and likely not in line with the intentions of Ogawa *et al.* (2018). It is used here solely to illustrate the class of non-Sybil attack resistant systems.

7.3 Conclusion

Blockchain systems that are *permissionless* (i.e. allow public participation without limitations), *resistant to Sybil attacks* (i.e. not vulnerable to attackers that skew collective decisions), and *free* (i.e. require expenditure of only negligible explicit monetary cost) are highly desirable for many applications; notably, for democratic self-governance. However, there is reason to believe that these three properties cannot be achieved in a blockchain system simultaneously. Therefore, we raised the question of proving their compatibility.

We have shown that no blockchain system can concurrently achieve the three desirable characteristics defined within the framework outlined in section 7.1, leading to the negative conclusion of this question within the defined paradigm. Furthermore, we categorised existing popular blockchain systems by the set of desirable characteristics they exhibit. For example, Bitcoin is *strongly permissionless* and *Sybil attack-resistant* but not *free*, whereas, Ethereum 2 is *Sybil attack-resistant* and *free* but not *strongly permissionless*. These findings have implications for the designers of future blockchain systems, who can apply them to address the tradeoffs between characteristics more consciously. A more deliberate treatment of these tradeoffs may enable systems that focus on two of the three dimensions and, thereby, achieve a better problem-solution fit. Our results also help to critically evaluate common claims that certain protocols are free and permissionless. A noteworthy limitation of our work is that the definition of *permissionless* is restrictive and excludes some systems that have been labelled permissionless by others, and this should be considered when interpreting this work.

Understanding whether given blockchain applications deliver on the promise of decentralisation remains a central challenge for the credibility of this emerging technology. While it was shown that systems that simultaneously exhibit all three desirable characteristics cannot exist under our formalisation, we recognise that blockchain technology offers great potential for applications, in self-governance and beyond, if the technological trade-offs are understood, managed, and communicated.

Conclusion 8

This thesis focuses on evaluating the hypothesis that blockchain technology can facilitate democratic self-governance. Recognising that consensus mechanisms are crucial for the democratic potential of a system, we undertook a systematic review of the consensus mechanism literature. We found that mechanisms with strong resistance to Sybil attacks, built on PoW or PoS, dominate the field, while other, less robust, mechanisms are applied to specialised use cases. Focusing on PoW, we examined its operational properties (foremost, its energy consumption) and suitability for self-governance. Our analysis revealed that, while PoW is effective against Sybil attacks, its significant energy demands make its suitability for democratic governance questionable. This finding emphasised the need for alternative consensus mechanisms (e.g. PoS) that could offer similar security benefits without the emissions associated with PoW. This led us to investigate users' awareness of the electricity consumption of consensus mechanisms. We developed a mathematical consumption model and applied it to various blockchains, confirming that the energy footprint of PoW is, indeed, substantial. Our findings highlighted the need to transition to more energy-efficient mechanisms, such as PoS, which have smaller energy footprints.

This reflects an ongoing trend in public blockchain systems, exemplified by the move from PoW to PoS by Ethereum. Building on this understanding, we explored whether cryptocurrency users would prefer less energy-intensive systems, if adequately informed about energy consumption characteristics. As a result of this exploration, we show that increased awareness of blockchain energy demand correlates with greater support for sustainability measures. This suggests that consumer education, i.e., making the technology properties of a given blockchain visible to potential users, could contribute to the adoption of novel blockchain practices.

In response to these findings, and to explore whether a more democratic consensus mechanism could be established, we proposed a novel consensus mechanism, IdAPoS, based on a reputation system. This mechanism was developed by combining the strengths of PoS with the democratic potential of reputation systems, with the ultimate goal of creating a consensus mechanism that is both energy-efficient and supports self-governance. Conceptually, the design showed promise to achieve this goal, indicating that IdAPoS could be a building block for more democratic blockchain systems. To validate IdAPoS, we simulated Sybil attacks of different magnitude using an agent based model. We found that IdAPoS could withstand attacks under some simplifying assumptions, namely, early detection and the identifiability of attackers. This demonstrated the potential of IdAPoS to create secure and democratic permissionless systems.

However, a deeper investigation with more restrictive assumptions revealed a fundamental challenge: the capability of decentralised systems to simultaneously achieve permissionlessness, Sybil attack resistance, and freeness. Our formal evaluation led to the conclusion that no blockchain system can simultaneously satisfy all three criteria, which presents a trilemma that systems designers must navigate. Through this exploration, we have answered the central research question 'Can reputation systems enable "one person, one vote" characteristics in permissionless decentralised systems?' in the negative.

Initially, we were optimistic that, in the absence of an effective consensus mechanism for permissionless democratic self-governance, IdAPoS might represent an unprecedented mechanism for democratic self-governance. However, the methods applied in this thesis revealed limitations: agent-based simulation showed a positive result only under simplifying assumptions. Ultimately, the proposed Sybil attack vulnerability trilemma demonstrated the incompatibility of democratic self-governance with permissionless systems (at least within our formalisation). The findings of this research show the obstacles involved in creating genuinely democratic blockchain systems.

8.1 Hypotheses

This thesis has examined the potential for democratic self-governance in permissionless decentralised systems by testing four primary hypotheses (see section 1.4).

Hypothesis 1 (*PoW consumes significantly more electricity than PoS.*) was confirmed by the mathematical model presented in chapter 3. This shows that PoW-based consensus mechanisms consume significantly more electricity when compared to PoS systems. The proposed model makes visible the significant energy use of PoW, caused by the power needed for mining processes.

Hypothesis 2 (*Users who are aware of the electricity consumption of PoW systems support measures to reduce their energy usage.*) was confirmed through survey research detailed in chapter 4. The survey results indicate that users who are informed about the high electricity consumption of PoW systems support measures aimed at reducing their energy consumption. This shows a positive correlation between awareness of electricity consumption and support for sustainable practices.

The work presented in chapter 6 confirmed hypothesis 3 (*PoS can model a reputation system, serving as the foundation for a democratic consensus mechanism independent of proof of resources.*) via the IdAPoS consensus mechanism. IdAPoS effectively integrates reputation systems within a PoS framework, demonstrating the feasibility of achieving a reputation-based consensus mechanism that does not rely on resource-intensive proofs.

Hypothesis 4 (*Using reputational signals as the foundation for a consensus mechanism enables a decentralised system to be permissionless, resilient against Sybil attacks, and free.*) was rejected by formal analysis as carried out in chapter 7. The analysis showed that, while reputational signals can improve the stability of decentralised systems, they are inadequate to simultaneously achieve permissionlessness, resistance to Sybil attacks, and the absence of explicit monetary costs. The formalisation highlighted inherent trade-offs, illustrating that it is impossible to simultaneously achieve all three properties at the same time.

8.2 Limitations

Even though this thesis conclusively answers the research question, it is subject to a number of constraints. The literature review, although comprehensive, does not cover emerging mechanisms[1]. Our focus on Sybil attacks meant that other attacks, such as 51 % attacks, double-spending, and

1: Works indexed after April 2021 were not considered.

smart contract vulnerabilities, were not analysed beyond chapter 2 subsection 1, despite their potential impact on permissionless systems. In addition, the energy model used in this thesis is based on available data, which may not fully reflect real-world conditions, considering the impact of hardware advancements, geographic distribution, and the market dynamics of blockchain networks. Also, the surveys conducted in chapter 4 and chapter 5 were confined to Nigeria and Romania, respectively, and lacked representative sampling, which reduces the generalisability of their results.

Certain areas of concern were ignored to maintain focus: the thesis focused on public, permissionless blockchains, excluding private and consortium blockchains, which have different security and governance paradigms. To focus on technical aspects, legal and regulatory issues were largely omitted from the analysis, although they significantly impact blockchain adoption. Issues related to interoperability between different blockchain systems were also not covered, despite their importance in the context of self-governance.

8.3 Future Work

The results, especially the rejection of hypothesis 4 (reputational signals do not enable decentralised systems to be permissionless, resilient to Sybil attacks, and free at the same time), provide numerous insights for future research. Since it is a common intuition that reputation systems can be used for consensus in decentralised systems, future work should first address *existing* systems. Subsequently, the knowledge gained in this thesis should be utilised to aid the design of *new* systems.

Vulnerability Analyses of Real-World Systems

As shown in chapter 2, many real-world systems apply reputation systems as a prominent component of their consensus mechanism (e.g., de Oliveira *et al.*, 2020; F. Gai *et al.*, 2018; K. Lei *et al.*, 2018). The assumptions applied, however, may be different to those in chapter 7, which means that our findings do not necessarily indicate that those systems are vulnerable. Nonetheless, investigations into the vulnerability of existing systems to Sybil attacks are strongly recommended. Such investigations may include analysing the risk of Sybil actors being admitted to the system (i.e., the preconditions which a given protocol establishes for obtaining an identity on the network), whether restrictions of Sybil identity growth are in place, and whether a given protocol is equipped to effectively revoke access of Sybil actors. Bringing the danger of Sybil attacks to the surface may make the technology risk of blockchain applications in investment contexts visible, which may aid further mainstream adoption of the technology (Azgad-Tromer, 2018).

Azgad-Tromer, S. (2018). Crypto securities: On the risks of investments in blockchain-based assets and the dilemmas of securities regulation. *American University Law Review*, *68*(1)

Considerably more work will need to be done to determine how interoperability between different networks with different Sybil attack resistance characteristics affects the trustworthiness of bridged assets. This is relevant not only in light of possible failures of bridges (Q. Zhao *et al.*, 2023) but also considering the trustworthiness of assets moved over trustless bridges, themselves a nascent area of research (T. Xie *et al.*, 2022).

Where networks operate using consensus backed by reputation systems, real-time vulnerability detection frameworks, leveraging AI/ML to predict potential Sybil attack vectors may be useful. Such systems have shown promise in the context of conventional cyber-attacks such as unauthorised access and data breaches (Ajala *et al.*, 2024). Investigating the effectiveness of such systems, potentially using agent-based techniques, as proposed in chapter 6, could produce relevant findings.

Guiding Systems Designers to the Right Consensus Mechanism

It is essential that a comprehensive framework is developed to assist in selecting appropriate consensus mechanisms for different use cases. Future studies should focus on creating methodologies that consider the specific needs and constraints of various decentralised applications, ensuring that the chosen consensus mechanism aligns with the desired outcomes and operational environments. Many use cases, especially those without public accessibility requirements (Platt, Bandara *et al.*, 2021), may not require a high degree of Sybil attack resistance and might, therefore, be served by permissioned systems (Capocasale *et al.*, 2023), which can apply central admissions processes to counteract Sybil attacks. Where systems designers require public/permissionless properties, PoW and PoS appear to be the archetypal tools at their disposal (see chapter 2), each with unique advantages and disadvantages along the dimensions of energy consumption, security, operating speed, decentralisation, scalability, and reward mechanisms (Pan *et al.*, 2021). A natural progression of this work is, therefore, to analyse hybrid consensus systems that combine elements of multiple mechanisms (e.g., combinations of PoS and PBFT) to optimise for trade-offs around security, decentralization, or scalability.

Quantification of Sybil Attack Resistance of Existing Schemes

Future research should aim to develop a quantification of Sybil attack resistance, which may be based on the concept of K-Sybil-proofness, introduced by Seuken and Parkes (2011).[2] This would provide a sliding scale of Sybil attack resistance, denoting, for example, the required effort an attacker would have to make to execute a successful Sybil attack. Such a framework should encompass metrics and methodologies to evaluate various consensus mechanisms under Sybil attacks of different intensities. This would mirror common cybersecurity metrics focused on protection mechanisms, such as penetration resistance (Gupta Bhol *et al.*, 2023). Such quantification may guide practitioners to navigate trade-offs between Sybil attack resistance and other non-functional aspects of blockchain systems (e.g. latency, throughput, deployment size, or energy demand).

2: The concept of 'K-Sybil-proofness' refers to a system's ability to remain secure and robust even when an attacker creates up to K Sybil identities.

The Sybil Attack Vulnerability Trilemma: A Call for Permissioned Systems?

Finally, permissioned networks with suitable off-ledger governance structures should be researched more deeply as an alternative to permissionless governance. Such research may involve studying how established governance frameworks can map to blockchain systems to improve security, efficiency, and democraticness. In addition, the legal and regulatory dimensions of permissioned systems might prove an important area for future research. Here, the regulatory implications of deploying permissioned blockchain systems in sectors that require stringent compliance (e.g., Finance and Health) should be investigated.

As Wüst and Gervais (2018) describe in the seminal paper 'Do you Need a Blockchain?', permissioned networks have advantages over permissionless ones where entities interact without mutual trust but need control over participation and privacy. However, determining when *permissioned decentralised systems* are more appropriate than *centralised systems* is non-trivial, as suitability depends on nuanced factors such as the availability of a trusted third party, trust in writers, and the need for public verifiability. Guidance is lacking on when a fully Sybil-attack resistant setup, vs. a permissioned network, vs. a centralised system (potentially with immutable characteristics Lupaiescu *et al.*, 2022) is most suitable: many industry initiatives apply a permissioned blockchain (Polge *et al.*, 2021) without a clear problem/solution fit, making this gap a promising avenue for future work.

Wüst, K., & Gervais, A. (2018). Do you need a blockchain? *Proceedings of the 2018 Crypto Valley Conference on Blockchain Technology*

In scenarios where permissioned systems are considered appropriate, there are different ways to anchor trust. Here, 'anchoring trust' refers to the introduction of a central entity that acts as a trust anchor within an otherwise decentralised system. Such an entity plays a key role in securing the network by verifying and controlling access. Specifically, in the context of Sybil attacks, anchoring trust could involve designing an admissions process in which only users approved or verified by the trust anchor are allowed to participate in the system's consensus mechanism.

The State as Trust Anchor

Governments have engaged in decentralised network design on many occasions, as evidenced by national blockchain strategies emerging around the world[3] and numerous government-led blockchain projects (Hussain, 2020; Leteane & Ayalew, 2024). Governments have also launched CBDCs, that resemble decentralised cryptocurrencies, such as the 'Sand Dollar' and the 'e-Naira'. Despite the limited adoption of the latter (Dowd, 2024), the possibility of the state acting as a trust anchor for decentralised systems warrants further investigation. We believe that, apart from designing systems that are centrally controlled by a state or government and therefore antithetical to the goal of the 'Cypherpunk' movement–the ideological home of blockchain technology (Beltramini, 2020)–future research should investigate opportunities to incorporate government identity assertions in network consensus. A promising development which may constitute the object of future studies is the introduction of electronic identity wallets (Podgorelec *et al.*, 2022), such as the 'EU Digital Identity Wallet' (European Parliament, 2024). Such sovereign initiatives may generate electronic credentials that credential holders are free to utilise in any form desired, without the knowledge of the issuer. This means that such credentials could be used as forms of authorisation required for validators of a network, thereby removing the risk of Sybil attacks. However, the trust assumptions and the inherent risk of centralisation around a particular credential issuer remain to be researched.

3: The British Blockchain Association (2024) names the United Kingdom, India, Germany, the United Arab Emirates, Ireland, Estonia, Australia, Bangladesh, Qatar, Malaysia, and Nigeria as countries with a 'national blockchain roadmap'.

Podgorelec, B., Alber, L., & Zefferer, T. (2022). What is a (digital) identity wallet? a systematic literature review. *Proceedings of the 46th Computers, Software, and Applications Conference*

The Commercial Sector as Trust Anchor

Equally in conflict with the original goals of the 'Cypherpunk' movement (Beltramini, 2020), but, nonetheless, an interesting topic for future work, is the idea of using the commercial sector, especially corporations, as trust anchors. Future research should examine if applying mechanisms of corporate governance, beyond the concept of 'one share, one vote' (Ratner, 1970) discussed earlier (see chapter 6) in consensus mechanism design would allow an increase in a system's robustness.

Furthermore, the impact of commercial incentives on trust should be investigated. Further investigation is warranted into how the commercial incentives of private enterprises (e.g., profit motives, market dominance) influence their role as trust anchors and the checks and balances that can be introduced to ensure they act in the public interest. In addition, the governance models of corporate consortiums operating decentralised networks might prove an important area for future research. Here, there is a need to evaluate how consortia can maintain fair governance models that align with decentralised values while navigating the tension between trustless collaboration (e.g. via smart contracts and DAOs) and the desire for organisational trust that can be observed in some blockchain consortia (Zavolokina *et al.*, 2020).

Scrutinising the Need for Permissioned Blockchain

Future research should focus on developing frameworks to help decision-makers assess when to opt for a centralised system rather than a private/permissioned decentralised one. Given that the use cases for permissioned blockchains often appear narrow, and centralised systems frequently offer more efficient solutions, there is a growing need for clear, evidence-based guidance. Ultimately, such work has the potential to help practitioners establish problem-solution fit and shift the popular perception of decentralised technology as a 'solution in search of a problem' (Tochen, 2019) to one where it becomes a legitimate tool that can be used for good.

Tochen, D. (2019). Blockchain technology: A solution in search of a problem—or a revolution? *The Brief*, *49*(1)

8.4 Final Thoughts

This thesis adds to the foundation of research in decentralised governance. It does so by integrating theory, empirics, and simulations. Its results can be applied to further the development of systems in democratic contexts. However, interdisciplinary dialogue will be required to create systems that are not only technically sound, but *fit for purpose*: may this thesis contribute to this goal.

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